

Close Statistical Reading:

Rhetorical Interpretation of Computational Analyses
of Texts at Scale

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Abstract

In recent years, computational text analysis has become remarkably accurate at text classification, such as in distinguishing texts by categories like genre based upon a text's word frequencies. In rhetorical studies though, something like genre would instead be distinguished in terms of the "social action" that it performs in the world, not differences in word frequencies. And while we might be able to imagine how rhetorical actions could link to specific word frequency patterns, the incongruence between these approaches to concepts like genre raises provocative questions about how rhetorical action is accomplished through recurring linguistic form. However, because many computational methods were developed with a goal of prediction, the research models around them are not necessarily built toward allowing this kind of interpretation – it would be infeasible to closely analyze a large corpus for all of the weighted features used in classification to understand why it works. Theories of textual interpretation are very much at the core of rhetorical studies, and as such, can offer a unique lens on this problem.

Toward this challenge, this dissertation takes a rhetorical approach to large-scale textual interpretation by offering a new hermeneutic, or interpretive theory and argumentative strategy, called "close statistical reading" that allows for defensible and insightful rhetorical interpretation. In developing interpretations of corpora, I argue that the goal is not to produce one "correct" interpretation. Rather, we can interpret tables of data about a corpus not by reading every text and data point, but by strategically analyzing how many different bits of data all point to the same understanding of a corpus. As such, the interpretive theory provides a way of understanding interpretation in a corpus context, but also an argumentative strategy the analyst needs for making the case for their interpretation. Compared to past work on computational hermeneutics, close statistical reading highlights the utility of synthesis of analysis toward understanding; it also takes a rhetorical approach to hermeneutics by foregrounding the argumentative strategies to justify interpretations of large corpora of texts. This new approach thus does not rely only upon the traditional mode of closely reading each text, but is instead conducive to working with large amounts of data. To demonstrate this theory, I offer an extended case study using stylometry methods to classify academic writing by discipline. This classification uses function word frequencies to classify disciplinary writing, and thus asks, where do function word frequencies fit into a conceptualization of disciplines as "ways of knowing and doing"? It synthesizes analyses of particular function words in action through different analytical techniques to develop rhetorical theory about how function words play a constitutive role in the creation of meaning, particularly in the context of academic knowledge representation. In particular, the analysis demonstrates differences between humanities and science writing in their focus on splintered interpretation versus thick description; the range of argumentation in the humanities and sciences from complex narrative to humanistic reasoning, and experimental to problem-solving science; and dimensions of rhetorical variation within computer science and history.

Keywords: interpretation, hermeneutics, corpus analysis, rhetorical analysis, stylometry, writing in the disciplines

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Chapter 1

Introduction

Natural language processing, an area of computer science, has been advancing the state of computational text analysis in ways that are incredibly useful and provocative for rhetorical studies and the humanities more broadly. The range of scholarship in this area is immense, though most frequently is rooted in the ability to classify texts in various ways with high accuracy. For example, machine learning techniques have been able to automatically predict the genre of a text (Stamatatos et al., 2000), an author’s age and political affiliation (Dahllöf, 2012), and even the grade assigned to student essays (Mayfield & Rosé, 2013). Still other techniques have been able to identify the actual author behind books published under pennames (Juola, 2013), or to point to an even more common experience, identify which emails we receive everyday should be marked as “spam” (e.g., Sasaki and Shinnou (2005)).

Looking further into the details of how these methods classify in the ways that they do, oftentimes the goals of analysis are not strongly rooted in a humanistic rhetorical approach. In a review in *Science*, natural language processing is described as “the subfield of computer science concerned with using computational techniques to learn, understand, and produce human language content” (Hirschberg & Manning, 2015, p. 261). In this context, understanding language is primarily application-based (for machine translation, spoken dialogue systems, information mining, etc.), and as such, interpreting language largely refers to the surface-level task of distinguishing the semantic meanings: for example, if mining newspaper headlines for information about trends in education, computers would want to interpret whether the sentence “Teacher strikes idle kids” is meant to indicate that a teacher is assaulting

idle children, or that teachers are on strike and this is boring for children (Hirschberg & Manning, 2015, p. 261).

A humanistic rhetorical approach to language, however, typically focuses on understanding language in more contextually-focused ways. Certainly, theories of language and definitions of rhetoric abound within rhetorical studies, but broadly speaking there is an attention to textual detail that does not necessarily serve applications (except, perhaps, in the teaching of writing), but rather serves some kind of fuller understanding of how people use language to perform actions in the world. For example, though a textual analysis of genre may pay close attention to regular textual form across groups of texts, the prevailing theories of genre focus instead on the “social action” they perform (Miller, 1984), distinguishing genres by the types of work that they do in the world where textual regularities contribute to that. Still, other approaches to rhetorical studies are even less steeped in attention to regular textual detail than genre can be; for example, a rhetorical analysis of Elizabeth Cady Stanton’s oratory is heavily situated in its particular context rather than looking to identify patterns across texts, in that it identifies particular emotional appeals she made when speaking to male legislators about women’s rights (Engbers, 2007).

Practically speaking, these distinctions in approaches to language instantiate themselves in the actual analysis of texts. A large swath of computational text analysis treats texts as “bags of words,” meaning that they see texts as lists of individual words (or perhaps, patterns and structures of those words) with frequencies. Thus, a sentence like, “The dog ran toward the yard.” might be represented as follows by a computer in a “bag of words” approach: [(‘the’: 0.29), (‘dog’: 0.14), (‘ran’: 0.14), (‘toward’: 0.14), (‘yard’: 0.14), (‘.’: 0.14)], where each word is connected to its relative frequency in the text. After computing frequencies, a computer might then be trained to classify. By contrast, that same sentence might be treated any number of ways in a rhetorical analysis – perhaps as a declarative statement; a subject, verb, and prepositional phrase; a narration in the context of a larger conversation. While there is a huge amount of variation and technique in both types of analysis – indeed, some computational text analysis out of areas like corpus linguistics are indeed interested in the humanistic aspects of language – in general, one of the major successes of computational text analysis has been finding remarkable accuracy in classification through these techniques, while rhetorical analysis has found remarkable insight into the roles of language in society.

Natural language processing work thus engages with similar textual concepts that we might engage with in rhetorical studies – genre, authorship, real versus fake news – but the approaches to those concepts can seem worlds apart. While we might be able to imagine how certain types of rhetorical action could link to specific word frequency patterns, the approaches are fundamentally quite different in focus and yet, both have been useful for various purposes. The incongruence between these approaches raises questions and has implications for both the humanities more broadly, as well as rhetorical studies specifically.

1.1 Textual interpretation and big data

Figuring out the place of the computer in the humanities is not a new problem: language scholars have been grappling with the affordances and use of the computer in their disciplines since at least the mid-20th century.¹ In 1965, John Perry Miller opened a conference on computers in the humanities by asking if the computer had a place at all in the humanist’s research (Pier-son, 1965, p. ix); in 2022, though, it clearly has found its place. At the very least, computers function as a word processor tool (though their utility and impacts were debated early on, e.g., Selfe and Wahlstrom (1988)). In recent years especially, computational research has begun making its way into more mainstream humanities journals (e.g., Kaufer and Parry-Giles (2017), Pollak (2021), Underwood and Sellers (2016)). Frequently, issues around computation in the humanities are now taken up in the digital humanities community, though not exclusively. Scholars in digital humanities have a wide range of concerns around using digital tools and data for humanities research, spanning broadly from automated text analysis to GIS data modeling to network analysis (Schreibman et al., 2016).

With computation and the humanities converging to some small degree, issues have arisen about how well quantitative analysis can contribute to traditional humanistic inquiry as well as the interpretability of computational analyses. Because many computational methods were developed with a goal of prediction, the research models around them are not necessarily built toward allowing interpretation. That is, a classificatory approach to texts is likely to produce some kind of weighted list of features, from which we might

¹Some small pieces of this section are drawn from my prior written work (Ringler, in press, 2021)

learn that one word has a slightly better chance than another word at predicting the class of a text. When that table reaches thousands of words and weights long, trained upon hundreds or thousands of texts, making sense of it becomes increasingly difficult. For computer scientists though, research goals have historically been framed in terms of accuracy in prediction, rather than what kind of knowledge we might gain about the things being predicted. In other words, there is not typically a strong interest in *why* this algorithm works well outside of improving its accuracy.² And although machine learning research in recent years has become interested in the problem of interpretability, it is largely considered to be a difficult challenge and there is debate over the worthwhileness, purpose, and feasibility of pursuing it (Blinkov & Glass, 2019; Doshi-Velez & Kim, 2017; Lipton, 2018).

In rhetorical studies, though, interpretation is very much at the core of the discipline. Interpretation, hermeneutics, and the methods of knowledge production more generally have a long and varied history that dates back at least to the ancient Greek Sophists. The history of hermeneutics even just in computational humanities work is too long and complex to treat thoroughly here (see van Zundert (2016) for an excellent review), but in rhetorical criticism more specifically, the human reader has been historically valued as an integral part of hermeneutics. From Richards’s (1929) close reading to Black’s (1978) rhetorical criticism, interpretation of texts has historically involved analysts reading individual texts with different frameworks and processes so as to better understand them and their contexts.

This focus on reading individual texts for interpretation began to be challenged and inched closer to the work occurring in natural language processing with Moretti’s (2000) argument for “distant reading” in literary studies. In this essay, Moretti argued that the vast amount of world literature before us is “a problem that asks for a new critical method” and that simply “reading ‘more’ hardly seems to be the solution” to getting a better grasp on world literatures than individual readings of hopefully-representative texts can allow (p. 55). He writes,

But the trouble with close reading (in all of its incarnations, from

²Computational linguistics has historically taken more of an interest in the details of language itself in regards to these techniques. However, in those cases, the focus of research might be more focused on a thick description of language as it appears, rather than rhetorical interpretation of how texts operate in the world and thus relate to categories like genre.

the new criticism to deconstruction) is that it necessarily depends on an extremely small canon. This may have become an unconscious and invisible premiss by now, but it is an iron one nonetheless: you invest so much in individual texts only if you think that very few of them really matter. Otherwise, it doesn't make sense. (Moretti, 2000, p. 57)

His solution is what he calls “distant reading,” or looking at entire swaths of literature, patchworking together literary historians’ research so as to achieve a further vantage point on the group of texts of interest.

While Moretti did not necessarily theorize corpus text analysis as the methodological solution to his concerns, his arguments are frequently cited as one of the key starting points for a quick growth of corpus methodologies in the humanities broadly, spurring theoretical debates about the place and value of human close reading within and given distant reading approaches (Hockey, 2000; Ramsay, 2011). Some scholars have argued in particular for the value of combining close and distant reading, arguing, for example, that “the macroscale perspective should inform our close readings of the individual texts” (Jockers, 2013, p. 28), that artificial views of a text “encourage a reader to read it differently” (Rockwell & Sinclair, 2015, p. 189), or that macro approaches can help us model and theorize about the relationships between specific textual and social variables (Underwood & Sellers, 2016). Thus the lenses that we develop of texts and corpora with computational methods should also be used to gather insight into the texts themselves — the analysis becomes a springboard for imagining new interpretive possibilities in readings of individual texts, as opposed to only looking at a big picture (this is what Breiman (2001, p. 199) refers to as the “information” goal in statistical data analysis).

To fully explore and more deeply understand what computational analyses of texts can reveal, it seems we need to grapple with the role of the human reader and close reading. Traditional interpretive techniques like close reading sit at odds with the amount of texts and data that computational text analysis engages with. With a computational view of texts, we might learn that a handful of particular features are useful for distinguishing classes of texts. But with very large textual corpora, serial reading for a particular feature quickly becomes an unmanageable task. Furthermore, some computational results may not easily provide just a small handful of meaningful features, but instead a very large list, which adds further complexity to the

interpretive task as we might traditionally undertake it with close reading. An analyst might serial read through a handful of texts and develop ideas about why certain textual patterns occur, but how do we know that the reasons scale? As corpora get larger, statistical differences become more stable but the analyst's ability to read each text closely wanes. When working with big data in rhetorical studies of texts, computational methods offer a unique insight into regularly occurring patterns that can act as lenses to shape our close reading, but we also face a verifiability problem in knowing that our interpretation scales appropriately to the entire corpus.

1.2 Rhetorical analysis and computational lenses on text

Advances in natural language processing not only create challenges for textual interpretation, but also pose some challenges to our fundamental rhetorical understandings of texts. Genre is a particularly good example, as a complex concept which has spawned an entire area of study in rhetorical genre studies, but can also be classified in some circumstances reasonably well based on word frequency (or at most, perhaps drawing on frequencies of grammatical labels like “noun” or “determiner”). These approaches to the same concept can seem fundamentally at odds with each other.

And yet, some textual analysis work out of rhetoric and English studies has begun exploring how computational text analysis can help us better flesh out how theoretical writing concepts are instantiated through specific, recurring linguistic form. For example, Brown and Aull (2017) used corpus techniques to demonstrate how higher-scoring English AP exam essays engaged in “elaborated specificity,” while lower-scoring essays engaged more in “emphatic generality.” Through keyness and correlation analyses, they were able to explicate how these concepts appeared linguistically through specific key terms and phrases. In another case, the text analysis tool DocuScope has been used for pedagogical purposes to identify the writing strategies that students use in first-year argumentative writing and to shape students' textual awareness in classrooms by highlighting how specific textual patterns are tied to broader rhetorical moves in writing assignments (Helberg et al., 2018; Ringler et al., 2018).

Moving from classificatory text analysis to rhetorical insight is a difficult

task, but we are starting to see that these types of analyses offer not simply more complex views of how to classify texts, but insight into interesting recurring textual patterns across large swaths of texts. The AP exam piece by Brown and Aull (2017) is one example of research that used classification to move toward rhetorical insight. This piece was able to identify key phrases that distinguished two corpora well in order to make claims about the types of rhetorical moves students engaged in for each type of essay. Brown and Aull (2017) facilitated this movement in part by calculating the keyness and effect size of part of speech categories rather than only individual words, making it easier to see patterns of usage across the corpora. Some computational techniques find high accuracy using just simple word frequencies though, and then the insights afforded by the analyses are sometimes more limited. For example, Underwood and Sellers (2016) built a logistic regression model to distinguish 19th-century English poetry that was considered prestigious or not, using word frequencies to do so. Their primary claim from this analysis is in regards to literary history, recognizing that there did seem to be some noticeable distinction between the corpora linguistically, and that what “prestige” looked like in poetry seemed to change over time. Looking more deeply at the model, they point out that the top words characterizing prestigious poetry are not immediately meaningful for describing the poems, though they are able to say that the prestigious poems seem in some way to be more negative. In this sense, while computational text analysis techniques seem promising for offering insight into how we understand texts and some of the key concepts we work with in rhetorical studies, it is not always clear how to fully move to that insight, especially as the unit of analysis (i.e., language category like Docuscope provides, part of speech tag, individual word) becomes more varied.

1.3 Contributions of this dissertation

Techniques from natural language processing seem to offer some kind of potential value to rhetorical studies, but interpreting computational analyses of texts and determining what kinds of insight can be garnered from them is a difficult challenge. The high-level question guiding this dissertation then is, how can a rhetorical analysis lens to computational data about texts lead to insight about those texts? While computational fields focus primarily on the statistical part of analysis and classifying/predicting with accuracy, the

field of rhetoric has the expertise in textual interpretation that is crucial to theorizing about how to effectively make sense of those outputs.

The primary contribution of this dissertation is thus a new hermeneutic, or interpretive theory and argumentative strategy, that I call “close statistical reading” for interpreting the outputs of computational analyses of textual corpora that allows for defensible and insightful rhetorical interpretation. In developing interpretations of corpora, I highlight that the goal is not to produce one “correct” interpretation, but rather explain that interpretations are ultimately arguments about meaning that are socially situated. As such, the interpretive theory provides a way of understanding interpretation in a computational and corpus context, but also an argumentative strategy the analyst needs for making the case for their interpretation. Compared to past work on computational hermeneutics, close statistical reading highlights the ability of synthesis of analysis toward understanding, rather than a narrowing in toward possibilities; in addition, it takes a rhetorical approach to hermeneutics by foregrounding the argumentative strategies by which we can justify interpretations of very large corpora of texts and data. The development of this theory thus grapples with the ways that we understand meaning-making in textual studies, and offers a new way of approaching this that does not rely upon the traditional mode of close reading each text, but is instead conducive to working with large amounts of data.

As a case study to demonstrate this theory, I use stylometry methods to classify academic writing by discipline. Stylometry methods are often used to classify texts, especially by author (two notable case studies of authorship attribution using these methods were identifying J. K. Rowling to be the author behind the penname Robert Galbraith (Juola, 2013) and identifying the authorship of several of the *Federalist Papers* (Mosteller & Wallace, 1964)). The methods work primarily by calculating “distances” between texts in a corpus, where texts with smaller distances are thought to be more stylistically similar and thus more likely to be written by the same author (or belong to the same time period, genre, etc). The distance calculations are typically based on the frequency of the most common words in a corpus, which are primarily function words (words like *the*, *and*, *of*). This is a surprising finding as function words are actually excluded from a lot of computational text analyses as being content-less and therefore not useful, but they seem to have a certain predictive power for text classification that is not well understood. Some have responded to this phenomenon with a suggestion that authors have some kind of “unique stylistic fingerprint” that is represented in func-

tion words (Rybicki et al., 2016, p. 124), or simply a recognition that we are still unsure why these techniques work and that the answer lies somewhere “in the communicative resilience of the language and the astonishing force of human individuality” (Burrows, 2002, p. 282). For this reason, stylometry methods are an especially useful case study for investigating how to make sense of computational analyses toward rhetorical insight of texts, in that they seem to be reasonably accurate in classifying but also not well understood.

Academic disciplinary writing offers a useful corpus for this case study, given its ability to easily break down into groups that can be classified and a rich scholarly literature that has investigated disciplinary differences in writing. Current research has demonstrated well why disciplinary writing might vary by organizational structure (Flowerdew, 2008), their method for signalling knowledge claims (Dahl, 2009), engagement features (Hyland & Jiang, 2016), or more explicitly content-related words like “titration” or “genre” (Wang et al., 2008; Yang, 2015), just to name a few. But when stylometry methods are applied to disciplinary writing, disciplinary writing can also be fairly neatly classified by function word frequency, which is more surprising. Differences in disciplinary writing are typically explained in terms of broader theories about disciplinarity and genre. Regarding disciplinarity, Carter (2007) argues for seeing disciplines as “active ways of knowing” rather than “repositories and delivery systems for relatively static content” (p. 387). This is also in line with wider-reaching philosophical theories of science that conceive of knowledge-making as a social process, rather than a type of linear progress (e.g., Kuhn (1962)). These conceptions of disciplinarity have some significant overlap with genre theory, especially when we consider their mutual focus on social action and norms. Berkenkotter and Huckin (1995) explain that genre knowledge in a discipline is a type of situated cognition, and that our ability to participate in a genre depends on our genre, procedural, and social knowledge that we acquire through enculturation into that discipline. Disciplinary writing, then, responds to recurring rhetorical situations that are specific to the “ways of knowing” in a discipline, and which accomplish some kind of “social action” in those communities through “relatively stable” types of language (Bakhtin, 1996; Miller, 1984).

So, given a conceptualization of disciplines as “ways of knowing and doing” and their texts as doing some “social action,” where do function word frequencies fit in this? As a major part of this dissertation, I posit that function words, and their frequencies, have a constitutive role in the action

of a text. That is, this type of analysis does not simply tell us that a word like *the* is an useful way to distinguish between corpora, but demonstrates how *the* is a signal of and piece of broader rhetorical moves related to academic knowledge representation. Apart from very few stylometric studies on particular authors, we overall do not have a strong understanding of the types of rhetorical roles function words play. The key questions guiding my dissertation research are as follows: How does function word frequency vary among disciplinary writing? How are “social action” and “ways of knowing and doing” constituted through specific linguistic forms and patterns of function words? How can investigating why function words distinguish disciplines enrich our theoretical understandings of disciplinarity and genre, and help us to better articulate how action is accomplished through specific linguistic resources? Moreover, how can we interpret the results of a computational analysis of texts in such a way as to gain rhetorical insight about those texts, given the large amount of data and texts involved? What kinds of insights do we gain about writing, and how granular might those insights be?

As a whole, this dissertation offers at least four key contributions:

1. A new hermeneutic, or interpretive theory and argumentative strategy, called “close statistical reading” for interpreting the outputs of computational analyses of textual corpora that allows for defensible and insightful rhetorical interpretation. In particular, I argue that we can interpret tables of data about a corpus not by reading every text and data point, but by seeing how many different pieces of data (from close and distant reading) all point to the same understanding of a corpus;
2. A sketch of a methodology for how to do close statistical reading with stylometry methods in particular;
3. Three different rhetorical analyses that analyze the corpus of academic writing at different scales to illustrate how the choice of analytical scale shapes rhetorical insight. These three chapters include both analysis, or “discovering what is in a rhetorical artifact” and interpretation, or “determining what it means” (Kuypers, 2005, p. 20);
4. Rhetorical theory about how function words play a constitutive role in the creation of meaning, particularly in the context of academic knowledge representation.

Through this work, I hope to offer a theory of interpretation for computational analyses of texts that can be applicable to textual studies more broadly, while illustrating how that theory can be operationalized in terms of a specific set of computational techniques and texts.

1.3.1 Chapter outline

The dissertation proceeds as follows: Chapter 2 fully develops the theory of interpretation offered in this dissertation, called close statistical reading. Chapter 3 outlines the construction of the corpus of academic writing, as well as the more technical details about the stylometric analysis and implementation of close statistical reading. The stylometry methods used are largely standard based on the field's literature, with some modifications to aid in interpretation.

Chapters 4 - 6 all offer interpretations of the stylometry results through rhetorical analyses conducted at different scales. That is, chapter 4 analyzes the highest-level split in disciplinary writing stylistically, between the humanities and sciences and uses the function words characteristic of each to characterize the broad rhetorical activities that are fundamental to their respective knowledge representation activities. Chapter 5 breaks apart the high-level split created in the previous chapter to investigate the span of the humanities and sciences, focusing on the differences between history and philosophy, and computer science and biology, as the endpoints of their respective metadisciplines. The chapter similarly investigates the splits between these disciplines, using the function words characteristic of each to characterize the rhetorical activities that occur within those disciplines. Finally, chapter 6 zooms in to just one discipline in each metadiscipline – computer science and history – to investigate the range of variation within one discipline, as represented by function word frequency. That is, though each discipline may be roughly separable from others on a large scale, each discipline also has a certain range of function word usage. This analysis in this chapter focuses on the different dimensions of rhetorical variation within each discipline that are represented by the function words with the highest amount of intradisciplinary variation in usage. In this sense, this chapter is less clearly comparative of two groups and focuses more on a nebulous space of variation. Chapter 7 offers reflections on the interpretive theory and the types of rhetorical insights it provided in this analysis, and speculates on areas for future work.

Chapter 2

Close statistical reading: an interpretive theory and argumentative strategy

Textual analysis has traditionally worked with either one text or a small set of texts at a time, meaning that the analyst can feasibly read through every text thoroughly with their chosen analytical lens. However, computational forms of text analysis that work with hundreds or thousands of texts seem to offer traditional language scholars something of interest – if something like a logistic regression model can distinguish between 19th-century English poetry that was seen as prestigious or not (Underwood & Sellers, 2016), shouldn't that be able to tell us something interesting about the cultural construct of prestige in 19th-century England? How can we account for something like prestige being identifiable by word frequency, in terms of theoretical and cultural discussions of the concept?

Doing so is not an easy task. In computational text analyses, oftentimes corpora are very large (hundreds, thousands, or even millions of texts) to the point that no single analyst could feasibly read through them all thoroughly. And even if a small army of analysts could, there is an additional challenge of parsing through an overwhelming amount of data about these texts. Results like those produced in Underwood and Sellers's (2016) prestige study consist of a list of hundreds of individual words, sorted by weight, that help to distinguish prestigious versus non-prestigious poetry. Should each text be read through carefully for each of these words? The task quickly becomes

impossible using traditional analytical techniques.

In this chapter, I introduce “close statistical reading,” an interpretive theory, or argumentative strategy, for creating and justifying an interpretation of the results from computational text analysis techniques, even though every text and piece of data cannot be examined with the same detail as might be seen in traditional text analysis.¹ How do we make sense of computational data about texts in a way that leads to rhetorical insight? What are the processes by which we do so? And how do we do that when there are too many texts and too much data to serially read through closely? I start by explaining that close reading is traditionally a key part of textual interpretation in rhetorical studies, but that adding distant reading complicates the interpretive process in ways that we have not fully theorized. I frame this problem in terms of key concepts like hermeneutics, interpretation, methodology, and rhetorical analysis, and then propose that we can develop defensible interpretations of computational analyses of corpora with large amounts of data not by explaining all the small minutiae of the results in-depth, but by focusing on developing a robust understanding of the corpus as a whole based on how many disparate pieces of data together accumulate to a broader rhetorical effect across texts. This approach thus centers synthesis, convergence, and triangulation rather than comprehensive verification. In doing so, we are ultimately able to find interpretability at scale, developing large granularity insights into textual corpora through computation.

2.1 The problem of close and distant reading in rhetorical studies

Close reading by a human reader has long been at the heart of rhetorical criticism. From classic studies of temporality in Abraham Lincoln’s inaugural speeches (Leff, 1988) to recent examinations of amplification in writing by Ida B. Wells (Libertz, 2021), rhetorical criticisms are steeped in close attention to textual detail to the point that Celeste Condit (1990) described this attention to close reading as “the unique ‘way of thinking’ that is enabled by rhetorical criticism” (p. 332).

One of the more recent and influential challenges to this focus on close

¹Most of the interpretive framework presented in this chapter, including some of the actual text that appears, is drawn from my prior written work (Ringler, 2021).

reading was Moretti's (2000) proposal of "distant" reading. Moretti was concerned with the focus of literary studies on a few particular texts or a canon given the vastness of world and national literatures, arguing that "you invest so much in individual texts *only* if you think that very few of them really matter" (Moretti, 2000, p. 57). He proposed reading at a distance, looking at broad trends across many texts (rather than individual representative texts) in order to make claims about groups of literature. The impactfulness of this idea has become a driving force behind the development of distant reading and corpus analysis in the humanities broadly, including partially driving the emergence of digital humanities as a field.

Distant reading approaches, however, differ fundamentally from close reading in that they often produce new kinds of texts in the form of tables and charts, and new lenses on entire (sometimes very large) corpora of texts. And while these new ways of looking at texts seem potentially insightful in the analysis of texts (either per Moretti for literary studies or like in the prestige study described earlier), *how* to read the outputs to get to that insight is often unclear. In other words, how do we get to analysis that "explain[s] how the rhetorical artifact works" (Kuypers, 2005, p. 18) rather than simply a description of how the text looks in terms of word frequencies? Heuser and Le-Khac (2011) open a piece of theirs with a brief vignette that describes this lack of clarity in process:

As humanists, we believe we are trained as expert readers, able to read almost any kind of text closely, deeply, and critically. But on 21 February 2010, the two of us sat staring at a computer screen dumbfounded by a kind of text that for once we had no idea how to read. Working with a corpus of thousands of digitized nineteenth-century British novels, we had just produced a plot, similar to figure 1, showing the usage trends of a massive group of abstract words relating to social values...What exactly were we seeing here? (p. 79)

As a response to this problem, some scholarship has argued for the combination of close reading with distant reading to lead toward interpretation of some kind. That is, while some have found distant reading on its own to be productive for answering certain kinds of questions (e.g., Underwood (2016)), considering how they might complement each other could aid meaning-making of computational analyses by allowing space for the interpretive insight of close reading. For example, in arguing for the potential

value of this integration, Jockers (2013) claimed that “the macroscale perspective should inform our close readings of the individual texts” (p. 28), and Rockwell and Sinclair (2015) suggested that artificial views of a text “encourage a reader to read it differently” (p. 189). Ideally, the models that we develop of texts and corpora with computational methods can (and in many cases, should) be used to gather some kind of insight into the texts themselves. Doing so would shift the focus of rhetorical text analysis from close reading for notable features that might be insightful for some purpose but not necessarily regular or recurring, to looking at an entire corpus in terms of particular features and patterns that recur statistically; these patterns, if noticed, marked as potentially meaningful (perhaps, for example, for their ability to classify texts), and interpreted productively, can lead to new kinds of insights into a text or corpus as close reading would traditionally provide, but are insights which close reading alone cannot easily provide on its own given the limits of the human serial reader to both notice and trace these features across entire corpora.

Others have recognized too that close and distant reading need not always be dichotomous and separate processes, but can indeed be integrated into the same analysis in a more cohesive way. For example, the text analysis tool Docuscope (Kaufer & Ishizaki, 2020) is sometimes referred to as “interpretation lite” (Ishizaki & Kaufer, 2012), tagging texts with rhetorical categories of language like “description,” “narrative,” or “academic terms.” In doing so, the distant form of reading itself is already infused through with rhetorical lenses derived from close reading, and thus shapes the distant read and interpretive acts of making sense of the results. When used in practice then (e.g., Kaufer and Parry-Giles (2017)), the analysis does not just move back and forth between close and distant reading, but rather is able to integrate them into the same process given the benefits of the tool itself operating through rhetorical lenses on language.

With distant reading as more traditionally undertook though – classifying texts well using frequencies of various textual features – one key challenge for fully exploring and more deeply understanding the patterns revealed by computational methods is the place of the human reader. The relationships and patterns that statistical analyses reveal can function as a sort of analytical lens, making present textual patterns that may have previously gone unnoticed but which the analyst can now read for with the knowledge that they statistically recur (Ramsay, 2011). For example, the analysis in chapter 4 shows that the frequency of *the* helps statistically differentiate one corpus

from another. This is a pattern likely to have gone unnoticed by the average serial reader, both because of how common (and thus, frequently unnoticed) *the* is, and how slight (even if regular) the difference between corpora might be. Once pointed out though, the analyst can return to close reading texts with a focus on this word to more deeply investigate.

With very large textual corpora though, serial reading for a particular feature quickly becomes an unmanageable task. An analyst might serial read through a handful of random or even strategically-chosen texts and develop ideas about why certain textual patterns occur, but how do we know that the reasons scale? As corpora get larger, statistical differences become more stable but the analyst's ability to read each text closely wanes. To this problem of interpretation through integrating close and distant reading, I would argue that the key methodological challenge is not so much the place of the human reader, but a question of *scale*. When working with large amounts of data in the humanities, computational methods offer a unique insight into regularly occurring patterns that can act as lenses to shape our close reading, and we may feel able to do this kind of close reading on a small handful of texts, but scaling this reading up to an entire corpus is unreliable.

2.2 Hermeneutics and computation

How do rhetorically analyze a corpus that is very large, and about which we have very large amounts of data that we want to make sense of? Clarifying a few key terms help us to better understand the nature of this challenge. At the heart of the practice of rhetorical studies is rhetorical analysis, or analyzing particular texts to better understand how they work. That is, we might analyze a text for something like pathos, looking for appeals to emotion in any form that they might occur in the text and thinking about how those appeals contribute to the overall work of the text. Kuypers (2005) explains rhetorical analysis as follows: "Analysis asks us to explain how the rhetorical artifact works; it provides a sketch of sorts, showing how the artifact is put together; what are its parts, how do they go together, and what the whole looks like" (p. 20).

The act of analysis itself can take many different forms, and is based upon (and consequently shapes) rhetorical theory. Drawing on techniques from something like discourse analysis, the analyst might look for specific textual features like evaluation or gradation (Martin & White, 2005). Other

times, an analyst might have a heuristic like *identification* that they are looking for evidence of, where a heuristic is a concept or certain “method of solving problems” that leads to some kind of insight (Young & Becker, 1965, p. 457). Typically, these particular heuristics or frameworks for analysis are based on rhetorical theories. Brummett (1984) explains that rhetorical theories provide a way of seeing or understanding the world – they help to build a model of the texts that explains features in a cohesive way – and that rhetorical theories are a way of choosing what pieces of the mosaic of information about the world to put together and how in order to make meaning out of them. The theory provides a method, a way of choosing mosaic pieces, where method is typically looking for those pieces that the theory lays out (Brummett, 1984; Nothstine et al., 1994). Of course, theory and analysis shape each other, where theory informs the analysis and new analysis can refine and develop theories.

While analysis lays out the particular parts and pieces of a text’s makeup, interpretation is then the work by the analyst to determine what those bits and pieces tell us. If analysis is “discovering what is in a rhetorical artifact,” interpretation is then “determining what it means” (Kuypers, 2005, p. 20). Hughes (1965) would describe this movement from analysis, where you lay out specific aspects of the text, to some interpretation of what they mean as a form of invention. The analyst experiences a “gradual evolution of a judgment out of disparate and embryonic evidence” (Hughes, 1965, p. 158). While rhetorical invention is traditionally considered in terms of constructing arguments (Aristotle’s *topoi*, for example, give strategies for developing new arguments), invention can also be thought of as an “epistemological heuristic,” or a form of inquiry for how we come to new understandings of texts (Young & Becker, 1965, p. 131). This framing of invention as useful for interpretation was a focus of Augustine of Hippo in interpreting scriptures, as well as Burke (1943) in discussing uses for his Pentad (Lauer, 2004).

As an inventive process, developing interpretations of texts is not neutral or objective, but rather finds its rigor in carefully-made decisions. There are many possible interpretations for any given text (Nystrand (1982) describes interpretation as “participation in a theatre of meanings” (p. 81)) and constructing particular meanings can be viewed as an interpretive process guided by choices (Richards, 1936). Yet unlike the harder sciences that find rigor and verifiability in agreed-upon methods and design principles, rhetorical studies relies upon “procedural norms of argumentation” to make strong cases that are reliable, flexible, and humane (Zarefsky, 2008, p. 634). In doing

so, rhetorical criticism seeks to open up possibilities for interpretation rather than closing them off; or as Lauer (2004) explains, examining alternatives is “inherent” in the very idea of invention as handed down from Aristotle (p. 6-7).

The carefully-made decisions about how to make meaning out of analysis are referred to hermeneutics, or interpretive theories, and these hermeneutics then shape methodology. Often referred to as the method of interpretation, hermeneutics has its etymological roots in the Greek god Hermes, the messenger between the divine realms, the underworld, and the mortal world. Palmer (1969) provides a succinct description of the nature of hermeneutics saying that it is the process by which, “something foreign, strange, separated in time, space, or experience is made familiar, present, comprehensible; something requiring representation, explanation, or translation is somehow ‘brought to understanding’ – is ‘interpreted’” (p. 14). Mailloux (1985) sees this process as distinctly rhetorical in nature, viewing hermeneutics not necessarily as a process but as a set of contextually-decided argumentative strategies for making meaning out of a text. In effect, hermeneutics – being both a guide for inventing interpretations and a set of argumentative strategies – refers to the process by which we come to and justify an interpretation. Having a hermeneutics does not guarantee a correct interpretation of a text (Zarefsky (2008) would say that a good interpretation of a text is not one that is provably correct, but simply one that can be argued for well), but rather supplies robust and communally-accepted strategies by which the analyst can come to and argue for a meaning. The process for coming to an interpretation is thus inherently connected to methodology, or the values and justifications behind a particular method. While method itself refers to the procedural steps of analysis, methodology refers to the justifications for the use of a particular problem-solving technique, perhaps arguing for a value of reasoning patterns or verifiability (Kaplan, 1964). That is, if hermeneutics provides the guide for inventing and arguing for interpretations, the justification of a particular method must be in line with that hermeneutic.

To turn back to the problem of computational text analysis at scale, what is fundamentally missing is a hermeneutics, or interpretive theory and argumentative strategy, and a methodology. We have methods by which to produce data about texts (creating, in a sense, an entirely new text in the form of data), but how do we make sense of those? Some computational techniques like DocuScope (Kaufer & Ishizaki, 2020) may restrain your interpretive possibilities in certain ways by providing concepts to analyze in

the texts and thus analysis can proceed in slightly more traditional ways, but others produce outputs like tables of words ranked by weight, which are more agnostic and thus harder to tackle. In traditional rhetorical analysis, we might start with a rhetorical theory and look for evidence of something in a text (perhaps pathos or identification) as our method, and our interpretive theory for making sense of that analysis is based in the tradition of rhetorical analysis. That is, we take those bits of analysis and argue “well” for an interpretation, perhaps providing particular quotes or other forms of textual evidence that allow our interpretation to be communally accepted.

In computational work, by contrast, we start with bits and pieces of data, and do not know what rhetorical theories might be useful to look for in that data. Furthermore, we are necessarily beholden to frequency and particular pieces of data as the focus of our analysis (being what computational results provide), and cannot choose only one particular aspect to focus on at a time. Finally, even if we could make sense of those highly frequent bits and pieces of data, how do we determine what they all mean and create an interpretation that we can argue for well? In order to address these problems, we need to develop an interpretive theory, argumentative strategy, and methodology for making sense of large-scale data about corpora, or for rhetorical analysis that allows for analysis at scale. An interpretive theory would provide a way of focusing the analysis in spite of the large amount of data, and an argumentative strategy would outline how we could argue for an interpretation despite not being able to read every text and piece of data, which would then ultimately shape a methodology that justifies how to go about that type of analysis in both a rigorous and feasible way.

2.3 Close statistical reading

Toward these goals, some productive guidance can be found in the ideas of a traditional text analysis theorist, Paul Ricoeur. Ricoeur was a 20th-century philosopher who engaged with text interpretation and hermeneutical theory, and who theorized textual interpretation as a multi-step process. Ricoeur (1981) described hermeneutics as “the theory of the operations of understanding in their relation to the interpretation of texts” (p. 43), and saw the understanding of texts as at least somewhat separate from the author’s intent and created by readers at different points in time. In particular, Ricoeur (1981) theorized a “hermeneutical arc,” or different phases that readers go

through in constructing their understanding of a text. Bell (2011) slightly adapted this arc for discourse analysis into what he called the “interpretive arc.” The ideas share a lot of similarities, and I draw my description of the “arc” that follows heavily from both sources, as Bell’s adaptation has a useful focus on textual interpretation as we might do in close reading.

Ricoeur describes the process of interpretation as having several steps that the reader progresses through. First, the reader experiences *distanciation*, meaning texts are necessarily estranged and separated from the author and their original intent through time and context (Ricoeur, 1981). Second, the reader experiences *pre-understanding*, which is the state of mind that the reader encounters a text with. For any text, a reader might have some knowledge or preconceived notions of the topic, author, and so forth, which shape their impression of the text before they ever read it. Next, the reader experiences *naïve understanding*, or their initial, even visceral and impressionistic, guess at what the text means (Ricoeur, 1976). Ricoeur talks about this initial understanding as a guess in that readers do not have access to what authors meant, and the understandings generated here are starting points and not the result of analysis. After developing these initial guesses, the reader moves into more analytic processes, which are the processes we are working towards with computational data in the humanities.

Following naïve understanding, the reader moves into developing explanation (or what Bell (2011) calls “analysis”) and then understanding. Bell’s reading of Ricoeur is especially useful here, as Bell envisions explanation/analysis as the goal of discourse analysis with texts. Bell (2011) describes these phases as follows: “Understanding is mediated by explanation/analysis, which is fulfilled in understanding. While explanation lays out the parts through analysis, understanding grasps the whole through synthesis” (p. 534). In other words, analysis of the text helps to explain bits and pieces of what the text means. Combined as a whole, the reader can then gain understanding, or “the ability to take up again within oneself the work of structuring that is performed by the text” (Ricoeur, 1991, p. 18). The processes of analysis may validate (or challenge) the reader’s initial guesses about what the text means, but are “the second-order operation grafted onto this understanding which consists in bringing to light the codes underlying this work of structuring that is carried through in company with the reader” (Ricoeur, 1991, p. 18).

Ricoeur developed this theory in regards to traditional text analysis, but turning back to the problem of interpreting data about texts, we can think

about reading computational data similarly to how Ricoeur describes reading texts. We are distanced from it in a sense, in that computational representations of texts are strange numerical representations of actual written texts. We may even come to these corpora with notions about what they might reveal. After we conduct computational analyses, we are then in a space to start making guesses about what those numbers mean. But at this point, we are only in Ricoeur’s “naïve understanding.” It is not until we return to reading individual texts that we can start the process of analysis/explanation and making sense of bits and pieces of data (for example, developing an explanation for why *the* is one of many weak discriminators in a stylometric analysis).

But how are we to know that the explanations of those bits and pieces scale? In other words, how do we feel confident in the understandings that we are forming and gathering bits of evidence to support? In traditional rhetorical analysis, the analyst may be interested only in one or two occurrences of a textual feature for the effect they create, and not have to engage with questions of scale (or if they do, they may hand code for features (see Geisler and Swarts (2019)) which becomes quickly unfeasible in a large-scale corpus analysis or even in a smaller-scale analysis when analyzing a very common word like *the*). In a computational analysis of a corpus though, regularity and scale must be accounted for. Ricoeur might address this concern by falling somewhat in line with Karl Popper’s (1959) falsifiability thesis: there are many possible interpretations, but the best interpretation is the one that is the least capable of being falsified (Ricoeur, 1976, p. 79). Bell further explains that, “as discourse analysts, we know that we are not actually able to prove that our reading of a text is the ‘right’ or even the best one. But we can try to demonstrate that competing readings are less valid or probable than the alternative” (p. 537). In sum, the more probable interpretation of a text is that which “takes account of the greatest number of facts furnished by the text” (Ricoeur, 1981, p. 175). Heuser and Le-Khac (2011) make a related point about computational text analysis, arguing that interpretation can be made “more rigorous by having to account for a wide set of related observations” (p. 85).

When arguing for an interpretation of computational data then, we can do so not by objectively proving our understanding is correct but rather by demonstrating how our understanding takes account of as many facts as possible furnished about the corpora through both computational and traditional analyses. The distinction between analysis and understanding

becomes useful here, as many different pieces of analysis might support one particular understanding. By showing how several disparate pieces of data analysis all support one understanding of the corpus (even if each analysis results from close readings of only a few texts), we account for more and more facts and strengthen the argument for both the analyses and that particular understanding. In this sense, when it comes to a strategy for choosing what pieces of the mosaic of textual data to focus our analysis on, we want to focus on choosing pieces that can be triangulated and synthesized together toward a cohesive understanding. While there may be many possible understandings and interpretations of computational results about a corpus, the strongest will be those that can account for large swaths of data at the same time.

This interpretive theory and argumentative strategy, what I am terming “close statistical reading,” allows rhetorical interpretation of large amounts of computational data about texts. Close statistical reading focuses on demonstrating how the analyses of many pieces of data support a particular understanding of a text or corpus. This is an especially useful strategy for a technique like stylometry in particular, where as Evert et al. (2017) remind us, each individual word is likely to be a relatively weak discriminator on its own. To gather these additional types of data, we can conduct additional computational analyses to create more lenses that help refine and cross-validate our understanding with new types of information. In doing so, the analytical process focuses on testing understandings rather than specific analyses. That is, by triangulating many different pieces of data and focusing on validating an understanding rather than explaining every possible data point, we focus our contributions on the humanistic insight we can gain from our corpora. Figure 2.1 visualizes what this interpretive theory might look like as an analytical process. As a whole through close statistical reading, we gain an ability to go beyond simple description of a corpus’s make up into explaining how the artifact works, or as Kuypers (2005) might say, a sketch of the parts, how they go together, and what the whole looks like (p. 20). In chapter 4, for example, we learn not simply that *the* is common in the sciences or even that therefore noun phrases are common in the sciences, but rather understand better how the sciences argue for knowledge claims through detailed physical and localized description, that linguistically involves heavy uses of definite noun phrases and thus the word *the*. In effect, this process make it possible to glean insight in a way that is feasible for the limited human reader while still having strategies to argue for (or against) certain interpretations.

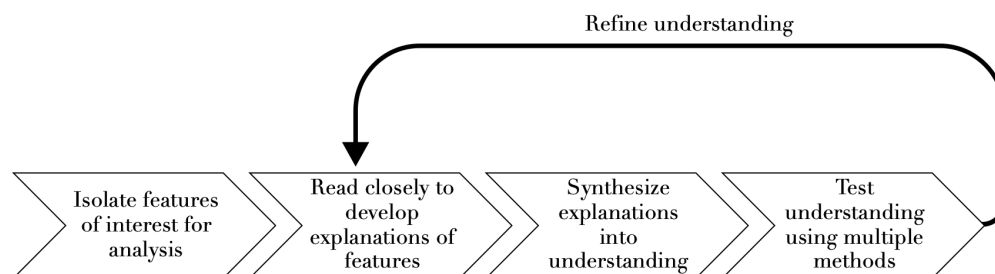


Figure 2.1: A visualization of “close statistical reading” as an analytical process

Lauer (2004) reminds us that interpretation in rhetorical analysis is generative, in that it opens up possibilities for understanding rather than closes them off. Even though working with computational analysis may seem at first to close off possibilities by focusing only on what is statistically relevant, I would also argue that close statistical reading is generative, although in different ways than rhetorical analysis might be traditionally conceived of as being. A statistical analysis of a text is likely to be capturing some aspect of the text, and the challenge of interpretation is to discover what that aspect (or many aspects) is. Traditional rhetorical analysis might start with a rhetorical effect and try to uncover how that effect is achieved through language, which opens up a wide range of possibilities in terms of effects to be analyzed and different ways they are achieved. Close statistical reading reverses that process by starting with the language features, and investigating what effect is being achieved by them. While there may be more than one effect being achieved by a set of features, it is significantly more limited.

Interpreting computational analyses of texts offers at least two new kinds of generative possibilities though. First, it allows for clarification and insight into rhetorical theories around classificatory concepts like disciplinarity, genre, time period, or authorship. When we computationally compare sub-corpora of texts in regard to these concepts, a method performing with some accuracy will be pointing to specific linguistic features that characterize those concepts. Literary and rhetorical studies already have complex and developed theories of these concepts that are often only partially based in specific linguistic form. Interpreting the results to ask *why does this work?* can help us to probe those concepts by asking how the theories of those concepts relate to the linguistic features. For example, in terms of genre, we can use com-

putational analyses to ask how particular “social action,” to invoke Miller’s (1984) definition, is achieved through particular linguistic form. This line of inquiry deepens and expands our understandings of genre theoretically, but at a large granularity about how linguistic features relate to the rhetorical moves in the genre as a whole rather than necessarily in particular detail about one specific text. That is, rather than developing deep insight into the particularities of one or two texts, this process allows for a larger scale rhetorical analysis that produces deep insights into how rhetorical moves are instantiated commonly and recurringly across a wider swath of texts with similar classifications.

Second, in doing this kind of interpretive work we also deepen and restructure our understandings of particular linguistic features. This will become especially apparent throughout chapters 4 to 6 (and is discussed more fully in the conclusion), which demonstrate how function words are not just lexicogrammatical features of the English language but actually have a complex epistemic role in the creation of meaning. This new view of function words is a product of the interpretive work on computational results, given the ways that the method probed the larger classificatory concepts of disciplinarity and genre.

Close statistical reading ultimately offers new pathways both in computational text analysis as well as more traditional rhetorical analysis. Computational text analysts have long struggled in learning how to glean insight from corpus analyses, at least partially because of the limitations of the analyst: that is, we cannot read everything. Especially as both corpora and data outputs get bigger in the rise of big data and machine learning, it is less and less feasible to read in our traditional ways for analysis. The approach to hermeneutics developed in this section has resemblances to Piper’s (2015) model of computational hermeneutics wherein a reader oscillates between close and distant reading to converge on an interpretation, as well as Guldi’s (2018) “critical search” in its winnowing away at a mass of data. In a way, close statistical reading does suggest a sort of oscillation, or a movement back and forth between the close and distant toward a fuller understanding. Rather than a winnowing or narrowing in thought, close statistical reading instead focuses on a synthesis of many disparate bits of analysis toward an understanding. In this way, it seeks to account for large swaths of data at the same time. In addition, close statistical reading takes a rhetorical approach to hermeneutics to ask not simply how might we come to a justified understanding, but how do we justify that understanding argumentatively?

Through the synthesis of analyses and combination of many pieces of evidence toward the same claim, this hermeneutic provides an argumentative strategy for justifying claims about computational analyses of textual corpora despite not being able to account for all of the data. Ultimately this theory opens new possibilities in rhetorical analysis by demonstrating how we can theoretically think about developing insight from entire corpora of texts apart from comprehensive close reading, what kinds of information (in particular, computational information) might let us do that productively, and what kinds of new generative possibilities we have to gain from this analysis.

What this interpretive theory offers in terms of new pathways for natural language processing, digital humanities, and rhetoric are larger theoretical points that I return to in the conclusion. In the following chapter, I outline the setup of a case study that uses this interpretive theory, including a discussion of how close statistical reading is implemented in the analyses in this dissertation. Then, in the three analysis chapters (chapters 4 to 6), I move further from theory to praxis and illustrate how this approach can lead to insight into corpora. In particular, the three analysis chapters analyze the corpus at different scales, or from different viewing angles. In doing so, I demonstrate how this interpretive strategy operates (including its strengths and challenges) at macro and micro scales and how it affords different kinds of insights into corpora as the viewing angle of the corpus changes.

Chapter 3

Study setup: corpus construction and stylometric analysis

This chapter provides the details of the computational analysis and implementation of close statistical reading in this case study. It describes the corpus construction and then walks through the computational analysis – here, stylometric analysis – that produces the data from which we can begin interpretation. Supplementary material for the methods of this project (including corpus metadata and code) can be found online at <https://github.com/hannahringler/function-words>.

3.1 Corpus construction

A study investigating how to make sense of computational data about texts could reasonably start with a wide range of corpora or analytical techniques. The analytical technique (described in section 3.2) is a classificatory technique, and thus a corpus with many different clear groups and previous research on those groups is useful for interrogating how the classifications and groupings developed in that computational model can be rhetorically insightful. In accordance with those goals, this analysis uses a corpus of academic writing across disciplines, which is already often a source of corpus and detailed discourse analysis in the English for Academic Purposes (EAP) and Writing in the Disciplines (WID) communities, and can more broadly speak to theoretical debates around disciplinarity and epistemology.

The corpus itself consists of the texts of 1,625 academic journal¹ articles, gathered equally across thirteen disciplines. The goal of the corpus construction was to obtain a random sampling of articles published in disciplines across academia, that was not heavily influenced by time or specific publication venues. For this reason, each discipline’s subcorpus was gathered equally across five top publication venues in the field,² and across the five-year span of 2013-2017. For example, in the education sub-corpus, five articles were randomly selected from all of the research articles published in each year (2013-2017) in the following journals: *Higher Education*, *American Educational Research Journal*, *Early Childhood Research Quarterly*, *Educational Researcher*, and *Teaching and Teacher Education* (five years * five journals * thirteen disciplines = 1,625 articles). A full listing of disciplines and the journals used in each can be found in Appendix A. In total, the corpus contains 13,604,023 words.

Each text was formatted as a .txt file and cleaned by hand to remove footnotes, endnotes, appendices, headers and footers, bibliographies, page numbers, and other unreadable characters. Image captions and section headers were retained because the text of these have some meaningful place in the argument of the text itself, and do not regularly serve as only citation-related or otherwise meta information (while footnotes or endnotes do sometimes provide detail that is pertinent to the text itself, depending on the citation style they sometimes provide purely bibliographic information and thus were not retained to be consistent). The goal was for each file to contain only the text of the article itself, and to remove other extraneous or meta-related information.

¹Computer science was the only discipline gathered from conference papers as opposed to journal articles. This is because computer science, drastically more than other disciplines, tends to value conference papers over journal articles (perhaps because of their timeliness) and tends to publish its higher quality and most novel research in them (Meyer et al., 2009; Vrettas & Sanderson, 2015). Substantively, the basic form and function of the conference paper is very similar to the journal article in other fields.

²The disciplines were chosen to span the range of fields represented at the four digit level of 2010 CIP codes provided by the National Center for Education Statistics (n.d.). For example, computer science accords to CIP code 11.07, chemical engineering to CIP code 14.07, and so on. Within each discipline, publication venues were chosen among those with the highest h-5 values for 2013-2017 as provided by SCImago (n.d.) and Google Scholar (n.d.) for the discipline, while also trying to balance choosing journals that spanned the range of scholarship produced in a discipline.

3.2 Stylometric analysis

In choosing an analytical technique for testing and demonstrating a theory of interpretation, I aimed to choose a technique that produced results which were not easily interpretable. That is, my goal was to create a results that did not have any kind of obvious or easily apparent interpretation or insight about texts but one which still seemed to be capturing something of interest about the corpus, so that the interpretive theory was really “put to the test,” so to speak, while still giving reason to believe it was worth interpreting.

Given these goals, this analysis uses techniques from stylometry. Stylometry is a particularly useful example of a computational technique that has been tricky to tackle in terms of interpretation. It has been most commonly used for authorship attribution, but is in general very effective at classifying corpora by groups (author, time period, genre, etc.). For the corpus of academic writing used in this project, it classifies by discipline remarkably well (see chapter 4 for more detail on this). Stylometry typically works by calculating stylistic distances between texts in a corpus with the expectation that texts with smaller distances between them are more stylistically similar and thus more likely to be written by the same author (or belong to the same group). The method is typically quite accurate, and has been used for cases like identifying the authorship of Federalist papers (Mosteller & Wallace, 1964), and suggesting that Robert Galbraith, the author of *A Cuckoo’s Calling*, was actually a penname for J.K. Rowling (Juola, 2013).

The distances calculated between texts with stylometry methods are based on the distribution of most frequent words, which are often function words like *the*, *and*, or *of*. Thus, interpreting the results of a stylometric analysis, or in other words asking, “why does this work?” or “what do we learn from discriminating authors?”, has been widely recognized as difficult because of the huge amount of data and variables at play (e.g. Mosteller and Wallace (1964); Craig (1999); Burrows (2002); Kestemont (2014)). How is it that the frequency of primarily function words is so accurate at distinguishing authors (or in this case, disciplines)? Evert et al. (2017) explain that the distances between texts (and determinations of authorship) are therefore based on many variables which are likely weak discriminators on their own, as opposed to a small number of stronger variables, making this problem of meaning-making a more complex one. While some have begun to explain why stylometry methods work from a mathematical perspective (e.g. Arg-

amon (2008)) or begun to analyze what they might tell us about authors or genres (e.g. Craig (1999); Argamon et al. (2008); Noecker et al. (2013)), we still lack a robust interpretive framework for thinking theoretically about how to make sense of the amounts of data produced by these methods and others like them. Certainly fields like data science or statistics have developed interpretive theories for certain purposes, but as van Zundert (2016) explains, the use of computation in textual interpretation and bringing together of disciplines requires new forms of hermeneutics, given the theoretical and methodological tensions in working with texts in particular.

Technically speaking, stylometric analyses work by calculating distances between texts. There are many techniques for doing so, but this analysis uses a metric called Delta (Burrows, 2002), which is one of the most popular and well-tested measurements. In simplified terms, the Delta measurement looks at every individual word that two texts have in common, calculates a number to represent how different the frequencies in those words are (taking into account the average frequency and standard deviation of that word in the corpus as a whole), and sums up each of those numbers for every word in common throughout those two texts to create a total distance between the texts. For example, if one text includes the word *be* 50 times and another text includes the word *be* 400 times, the calculated distance for *be* between those two texts might be 10 if the average of *be* in the corpus is 47 and it has a low standard deviation, or it might be 2 if the standard deviation was very high. The total distance between two texts is then the sum of all of those numbers for each word that the texts have in common (or for a specified number of top words, often the top 100-1000), meaning that texts with similar frequencies of common words have low distances between them. In slightly more technical terms, Delta calculates the sum of the absolute value of differences in z-scores for each word between two texts. For a whole corpus, this technique will produce a distance between every pair of texts, which can then be visualized. For reference, the full formula for calculating the Delta distance between two texts is as follows:

$$\Delta_{(A,B)} = \frac{1}{n} \sum_{i=1}^n \left| \frac{A_i - \mu_i}{\sigma_i} - \frac{B_i - \mu_i}{\sigma_i} \right| \quad (3.1)$$

where, n = number of frequent words (or other features) being measured, here, function words; A, B = texts being compared; A_i = frequency of a given feature i in the text A ; B_i = frequency of a given feature i in the text

B ; μ_i = mean frequency of a given feature i in the corpus; σ_i = standard deviation of a given feature i in the corpus.

Traditionally, stylometry analyses will calculate distances between texts using the top several hundred words, which in practice are typically function words. For this particular analysis, I chose to calculate distances between texts using only a list of 211 function words, so as to focus the analysis on words more clearly related to form and style rather than content (though the two are not neatly separable) as well as to demonstrate methods of interpretation that will be more easily transferable to other stylometric analyses. In practice, using only function words or the top 200 most frequent words produced very similar results. The list of 211 function words used in this analysis is derived from the *Longman Grammar* (Biber et al., 1999), and excludes pronouns given their close connection to subject matter (for reference, the full list of function words can be found in Appendix B). For this analysis, the distances were calculated using the “Stylo” package in *R* (Eder et al., 2016), and visualized using *Gephi* (Bastian et al., 2009) per the techniques described by Eder (2017).³

The computational model itself produces a table of distances between every pair of texts, which can be visually mapped onto a 2-D plane. But in order to move toward making sense of the model, we must move beyond just classifying to ask, “what makes up those distances?” In more specific terms, the distance between any pair of texts is a summation of many smaller distances contributed by differences in individual words. To understand what is causing those distances, we can break apart the total distance into its pieces and look for which words contribute the most to the total distance.

At this point, we have a table of distances between texts that can be visualized so as to see how stylistically close texts are to each other. Each piece of the case study proceeds slightly differently at this point, though all engage in close statistical reading as the basis of study. The study as a whole has three parts, wherein the overarching goal behind the entire case study is to investigate what we can learn from interpreting this computational model, or table of distances. Each part becomes progressively narrower in focus, with part one looking at the classifications in the corpus at a very broad level and part three looking very narrowly. The following section provides the details of the analysis and interpretation for each part.

³The visualizations for this corpus appear in chapter 4 in Figures 4.1 and 4.2.

3.3 Application of close statistical reading

Chronologically in terms of the method of this case study, we now have tables of words ranked by how much they contributed to a “distance” between subcategories of the corpus. As such, the analysis has started isolating features of interest for analysis. To further narrow down features of interest for each chapter, I isolated the top four words which contributed most to the total distance between each discipline or subcorpus to focus the analysis on. The choice of number of words is a bit arbitrary; each additional word contributes to a fuller and more complex understanding of the distinctions between groups, but for the purposes of simply demonstrating how the analysis works and the types of insights we might gain, I found that four offered enough to develop a complex understanding without becoming overwhelming.

For each word of interest, I then identified a small handful of representative sample texts (see the following subsections for more specific detail) that I could use to investigate how the words were used in the disciplines. My goal in this step was to have some sample texts that used the word of interest in normal, common ways. An analyst might also choose to look at the distribution of word frequencies in a discipline, which may suggest subgroups in that discipline; though this could be an interesting and fruitful line of inquiry, the analyses I conducted here were interested only in high-level comparisons. For all three chapters, once the word was chosen, I first read through each of the texts to get a sense of the general argument and structure of the text. I then returned to the texts with a concordance table of the word of interest, and started taking notes and developing hypotheses about how they were being used and for what purposes (paying attention to things like grammatical structure, where they appeared in the text, what kinds of words appeared around them – the particular features of note depend on the particular function word).

At this point I had explanations that seemed to be consistent for the couple texts that I was reading, but I did not know if those explanations scaled. That is, I needed these explanations to be able to account for a wider set of observations about the texts. To do this, I began developing ways to test my explanations.⁴ The precise techniques varied depending

⁴Here I list several different ways to test explanations, some computational and some traditional. The possible list of ways to probe explanations could be endless, and should

on the word and explanation I had formed, but I regularly drew on part-of-speech analysis, a dictionary-based text analysis tool called Docuscope (Kaufer & Ishizaki, 2020), and even other scholarship on grammar (e.g., Biber et al. (1999)) and textual theories like intertextuality (Bakhtin, 2010) or engagement (Martin & White, 2005) to check and add strength to my explanations of particular words. For example, when probing my explanation of *the* as naming specific objects in scientific experiments, I looked first at the general makeup of the science texts with Docuscope and found that they had a much higher prevalence of descriptive language than the humanities (5.4% vs 3.6% of the total text). This supported my hypothesis that *the* had a role in heavy physical description of experiments. But to probe this further I used part-of-speech tools from the Python package spaCy (Honnibal et al., 2020) to isolate all noun phrases that started with *the* in both the sciences and humanities, and then used Docuscope to learn that of the words that followed *the* to make up the noun phrase, almost half of those words were academic or descriptive terms in the sciences whereas that number was closer to one-third in the humanities. Humanities *the*-phrases instead showed much higher occurrences of character and narrative terms in their noun phrases.

All of these bits of data together – readings of individual average texts, data about the rhetorical makeup of the texts as a whole, and specific probings of particular grammatical forms – combine to suggest one understanding of what a particular word is rhetorically doing in a text. I then synthesized the understandings of each individual word and developed an explanation for what was being captured with the distances as a whole, based on the top-contributing words to those distances and ensuring that the understanding matched with a wide variety of different types of data. To argue for this understanding (in other words, what appears in the analysis chapters), I first present the overarching understanding and then construct an analysis that integrates the variety of different forms of data as well as specific quotes that illustrate average usage as I saw in my close readings. This analysis ultimately does not seek to be comprehensive in explaining every particular usage of a function word, but rather seeks to pull out what usages are com-

be decided upon based on the particular explanation being probed. In this case, part-of-speech tagging was especially useful in many cases because I was working with function words which are typically tied to unique grammatical patterns. If an analyst was instead working with all words in a corpus, grammatical form may be less useful than hand-tagging a representative sample for identification techniques, for example. Each of these are perfectly valid approaches, so long as they help the analyst to refine their explanation.

mon and appear to be contributing to the overall distance between groups as seen in the computational results. In doing so, the methodology for these analyses values how many pieces of data all point toward the same understanding of the corpus, which provides insight into the corpus itself, helps to refine the theoretical concepts of disciplinarity and genre, as well as to develop rhetorical theory around the rhetorical work of function words. The three analyses each provide different understandings and insights, demonstrating as a whole how changes in the viewing angle of a corpus cause shifts in the interpretive processes and insights.

3.3.1 A high-level view (chapter 4)

Chapter 4 takes a high-level view of the corpus as represented in the stylistic model, to investigate what we learn by looking broadly at how the results divide academic writing. In the results (presented in chapter 4) it becomes clear that the highest-level distinction between disciplines made in the corpus is between the humanities and sciences. This split is then used a sort of proxy for understanding the high-level distinctions in disciplinary writing that allow for such accurate classification based on function word frequency.

Given this split, we can begin interpretation by looking at what makes up the distance between the groups. In other words, which words are driving the total distance between the humanities and sciences? An alternative way of framing this question is, if I took the distance between every possible pair of humanities and science texts⁵ and added them all together, which

⁵While the humanities and sciences texts divide relatively neatly into two groups in the results, it is not a completely clean divide: some texts from political science intermingle in the visuals with texts from statistics, for example. For this analysis, I created the humanities and sciences groups based on their disciplinary classification (where the humanities group includes education, history, linguistics, literature and literary theory, performing arts, philosophy, political science, and sociology, and the sciences group includes cell and molecular biology, chemical engineering, computer science, physics, and statistics), using the highest-level split seen in Figure 4.2 to guide the groups. In practice, because this corpus splits relatively cleanly, using groups based on disciplinary classification rather than the precise divisions in texts made in Figure 4.2 is likely to have little to no effect and allows for claims to be made about the humanities and sciences rather than group-that-is-largely-humanities or -sciences. If the corpus did not split so neatly into different groups with pre-existing classifications, it would be advantageous to investigate each group as it appears at dendrogram-level (or to be more complex but also more nuanced, cluster-level)

words are making up the largest pieces of that distance (recall that the total distance is a summation of many small distances)? Mathematically, we can calculate how much distance (in total) is contributed by each function word to the distance between every possible pair of humanities and science texts. The original Delta measurement produces a distance between a pair of texts. Below is a modified Delta, which instead calculates the distance contributed by each word to the total distance between two groups of texts. It reads as follows:

$$D_{(i)} = \sum_{c=1}^P \left| \frac{A_i - \mu_i}{\sigma_i} - \frac{B_i - \mu_i}{\sigma_i} \right|$$

where, D = the total distance contributed by a function word i summed over every pair of humanities and science texts; i = the function word being analyzed; c = the combination of humanities and science texts being compared; P = the total number of unique pairs of humanities and sciences texts (for this particular corpus, that is 625,000 possible unique pairs); A , B = texts being compared; A_i = frequency of a given feature i in text A ; B_i = frequency of a given feature i in the text B ; μ_i = mean frequency of a given feature i in the corpus; σ_i = standard deviation of a given feature i in the corpus.

This calculation produces, for each function word, the total distance that it contributes to the space between the humanities and sciences. The differences between the two version of Delta, and what the modified Delta is producing, can be visualized as in Figure 3.1. The words with the largest distances appear in Table 4.1 in chapter 4. To begin analysis, I isolated five texts for each discipline being compared which were closest to the average frequency of the word in the discipline, and then proceeded with the reading, analysis, and interpretation. For examples, for this chapter, where *the* is frequent in the sciences, I isolated the five science texts that had frequencies of *the* closest to the average of *the* in the sciences, and the same for the humanities.

3.3.2 Metadisciplinary span (chapter 5)

Chapter 5 takes a similar approach methodologically to chapter 4, but refines the broad framework developed in chapter 4 by investigating what makes up the “span,” or range of variation within the humanities and sciences. As

splits.

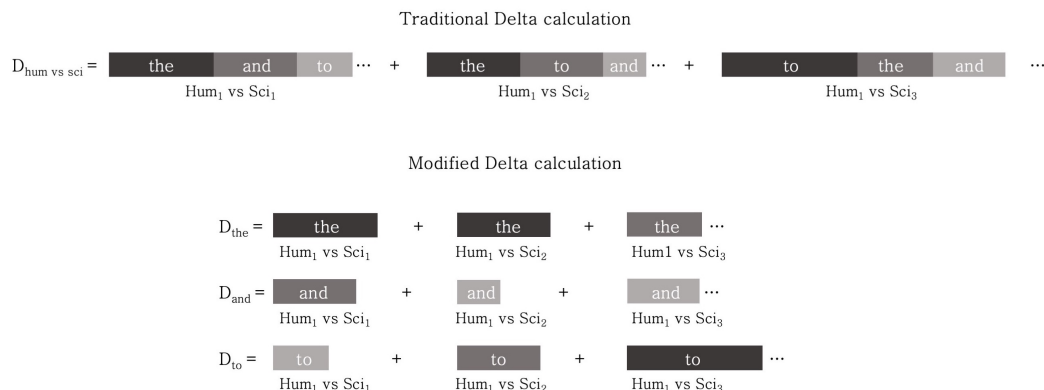


Figure 3.1: A visualization of how the traditional Delta calculation has been modified to produce distances contributed by function words between two groups, as opposed to distances between texts.

such, it compares the two disciplines that sit at the far ends of both the humanities (history and philosophy) and sciences (cell & molecular biology and computer science).

With two pairs of two groups to compare, the calculations proceed essentially the same as outlined in section 3.3.1, including the use of the modified Delta formula, but with the substitution of the disciplines for humanities and sciences. These calculations produce the total distances that each function word contributes to the space between disciplines. The words with the largest distances for each pair appear in Tables 5.1 and 5.2 in chapter 5, and are the starting point for analysis.

3.3.3 Intradisciplinary variation (chapter 6)

Chapter 6 is not quite as fundamentally comparative as the earlier chapters, in that it looks instead at the range of variation within two disciplines respectively (computer science and history). To do so, it uses the modified Delta formula to calculate which function words contribute most to the total distance between every possible pair of intradisciplinary texts. These words do not necessarily point to the same dimensions of variation as they do in chapters 4 and 5, but instead each point simply to words that vary the most (in regards to the Delta distance they contribute) within each disciplines.

This chapter also calculates the correlation between the top words produced for each discipline, using Pearson's R value, in order to identify if any of the words do happen to be pointing to the same dimension of variation (in practice, just one pair of words in computer science correlate). These values all appear in Tables 6.1, 6.2, 6.3, and 6.4. Because this chapter focuses on only one discipline, I isolated ten texts for each word – the five with the highest and lowest frequencies of each word – in order to probe at the range of word usages in the discipline.

Chapter 4

A view from 30,000 feet: the humanities and sciences

This chapter begins a three-part extended case study to illustrate close statistical reading in action. In particular, this case study interprets a model of a corpus of academic writing created using stylometry methods. Stylometry is a well-established text analysis technique that typically is used to classify texts in corpora categorically, for example by genre (Gianitsos et al., 2019) or author (Craig, 1999; Mosteller & Wallace, 1964). The method has been shown to be remarkably accurate in classifying texts, and traditionally these classifications are based on the frequencies of the most common words in texts, which are overwhelmingly function words. Function words – the “little” words of the English language like *the*, *and*, or *of* – are not the types of words we might expect would distinguish texts by genre or author based on more theoretical understandings of those concepts, but computational linguists have found that they do, in fact, work remarkably well.

In applying stylometry methods to a corpus of academic writing across disciplines, the technique – perhaps surprisingly to rhetoricians and unsurprisingly to computational linguists – classifies texts by discipline well (see Figure 4.1). But even though we can classify texts by discipline well, what do we learn about method and the textual instantiation of disciplinarity by looking at how the computational analysis divides disciplinary writing? This question guides the entire case study. This chapter, as part one of the case study, focuses on interpreting the classification of disciplines by function words at a very large scale – at the level of metadisciplines – while chapters

5 and 6 dive deeper into inter- and intra-disciplinary distinctions to develop nuance in both methodology and rhetorical insights about disciplinarity in texts. This chapter first demonstrates how well function words classify the corpus, and then uses close statistical reading to investigate what those results mean. That is, though we might be able to classify by *metadiscipline* well, what do we learn? Regarding a rhetorical lens to academic writing, how do function words contribute to the ways academic texts perform disciplinarity and represent knowledge claims? How does an understanding of how function words distinguish metadisciplines match with our theoretical understandings of disciplinarity and representation of knowledge claims as made in texts? What kinds of rhetorical and linguistic insight does this analysis provide at a large scale, and what does it not? These are the questions that guide this chapter.

4.1 Disciplinarity, knowledge production, and linguistic representation

The analysis in this chapter¹ takes a high-level view of a corpus of academic writing to look at the large-level split in disciplines that classifying by function words creates (in particular, the classification creates a divide between the humanities and sciences broadly). In taking this 30,000-foot view, the analysis engages with the overarching ways in which scholars argue for knowledge claims across the academy. Knowledge claims and how they are constructed and argued for has been a topic of philosophical debate since at least ancient Greece, with Plato critiquing definitions of knowledge in *Theaetetus* and investigating the relationship between truth and rhetoric with the Sophists in *Gorgias*. In these kinds of debates, philosophers grappled with the kinds of processes that led to the creation of new knowledge: some philosophers from Descartes through the early 1900s might have responded with something about the role of verifiability and logic in developing scientific fact, especially with the development of the logical positivist movement. Mid-twentieth century thinkers like Kuhn (1962), Toulmin (1972), or Lakatos (1978) would have instead argued for the social elements involved in scientific progress and knowledge production.

But while philosophers of science have theorized about the social influ-

¹Some of the analysis that appears in this chapter also appears in Ringler (2021).

ences over what gets counted as valid research or what kinds of studies might be undertaken or interpreted in what way, rhetoricians and language scholars would point us to the role of texts for socially constructing and transmitting that knowledge. Bazerman (1992) in particular argues for the importance of focusing on texts when considering knowledge production in the academic disciplines, claiming that the text is “the medium through which knowledge is transmitted and is the matter in which knowledge is embodied outside the consciousness of any individual” (p. 31). He cautions against conceiving of texts and knowledge completely in terms of social constructivism, while still recognizing that texts gain meaning only through social acts of reading and negotiating interpretations between author and reader. The text itself has a critical role in producing knowledge for a discipline in that it “places itself within its multiple heterogeneous contexts and creates a temporary stable place within which the text exists and within which the reader may confront the text and construct a meaning” (p. 35).

Texts, then, are locales for socially transmitting and arguing for new knowledge claims. While the act of writing itself has a role in fully producing and constructing knowledge claims (e.g., Flower and Hayes (1981) and Knoblauch and Brannon (1983)), the final written product of the academic text rarely represents the entire inquiry process that led to the new claim: for instance, in many cases, disciplinary experts see writing the text as only representing the thinking that came *before* the act of writing itself (see, Kaufer and Young (1993)). For the purposes of this analysis then, texts will be analyzed in terms of the how they argue for the claims they make rather than assuming that those claims represent the actual knowledge-making processes behind those claims. Arguing for knowledge claims in disciplinary communities is a careful and social process: in crafting a text, authors represent the processes and standards that their disciplinary community needs for accepting their argument. For example, a biologist who outlines the details and results of an experiment or a literary historian who recounts the historical context around a novel are both supporting their final claims by narrating particular lines of inquiry and knowledge-producing processes.

These lines of argument are plural and many though as we consider the wide range of disciplines and types of knowledge in the academy (and indeed, vastly more so when we look outside the academy). Universities often distinguish disciplines by colleges and then departments, wherein you may find a humanities college with English, History, and Philosophy departments. But rather than distinguishing disciplines in terms of the specific content mat-

ter that they engage with, writing theorists like Carter (2007) have found it more productive to conceive of disciplines as “ways of knowing and doing” rather than static repositories of knowledge. For example, Carter categorizes disciplines like animal science, accounting, and engineering as “problem solving” disciplines, and other disciplines like biology, chemistry, and sociology as “empirical inquiry” disciplines. In a similar way, genre theorists like Miller (1984) conceptualize genres not as specific forms, but as groups of texts that do particular types of “social action.” Especially because of the constantly shifting state of knowledge in a discipline as it grows over time, thinking of a discipline as a particular way of knowing and doing around an idea, or a group of texts that do certain social actions, allows for a more stable description of the work that the discipline does.

These particular ways of knowing and doing are precisely the knowledge-making processes that happen in a discipline, and shape the ways that knowledge gets represented in texts. Carter (2007) describes different kinds of metagenres like the scientific article or literary criticism, and explains that they represent the epistemic commitments of the metadisciplines they are tied to. For example, the scientific article is modeled around the process of empirical inquiry, which is heavily present in disciplines like physics or chemistry. Certain generic features and regularities within texts or disciplinary writings thus give a window into certain regularities within the social acts of what disciplines do and how. MacDonald (2010) echoes this sentiment, claiming that “if academic writing is a form of knowledge making, then differences in knowledge problems or ways of addressing such problems should account for much of the variation among the disciplines” (p. 21).

4.2 Function words and the humanities/sciences divide

Investigating the textual variation in disciplinary writing offers a window into how particular types of knowledge claims are socially made. As such, language scholars, and linguists especially, have used a wide variety of methods to analyze disciplinary writing, each acting as a unique lens on the argument structures and commitments across disciplines. A full review of this kind of work would be far too extensive to undertake here, but representative examples include distinguishing common topoi in writing about literature

(Wilder & Wolfe, 2009), analyzing how citation practices vary grammatically in biology writing (Swales, 2014), and describing how register varies across disciplines (Gray, 2015). This analysis uses function word frequency as one particular lens into the differences in knowledge claims. Figure 4.1 shows how disciplinary writing clusters together based on the frequency of function words.²

Figure 4.1 demonstrates that, based on the frequencies of function words, disciplinary writing appears to cluster together. In other words, texts within the same disciplines appear to have similar frequencies of function words and therefore small distances and thicker lines between them. The visual shows two large groupings of disciplines on the left (sciences) and right (humanities), suggesting that similar frequencies of function words occur between groups of similar disciplines, or metadisciplines (Carter, 2007).

The split between the humanities and sciences becomes even more pronounced when the distances are visualized on a traditional dendrogram using cluster analysis, as in Figure 4.2. The dendrogram in Figure 4.2 is created by drawing connections between whichever two texts have the smallest distance between them, and then working up this way to build the whole tree. In this sense, it privileges the first-nearest neighbor when mapping how close texts are. The diagram in Figure 4.1 (created using a technique described by Eder (2017)) is especially informative because the connections depicted take into account not only the first-nearest neighbor, but also the second and third. In that sense, Figure 4.1 gives a more nuanced view as to the many relationships between texts, while Figure 4.2 allows for an easier visualization of groups.

From these figures, we can see the the computational analysis here is working well in terms of its classificatory functions: the disciplinary splits are obvious in Figures 4.1 and 4.2.³ But what do we learn about disciplinary writing by looking at these results? Focusing in on the largest split in the figures between humanities and science writing, Table 4.1 shows the top four

²The measurements used in Figure 4.1 were calculated using methods from stylometry. See chapter 3 for the full technical details on these calculations and how they were implemented to create the data used in this chapter.

³Evaluating statistical clusters and being able to talk robustly about the significance of statistical clustering is a known difficult problem (Feldman & Sanger, 2007). Because this dissertation does not look to develop new statistical techniques for clustering, and significance is not critical to the results of the analysis, we can reasonably rely upon human judgment to visually recognize a clear similarity between the pre-determined labels and the cluster visualization.

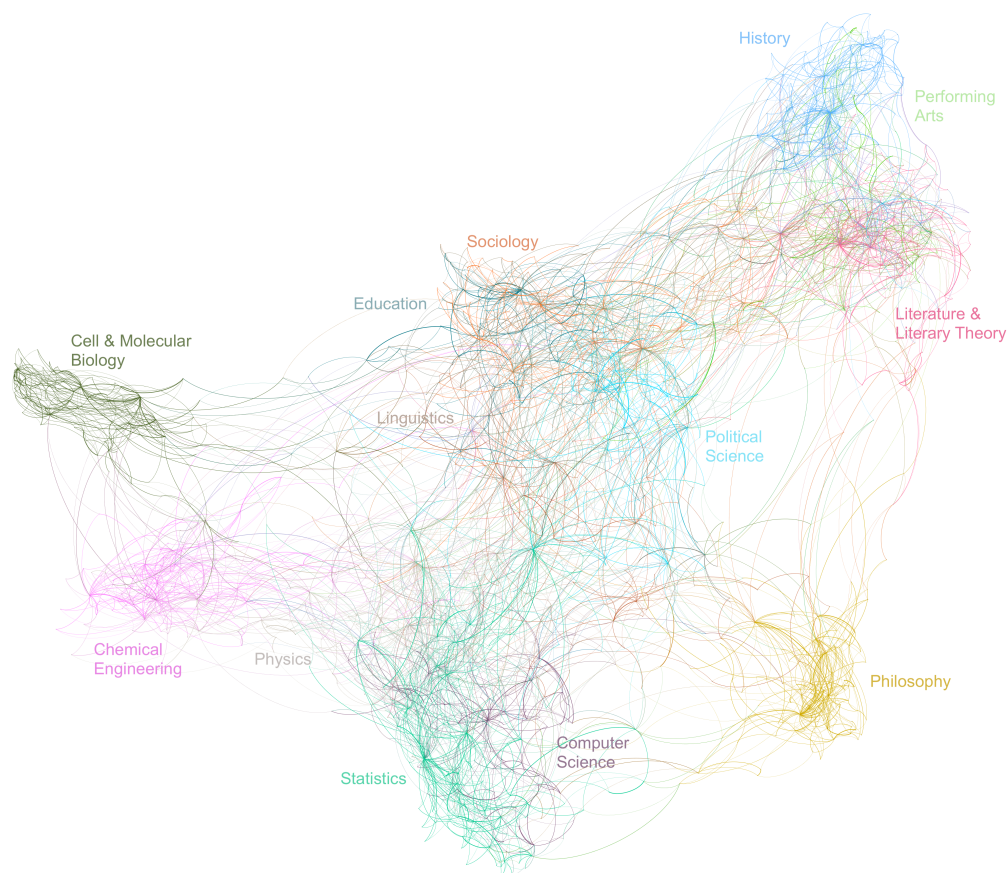


Figure 4.1: A visualization of the distances between academic journal articles from 13 various disciplines, with distances calculated using Delta (Burrows, 2002). Every point represents one unique article, and thicker lines and smaller distances between texts indicate that the texts are more closely related in terms of their distributions of function word frequencies (for more detail, see Eder (2017)). For reference, each text is colored according to its discipline.

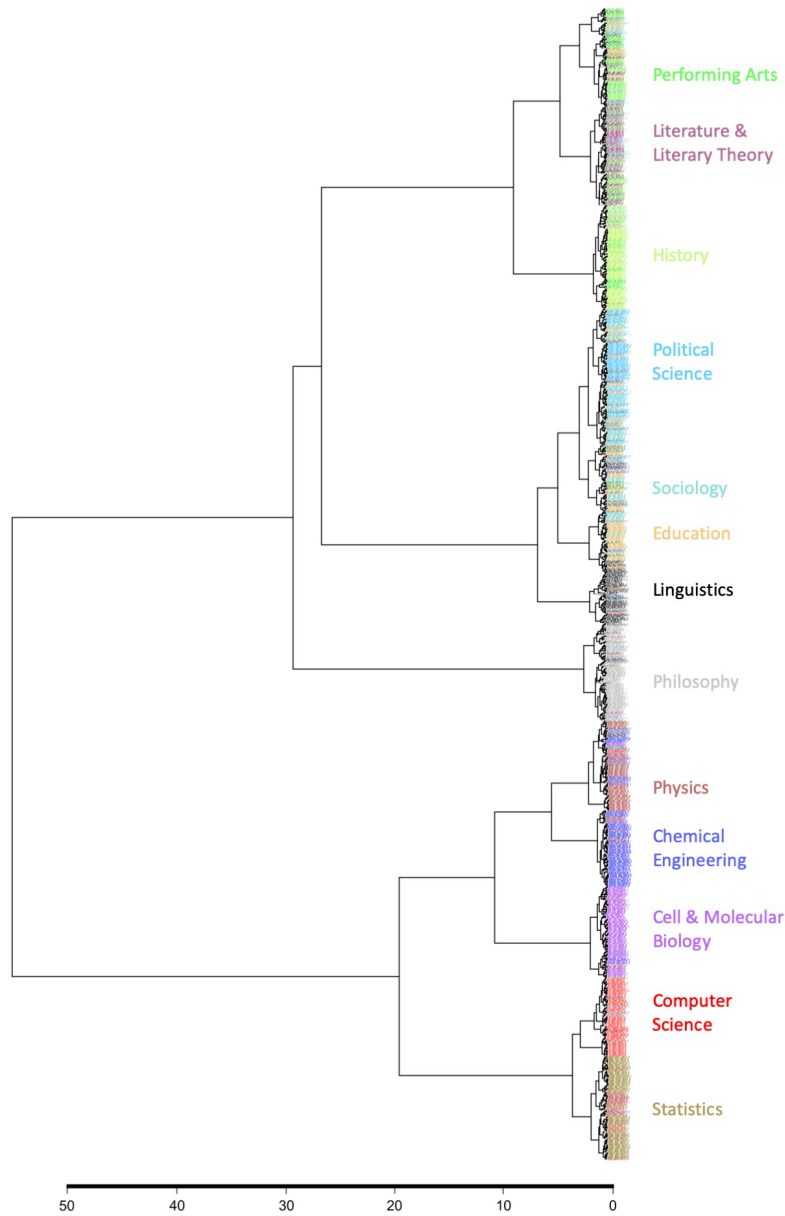


Figure 4.2: A dendrogram created using agglomerative hierarchical cluster analysis, with distances calculated using Delta (Burrows, 2002). Each end-point represents one text, which are colored according to their disciplines.

	Delta distance	Discipline	Log- likelihood	Effect size	N
<i>the</i>	794,889.6	Sciences	286.41	0.053	864,090
<i>with</i>	786,419.2	Sciences	1,367.87	0.347	98,942
<i>for</i>	784,973.8	Sciences	1,599.56	0.319	137,746
<i>but</i>	778,957.4	Humanities	5,001.60	1.372	32,883
<i>by</i>	759,898.5	Sciences	385.97	0.205	81,460
<i>not</i>	756,177.8	Humanities	5,471.20	1.045	60,362
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>who</i>	751,306.2	Humanities	9,514.77	4.188	15,109
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>to</i>	744,783.4	Humanities	7,502.43	0.491	304,778

Table 4.1: The top 4 function words per discipline that contribute most to the distance between the sciences and humanities (note: the table is sorted by Delta Distance, but edited to only show the top 4 words per discipline). The Delta distance column shows the sum of the Delta distances between each science and humanities text for that particular function word (using the modified Delta formula described in chapter 3). The discipline column shows which discipline (sciences or humanities) that the function word is most common in, based on which discipline has the highest average z-score for that word. The table also shows the log-likelihood measure of the word for the discipline it is most common in (using the less-common discipline as a reference corpus), the effect size of the word for the discipline it is most common in using log-ratio (Hardie, 2014), and the total number of instances of the word across all 125 texts for that discipline (N).

function words per metadiscipline that contribute most to the entire distance between humanities and science writing as represented in both Figures 4.1 and 4.2.

If we think of regular, recurring linguistic features as lenses into how disciplines represent knowledge claims, function words suggest a unique and statistically robust insight into these differences. That is, these words seem to capture something about what we normally conceive of as a discipline. But our theoretical conceptions of disciplinarity or genre have more to do with “ways of knowing and doing” and “social action,” rather than the frequency of *of*. What exactly do the function words in Table 4.1 reveal about the work

of these larger disciplines in terms of their sociorhetorical action, their “ways of knowing and doing”? What theories of knowledge representation across the sciences and humanities do they support?

In the rest of this chapter, I focus on a few top words from Table 4.1 and some of their common uses. It is not my goal to explain how every individual function word works toward disciplinary goals, nor is it my goal to explain every way that one particular function word might work in a discipline’s writing. Rather, by focusing on some of these top words and some of their regular uses, I hope to illustrate how these words might be made sense of and interpreted as linguistically undergirding larger rhetorical moves, and then use these interpretations to theorize more broadly about the types of social actions and argument processes happening in the disciplines.

4.3 A theory of humanistic and scientific knowledge claims

4.3.1 Humanities

In analyzing the function words indicative of the humanities, we ultimately learn that arguing for knowledge claims in the humanities is driven by interpreting and analyzing human action. This means that the analysis typically starts with stories about people. As such, humanities papers are filled with narrative language: many humanities papers have large sections or scattered chunks of the paper that are recounting what people, characters, or groups did. On average, humanities texts are made up of at least 2.9% explicitly narrative-based language (as opposed to only 1.6% in the sciences)⁴. These stories then form the basis of analyses, providing human actions and decisions to explain and interpret. Using narrative stories as the basis for analysis is especially supported by the word *who*.

Who as a relative pronoun in the humanities supports storytelling from the past. As a whole, humanities texts spend a large amount of time talking about people – in fact, on average, humanities papers have roughly three times the amount of character language as science papers (4.3% vs. 1.3% of the text), with most of that space being used to name specific character

⁴These percentages, and others like it throughout the analysis, are calculated using Docuscope (Ishizaki & Kaufer, 2012; Kaufer & Ishizaki, 2020)

types (1.9%) or proper names (0.9%). As a relative pronoun, *who* refers back to these specific names and characters in order to continue discussion about them. In particular, *who* allows isolation and naming of specific groups of people, or adding relevant detail about people that is useful for analysis. For example, in a history paper analyzing the collection and exhibition of photographs during the WWII liberation of Paris, the author writes the following about photographers and German soldiers:

- (1) Almost every photographer who was active during the Liberation captured at least one barricade. (Clark, 2016, p. 834) [History]
- (2) But while public photography was thus restricted for the French, the German soldiers who visited Paris came armed with cameras. (Clark, 2016, p. 840-841) [History]

The *who* in each case clarifies the specific group that the author is talking about: not all photographers, but photographers who were active during the Liberation; not all German soldiers, but German soldiers who visited Paris. Naming specific groups of people by specifying the subgroups with *who*-clauses allows for discussion about subgroups who did interesting things for the analysis. By using these constructions, the author is able to highlight an important characteristic for the analysis. These allow for further discussion on why the photographers captured photos of barricades, and why the German soldiers in Paris had cameras and what they did with them. Casting various people and groups as part of larger groups supports a sort of movement between individuals, subgroups, and groups that allows adding detail to the specific people in question in order to put them into a narrative structure where they do actions.

Other types of similar constructions do not necessarily cast the person as part of a group in the same explicit way, but add some kind of relevant analytical detail about the people being discussed and also allow more complex storytelling by connecting separate clauses together with the same character. For example, in a theatre research article about the Shigang Mama Theatre Group in Taiwan and how their theatre explores their own postcolonial cultural identities in the wake of an earthquake, the author describes scenes from a particular performance:

- (3) The second scene, “The Raining Sound of the Temporary House,” depicts the childhood memory of a Hakka woman, Yu-Chiung, who,

along with her grandmother, was evacuated from their house after it collapsed during the earthquake. (Hu, 2013, p. 452) [Performing Arts]

The use of *who* here allows this sentence to have two main statements of action: the second scene depicting the memory of a Hakka woman, and the woman being evacuated from her house after the earthquake. The statements of action are intimately connected by Yu-Chiung, who serves a vital role in both. The *who* serves as a relative pronoun that restates “a Hakka woman, Yu-Chiung” that was established in the first main clause, in order to use that same character as the main subject in the second clause. These kinds of constructions support a storytelling that is more complex because it allows both the movement of characters from one statement of action to another, as well as the intimate connecting of those two statements of action. In this case, it is clear that the memory being depicted in the second scene is related to the evacuation of Yu-Chiung and her grandmother. This relation is clear primarily because of the connecting of the two sentences through the relative pronoun; if the clauses were split into two sentences, the author would need to do more work to explain that Yu-Chiung’s evacuation is related to the memory, and not just a separate detail about her given for context. In this way, the relative pronoun is supporting more complex storytelling through allowing the movement of characters throughout the story’s many actions.

While *who* supports the telling of stories and description of characters in them, *to* supports analysis of those characters and the actions they take in the stories. In many cases, *to* is functioning as an infinitive particle that connects two verbs. Connecting verbs in this way supports both building complexity in actions and explaining what those actions mean and do in the world. For example, in the same theatre article, the author writes frequently about what these women did, pairing two verbs to do so:

- (4) In the play titled *Mama’s Huo-Fang* at the end of this stage, the Shigang women attempted to enact the issue that the reconstruction of the Hakka *huo-fang* created between four couples in the Liu family in the form of forum theatre to portray how they handled the benefits, emotions, and land rights within the family. (Hu, 2013, p. 450-451) [Performing Arts]

The first *to* in (4) connects *attempting* and *enact*, forming a connection between the Shigang women and their enactment of the issue. In this construc-

tion, the Shigang women do not simply enact the issue, but they “attempt” to enact the issue. Similar types of constructions appear in an education research article about childhood assessment interviews:

- (5) During Tyrell’s spring interview, the first author was highly focused on Tyrell’s comfort level, not wanting to create for him the discomfort that he experienced in the fall. (Parks & Schmeichel, 2014, p. 527) [Education]

Again, the *to* in (5) creates a connection between *wanting* and *create*, or between the first author and their potential creation of the student’s discomfort. In this sentence, the first author is not *wanting* to create rather than simply not creating.

The *attempted* and *wanting* in (4) and (5) hedge the successfulness of those actions. These construct an analysis that is postulating about the intentions of the actors, and shifts the focus of the sentences to explaining the actions rather than just describing or recounting them. That is, the author is not saying whether or not the Shigang women were successful in enacting the issue, but she is drawing a link between the action that the women took and the effect they wanted to have in the world; similarly the education authors draw a link between the action that the first author took and the effect that she wanted to have on Tyrell. In doing so, the authors frame their recounting of these events as analysis that postulates about what human actors try to do with their actions, rather than a simple narrative or a blanket statement about what the actors did or did not achieve. This points to some of the contingency around human actions, and allows space for and explicitly signals their interpretations.

This focus on interpreting human actions and the effects they have in the world is signaled more explicitly by *to* in cases where *to* is a preposition that functions as a shortened form of the phrase *in order to*. The second *to* in (4) functions in this way, connecting the actions of the Shigang women to their purpose: the Shigang women enacted this issue *in order to* portray how they handled issues within the family. Similar forms occur in (6), from the same theater article:

- (6) In the fourth scene, “All Their Families...,” women burn incense to honor Hakka ancestors, to symbolize the class restrictions in Hakka families, and to depict the experience of one non-Hakka mother, Yue-Shia, whose marriage was arranged by her future family-in-law to

improve the farming business of a Hakka family. (Hu, 2013, p. 452)
[Performing Arts]

The first three *to* constructions in (6) all point to various symbolic purposes of the women burning incense on stage, and the final *to* signals the reason that Yue-Shia's marriage was arranged. In all of these cases, the *to* functions as a link between an action and a reason why. While *to*'s use as an infinitive particle in (4) and (5) signals some contingency around the success of an action and the actual actor's purpose, *to* used as a preposition of purpose akin to *in order to* allows the author to speak about the interpreted purpose of those actions. That is, both functions of *to* support discussion about the purposes of actions, whether as intended by the agent or as interpreted by the author. By creating these discussions in their analyses, the authors of these pieces work towards creating humanistic interpretations of their subjects.

As a whole, *to* creates links between actions that people take, and what those actions mean. The analyses, or interpretations of meaning, are also characterized by their complexity, supported in part by the high frequency of the word *but*. *But* as a conjunction introduces a degree of contrast to whatever was said before. In humanities research, this contrast is especially useful for recognizing the multiple meanings or interpretations that could be present in an analysis. For example, in a history paper analyzing the collection and exhibition of photographs during the WWII liberation of Paris, the author discusses the significance of barricades in the photographs:

- (7) Barricades became places not just of the assertion of popular sovereignty, but also of the construction of communal solidarity during a joyous yet uncertain time. (Clark, 2016, p. 833-834) [History]

Later the same author discusses the historical significance of the liberation story told by photographs:

- (8) By harking back to the city's long history of revolution, the myth of the Liberation would work not only to define the significance of this event long into the future, but also to redefine the historical events to which it referred. (Clark, 2016, p. 830) [History]

In (7) and (8), the author is adding layers of interpretation by attaching the final clauses with the conjunction *but*: barricades assert popular sovereignty and construct solidarity, the myth is both defining significance and redefining

historical events. Adding multiple layers of interpretations speaks to the complexity of the events being analyzed, and allows the author to mark both of those in the analysis. That is, *but* creates a sort of analytical (as opposed to descriptive) complexity.

The choice to use *but* rather than *and* is an interesting one, though. In (7) and (8), *but* pairs with the *not just* and *not only* that precede the *but*-clauses. Framing the two interpretations in this way downplays the first one and highlights the second. This construction allows Clark to mark that the first interpretation is valid and important, though perhaps not the most interesting or nuanced piece of the analysis. Using *but* in this way makes space for the author to create complex layers of interpretation, but also to signal the important pieces of the analysis in a way that *and* glosses over by creating equality between pieces.

But takes a slightly different form in (9), where it appears in a literary theory paper about the representation of minority family relations under neoliberalism in Jhumpa Lahiri's memoir *Unaccustomed Earth*:

- (9) The structure of feeling represents what is emergent in a society, not yet realized in institutions or ideologies, but nonetheless able to “exert palpable pressures and set effective limits on experience and on action” (132). (Koshy, 2013, p. 349) [Literature & Literary Theory]

The sentence in (9) is not yet analyzing the story, as the previous instances of *but* did, but is setting up the theoretical significance of “the structure of feeling” which is something that will be analyzed later. Here the author is first setting up a non-feature of “the structure of feeling” by noting that it is *not* realized in institutions or ideologies, and then using *but* as a contrastive conjunction to explain what it can do. This construction contrasts what the feature does versus something it may be assumed to do: by stating that “the structure of feeling” is not yet realized, the author creates a sort of dialogic negation that constructs its possible realization (Martin & White, 2005, p. 118). In this case, the *but* does not add two layers of analytical complexity, but counteracts a possible interpretation or feature of the subject. This is, in a sense, a type of complexity, though one that reacts more explicitly to possible beliefs an audience might hold about a term, rather than building up two interpretations provided exclusively by the author.

But as a contrastive conjunction that builds complexity that responds to other beliefs is more explicitly apparent in philosophy, where it serves a

role more closely aligned with logic. (10) is drawn from a philosophy paper that argues against previous claims that the conservativeness of deflationist theories of truth leads to objections when applied to arithmetic:

- (10) Ketland seems here to say that the mere existence of a non-conservative theory of arithmetical truth that proves G poses a threat to deflationism. But this line of thought only goes through on the premiss—which is the crux of the current dispute, and which I take it that Tennant will deny—that the relevant theory of arithmetical truth (in this case, PA_T^+) is properly understood as deflationary. (Waxman, 2017, p. 466) [Philosophy]

In (10), the author is responding to a claim by Ketland by first stating Ketland's claim, and then using *but* as a contrastive conjunction to state that Ketland's claim relies on a certain premise that he does not adhere to (the next several paragraphs go on to explain why). This form of *but* responds more directly to other people's possible interpretations or claims by contrasting another line of thought with them. (11), from the same philosophy paper, shows this same form of *but*, but where the other possible interpretation is sourced from the logical buildings of the author himself:

- (11) I am much more doubtful of (ii). The best reason I can think of for endorsing (ii) appeals to the claim that proof-theoretic derivability is a necessary condition for mathematical or logical explanation. But—to anticipate issues that arise in §5—if we countenance a notion of logical consequence richer than proof-theoretic derivability, this condition will appear unmotivated. (Waxman, 2017, p. 466) [Philosophy]

In (11), the *but* contrasts a reason given by the author himself in order to introduce another piece of a logic puzzle. Using *but* as this kind of contrast to other interpretations allows the author to counteract other possible interpretations with ones of their own. Whether used in this way, or in a more purely conjunctive form to add another layer of interpretation that is not strongly negated, *but* supports a building of analytical complexity and recognition of multiple interpretations in an analysis.

Kaufer and Butler (2000) explain that argument “splinters the inner world of representation into alternatives, branching paths that define a decision space” from which the writer “systematically explores these alternative paths,

evaluates them, and either recommends an existing path or invents a new one to recommend” (p. 237). In this sense, argument is based on having multiple possible representations, and bringing the reader to a decision out of them. *But* functions in several nuanced ways in these examples, but all of which create nuance and complexity to the writer’s arguments; they allow for discussing multiple interpretations and contrasting them against each other, so that the writer can build towards a final argument. In humanities writing, *but* is a grammatical hallmark of argumentative writing in that it allows space for multiple interpretations and analysis that is necessary for analyzing the complexity of human action so that the writer can navigate through them to their final argument.

4.3.2 Sciences

Turning to an analysis of the function words indicative of the sciences, we ultimately learn that arguing for knowledge claims in the sciences is often very localized to describing and theorizing about the physical and technological worlds. As such, science papers are frequently filled with in-depth descriptions of physical objects or processes: biologists may describe a particular cell, or computer scientists might recount a particular process for data to be calculated computationally. On average, at least 5.4% of science papers are taken up with descriptive language, as opposed to 3.6% in the humanities. Large parts of the papers are full of specific descriptions, which are then used to theorize more broadly about those objects and processes. The heavy attention to physical description is most noticeable through the regular use of the definite article *the*.

The in the sciences supports description of particular objects, as opposed to abstract or general objects. It allows for naming specified, definite tools, processes, data, and so forth. In fact, in all of the noun phrases in the sciences corpus that start with *the*, 45.1% of the words after the determiner in these noun phrases are either academic terms or descriptive words (as opposed to 35.4% in the humanities – humanities *the*-noun phrases have much higher occurrences of character references, narrative terms, and public entities). For example, in a physics paper on the photodissociation dynamics of H_2O using a VUV-induced pump-probe approach, the authors write the following in the introduction when briefly summarizing their study:

(12) The interpretation of the delay-dependent dynamics of the water

isotopologues is supported by a mixed quantum-classical approach to calculate not only the molecular trajectories in the dissociative $\tilde{A}_1B_1$ state, but also investigate the influence of different final electronic states, accessible by photoionization with a single probe photon, on the observed delay-dependent yield. (Baumann et al., 2017, p. 2) [Physics]

The first instance refers to the interpretation they do in their own paper, but the remaining uses all specify particulars: the authors are interpreting *the* specific delay-dependent dynamics, investigating *the* particular water isotopologues, studying specific molecular trajectories in a particular state, investigating *the* exact influence of states on a specific yield. Each of these uses of *the* implies that there is only one possible instance of each. In other words, there are not multiple types of or possible delay-dependent dynamics, nor conceivable other $\tilde{A}_1B_1$ states, nor possible other observed delay-dependent yields: the text itself makes it impossible. By determining each of these with *the*, the authors restrict the possibility for other possible interpretations or meanings of those specified objects, and create only one true, possible object. Similar types of patterns show up in the results section of the same physics paper:

- (13) In the excitation scheme employed in this pump-probe experiment, the first VUV photon (7.7 eV) excites the molecule from its electronic ground state (black) into the $\tilde{A}_1B_1$ state (red). Since this state is unbound in the direction of the asymmetric stretch vibration, the molecule starts to dissociate. (Baumann et al., 2017, p. 4) [Physics]

The process described here of a photon exciting a molecule is a specific, definite one: there is one first VUV photon, a specific molecule, state, direction, and vibration. This same pattern of specification occurs not just when naming physical objects, but also when referring to specific methods and processes, as in this statistics paper:

- (14) The optimal testing rule is derived in two steps: first the hypotheses are ranked optimally and then a cut-off is chosen along the rankings to control FDR precisely. The optimal result on ranking is obtained by connecting the multiple-testing problem to a weighted classification problem. (Sun et al., 2015, p. 65) [Statistics]

(14) especially demonstrates the usage of *the*, as opposed to the indefinite article *a/an*, because this statistics example is actually referring to an abstract type of problem rather than the exact one they are actually engaging with. This sentence could be easily written using *a*, but the authors choose *the*. The article thus functions as a signal of the relative lack of contingency in scientific writing as it engages with more regular, predictable objects and processes. In other words, specifying with *the* in this way might be thought of as closing off dialogic space to imagine other possibilities, to invoke theories of dialogicality from Bakhtin (2010) – that is, the authors in (14) might be speaking about an abstract hypothesis or result, but are simultaneously creating the one possible hypothesis or result that they are using in their work.

This function of *the* might be seen more clearly when contrasted to article usage in the humanities. While *the* of course exists in humanities writing, it is less frequent because objects are often framed as plural and many. For example, this sociology paper describes models created to represent how separated parents' custody situations influence the time pressure experienced by those parents, and describes them in the results section:

- (15) In line with our expectations, sole-resident mothers experience more time pressure than mothers with shared residence in Models 1 and 2 (Table 2), although the difference is slightly smaller when conflict is controlled for. When we alternate the reference category, we observe that sole-resident mothers also experience more time pressure than nonresident mothers in Models 1 and 2 (see the notes to Table 2). (van der Heijden et al., 2016, p. 476-477) [Sociology]

The instances of *the* here all refer to the study itself: the difference observed in models, the reference category, and the notes to Table 2. *The* is notably absent, however, when referring to sole-resident mothers, time pressure, and nonresident mothers. In other words, the authors do not write, “In line with our expectations, *the* sole-resident mothers experience...” and therefore do not localize sole-resident mothers to this particular study. Neither do they write, “*the* sole-resident mother experiences...”, creating one model sole-resident mother (as we saw in statistics, (14)). Instead, the authors pluralize the subject, recognizing the wide and varied group of possible sole-resident mothers being represented by the statement. They also make the statement more generally applicable by not discussing only the mothers in this study.

The pluralizing of subjects and not using an article is a notably different approach from that of the physics paper, which referred to *the* specific dissociative $\tilde{A}_1B_1$ state of *the* molecules in that study. For the physics paper, though, we would not expect similar molecules in another similar study to behave differently, nor to appear in a different dissociative $\tilde{A}_1B_1$ state, because such a state is not conceived of as having a possible existence. And given possible changes in the experiment, it is entirely likely that the molecules would behave differently, though would likely not create a new form of the dissociative $\tilde{A}_1B_1$ state. By describing just their particular experiment, there is an implicit recognition that a similar setup would cause molecules to behave similarly because of the regularity and consistency with a non-animate object like molecules. As such, scientists appear to spend large amounts of the paper describing their very particular experiments and setups that were observed in their studies, which allow the use of the definite article *the* to refer to those objects.

And while the sociology paper has reference to these kinds of definite objects like “the reference category” in that experiment, sole-resident mothers lack a localization to that particular study. In doing so, the sociologists are able to extend their findings to other possible sole-resident mothers by speaking in the abstract plural, while still recognizing the contingency and unpredictability of human behavior by not creating a model sole-resident mother. Given that this example comes from sociology, which falls in the social sciences somewhere between the extremes of the sciences and humanities in Figure 4.2, it demonstrates some of *the*’s scientific usage in its reference to the localized study itself, while eliding articles altogether in a more humanistic fashion when discussing the subjects. It is precisely because of the contingency of human behavior that it is advantageous for the authors to extend these findings linguistically by writing about all sole-resident mothers, because it signals a level of confidence that their findings are generalizable to a degree and not specific only to the mothers being studied. This is modulated though, by avoiding the creation of one model mother and thus recognizes some of the variability around human behavior.

The, then, allows for naming objects that are particular to the study, and is complemented by the use of *with*, which allows for adding complex detail to those specific objects. That is, in describing a very localized, particular experiment, scientists also need to be very detailed about naming those objects so as to be precise about referencing the exact objects in that particular study. This particular move is supported by *with*-prepositional

phrases, which allow for creating specificity through extended noun phrases that add detail to the description of certain processes. For example, in this chemical engineering paper on using a specific nucleating agent to create a certain crystal structure, the authors use *with* to specify noun phrases in the introduction:

- (16) In the present work, the DMC-DBH with the shish structure was expected to manipulate the crystal morphology (kebab) of the PBA, which probably further modulates the polymorphism and biodegradation of the PBA. (Yang et al., 2017, p. 7911) [Chemical Engineering]

With allows for the modification of DMC-DBH by specifying its particular structure. Similar patterns appear frequently to modify nouns when describing the results of this paper:

- (17) However, no fractioned crystallization can be observed in the non-isothermal crystallization process (Figure 1a) of the PBA with the weight fraction of 0.1% DMC-DBH. (Yang et al., 2017, p. 7913) [Chemical Engineering]

In (17), *with* allows for modification of the PBA (a polymer) being studied in terms of its weight of DMC-DBH (the nucleating agent). This kind of structure allows the authors to add a lot of detail to nouns without turning them into clunky pre-modifiers like “the shish-structured DMC-DBH” or “the 0.1% DMC-DBH weight-fractioned PBA.”

With also allows for the modification of methods, and for drawing connections between data. In the same chemical engineering paper, the authors describe a band in one of their results figures:

- (18) The 1602 cm^{-1} band is assigned to the -C=O group of neat DMC-DBH, which shifted to 1598 cm^{-1} after blending with the PBA (Figure 4a). (Yang et al., 2017, p. 7915) [Chemical Engineering]

The *with* phrase in (18) allows for the act of blending to become a modifier by specifying what the DMC-DBH was blended with. In a statistics paper, the authors use *with* to show how two concepts are connected:

- (19) Denote by $\mathbf{X}^n = (X_1, \dots, X_n)$ the collection of random variables that are associated *with* locations $s_1^*, \dots, s_n^*$. (Sun et al., 2015, p. 69) [Statistics]

tics]

With in (19) allows the authors to connect variables and locations by drawing a one-to-one relationship between them. Similar types of patterns including *associated with*, but also phrases like *increased with*, *aligned with*, or *in agreement with* occur frequently throughout the sciences, especially when describing results. These constructions allow the writers to show how results are interconnected and dependent on each other, building those kinds of interpretations of results.

Finally, while *the* and *with* support heavy amounts of descriptive detail that specify the exact objects in the study, scientific writing ultimately makes moves to use that description to develop more widely applicable theories. Remembering how Kuhn (1962) conceives of normal science as working to fill in gaps and develop theories under certain paradigms, the large majority of normal scientific writing aims to describe a very specific experiment that was undertaken, and then use the results of that experiment to make some slightly broader contribution to the understanding of major questions in the field. In other words, the descriptive work of experiments serves the purpose of allowing the authors to talk more abstractly and theoretically about some particular concept, and to further develop knowledge around that concept.

The movement from description of a study to theory is especially evident in the frequent use of the preposition *for*. In particular, *for* is used frequently in service of developing models from concrete, specific observations. For example, in this computer science paper, the authors use *for* when describing how to notate and calculate work and span in their extended dag model:

- (20) For a foreground block f (which, recall, is itself a subdag of the overall dag of the computation), we write $W(f)$ and $S(f)$ for the work and span, respectively of the block. For a graph $g = (s, t, V, E, F)$, the *foreground work* $W^\circ(g)$ and *foreground span* $S^\circ(g)$ are the sum over all foreground blocks:

$$W^\circ(g) \triangleq \sum_{f \in F} W(f)$$

$$S^\circ(g) \triangleq \sum_{f \in F} S(f)$$

(Muller et al., 2017, p. 679) [Computer Science]

All three instances of *for* in (20) are allowing for a movement between a specific object and a model. These uses of *for* establish an abstract foreground

block, work and span, and graph – abstractions that don’t necessarily exist in real life, but are modeled by variables. That is, *for* links a variable (f) to what it represents (“a foreground block”). The user of this model could then create an actual foreground block, work and span, or graph, and expect them to be represented by the model’s f , $W(f)$ and $S(f)$, and g , respectively.

The use of *for* to move between specific instance and model is more explicit in (20), but also happens in less explicit ways. For example, in this sentence from a physics paper, the authors discuss the importance of choosing a model function:

- (21) Even when the instrument response function cannot be determined precisely, the choice of an appropriate model function for the delay dependence of the observable is necessary – in this case the time molecule needs to exit the ionization window. (Baumann et al., 2017, p. 6) [Physics]

In this sentence, the model function helps to describe or support the delay dependence. A similar pattern appears in the computer science paper, where the authors describe certain rules that are part of their cost semantics:

- (22) The rule for $\mathbf{fg}(e)$ evaluates e to a background thread and also gets a handle to the sink of the cost graph for the thread’s expression. (Muller et al., 2017, p. 683) [Computer Science]

All three uses of *for* in (21) and (22) point to some feature or characteristic of whatever follows the *for*. However, these expressions are not easily replaceable by a word like *of*, which also can describe features of something as in a phrase like “the cost semantics of the model.” While *of* creates a relationship of ownership (the cost semantics belongs to the model), *for* creates a relationship of support or service (the rule supports $\mathbf{fg}(e)$ but is not necessarily a part of it). The distinction is slight, and muddier in a case like “the cost semantics of the model,” which could easily read as “the cost semantics for the model.” But examples like (21) and (22) help to make this distinction clearer, by demonstrating the supportive relationship that *for* creates that cannot be easily replaced by one of ownership.

The relationship of support and service afforded by *for* allows the authors to create models where various pieces interact: the delay dependence is represented by a model function, $\mathbf{fg}(e)$ is bounded with a rule, the thread’s expression can be represented with a cost graph. Insofar as *for* supports

description in these cases, it is helping to describe the relationships between elements of the models that are being constructed. These models ultimately form some of the larger contributions of many scientific papers, helping the authors to clarify relationships and characteristics of the physical and technological worlds.

4.3.3 A theory of knowledge representation across the academy: Complexity in thought and grammar

The analysis of both metadisciplines ultimately allows for creating an overarching theory of knowledge claim-making in the disciplines that is supported by function words. The analysis ultimately reveals that the words from Table 4.1 are supporting the development of complexity in texts, which takes different forms and is largely predicated on the presence of people.

In the sciences, research frequently starts with cases that are very localized to the physical or technological world: a biologist may investigate a very specific reaction that happens in a cancer cell, or a computer scientist might build and test a particular program or programming language operation. Because of the focus on physical and technological objects, the bulk of the rhetorical work of science writing is in describing particular objects and processes that are localized in the world. From these, scientists build specific and often limited models, but ones that are more easily generalizable to similar situations. We might conceive of this in terms of what Kuhn (1962) calls “normal science,” in that scientists are making often very small contributions that are limited in scope, but are likely part of a larger research agenda (though this research agenda might not always be easily apparent in the text itself). As a whole, science appears to operate in very particular areas and make limited yet precise contributions towards theorizing about those particular objects of study. Because of the relative regularity of the objects of study, these kinds of theories and descriptive work can be productively used in future work, and have a sort of predictive function: for example, a description of how a particular type of atom behaves under certain conditions may be used by later physicists to set up later experiments using that atom.

In representing this work, *the*, *with*, and *for* create a complexity through specificity. *The* supports a naming of physical objects, while *with* supports naming features of those particular objects. *For* allows authors to then build abstractions from those objects to theorize about objects, as in a phrase like

“For a value *a*, ...”. Each of these words support grammatical forms that allow for large amounts of precise detail, and connecting those details to models and theories. In one sense, we might think of these words as allowing work at the stasis level of definition, in that they establish what *is* in the world. By allowing this type of specificity, these function words create a descriptive complexity that points to the process of knowledge creation that is distinctive of the sciences given their focus on the physical and technological worlds and types of claims they make.

The humanities appear to operate quite differently, in that they are fundamentally interested in grappling with and understanding the human experience. They might make larger theoretical claims about what humans do and how they create meaning and act and why, ranging from how social groups behave under certain conditions to how art fits into the cultural milieu of a time period. In doing so, humanists are necessarily dealing with the incredible contingency and complexity that surrounds the human experience. The claims that are made then, are largely made in the service of understanding and interpreting human action, as opposed to prediction or direct future utility. For example, a literary theorist would not necessarily offer an analysis of a piece so as to predict future pieces, but so as to better understand the cultural or social forces that created that piece. Wrestling with understanding the human experience as part of academic inquiry dates back to at least the ancient Greeks, and we find the continuation of that long line of work most clearly in the work of the humanities today.

Because of this contingency and difference in overall goals that is founded on the presence of people, the process of academic inquiry and claim-making in the humanities looks quite different. *Who*, *to*, and *but* create grammatical structures that support this type of claim-making through allowing for complexity in interpretation. In particular, *who* supports naming and re-naming of people with various characteristics and features, and *to* allows for an analysis of the actions of those people (as in, “she acted to represent her people”). *But* supports the building of complexity in those actions, recognizing that one action can mean one thing, *but* also another. For in building an argument, a writer explores multiple paths and alternatives to come to a decision space (Kaufer & Butler, 2000), and *but* supports that type of exploration. In offering these many forms of interpretation, we may think of the humanities as operating frequently at the stasis of definition in its descriptions of what things mean (as opposed to existence in science). Fahnestock and Secor (1991) analyzed writing in literary studies and found

that the discipline collectively values complexity, which they explain as being in part achieved through recurring topoi. The function word analysis here supplements this line of thinking by showing how complexity is achieved not just through general argument structure, but through the specific grammatical constructions that make up each sentence. That is, by grammatically allowing complexity through layers of interpretation, these function words point to a line of argument in the humanities as one of description of people and analysis of what they do and why in all of its complexity and nuance.

4.4 Reflections on a view from 30,000 feet

Humanities and science writing can be classified fairly robustly using function word frequency, with Table 4.1 showing the words contributing the most to that distinction. Ultimately, the function words point us to how complexity in grammatical form allows for complexity in textual representations of knowledge, and a difference in fundamental ways of arguing for knowledge claims that is predicated on the relative presence of people. The humanities name people, describe them, explain what they do in the world, and argue for interpretations of those actions. The arguments necessarily grapple with incredible amounts of complexity and nuance that is fundamental to human activity, and in doing so, make claims that explore the human experience in its many forms. Writing in the humanities thus involves engaging in narrative writing and story-telling as a significant part of the text's volume and landscape, evidenced through the highly-frequent function words connected to these moves. The large amount of text that does this work then supports a movement and navigation through complex decision spaces toward a final interpretation and claim about how to make sense of some piece of the human experience. The sciences, in contrast, theorize, model, and build the physical and technological worlds. They describe objects and processes in detail as they are, and theorize up to a more abstract understanding. Their arguments for knowledge claims involve description and modelling, specific and precise in form to as to most clearly theorize about the less-contingent objects under study. In this sense, writing in the sciences involves engaging in heavy description as a bulk of the text itself, which then supports the building of those models by providing its evidence, background, and process by which it came to be.

To turn back to the initial driving question, what do we learn by looking

broadly at how a computational analysis based on function words divides disciplinary writing? In part, we gain a theory of how the humanities and sciences fundamentally argue for their knowledge claims. But part of this theory itself may not necessarily be surprising: *humanities* involve humans, and the sciences are specific and detailed. What is perhaps surprising is that function words tell us this. Function words are unique in that they are highly frequent and support the entire grammatical structure of a text by allowing (or in their absence, disallowing) certain types of language patterns. The patterns evidenced by function words are rarely the ones that stick out to a reader as most significant in a text (indeed, function words are often talked about as “the filler words of the language” (Mosteller & Wallace, 1964, p.17), or going unnoticed by writers and readers (e.g., Kestemont (2014))), but they are the patterns that both make up and support the actual bulk of the text’s volume.

Studies of academic writing across disciplines tend to focus on more noticeable features like novelty moves, modality in claims, citation practices, and so forth that are more obvious hallmarks of the work that the genre performs (e.g., MacDonald (2010); Hyland and Bondi (2006)). But by focusing on the most highly-frequent function words, we gain a lens not into the most noticeable but infrequent sentences involving claims or framing, but into the overall bulk of the text and the type of patterns that make up the landscape of how those major claims are produced across pages and pages of words. In the sciences and humanities, heavy and regular description and narrative respectively undergird the production of those claims by allowing for developing descriptions into models or theories that describe more general phenomena, and considering multiple interpretations and crafting an argument through the complexity that arises in the narratives. These moves take place at the sentence and phrasal levels, made possible by the function words that give them grammatical form, and are the very moves that make up the work of the disciplines in arguing for their new knowledge claims. This analysis ultimately demonstrates how specific function words support specific types of grammatical and phrasal forms, and illustrates how those forms work en masse to do the sociorhetorical descriptive and narrative work that help readers navigate towards a new claim. Moving forward, this insight contributes to WID research by demonstrating how complexity linguistically occurs and functions differently across disciplines, and can help writing instructors to think critically and in new ways about how these language-level moves toward disciplinary goals might help students to better grasp writing

in their own disciplines.

In terms of the value of close statistical reading to this case, what kinds of rhetorical and linguistic insight does this analysis provide at a large scale, and what does it not? This analysis represents the humanities/sciences split as a bit of a binary so as to more clearly see the fundamental differences between them and make sense of the “space” between them in Figure 4.1. In doing so, we are able to produce a broad framework for considering the differences in representations of knowledge claims between the humanities and sciences. It becomes clear through the synthesis of function word analyses how these differences are predicated on the bulk of the texts consisting of the larger rhetorical moves of narrative and description, and how and why those moves happen through specific grammatical form and thus word frequency.

In reality, this split is more of a spectrum. Figures 4.1 and 4.2 demonstrate this, and if Table 4.1 were broken down by discipline, it would become clear that the sciences and humanities each have massive variation within them. While on the average humanities are higher in words like *who*, *to*, and *but*, if we look closer, we find that cell & molecular biology and performing arts actually have very similar average frequencies of *to*. Taken as a whole, *to* creates a large distance between the sciences and humanities, but in reality, these distinctions become messy and there is incredible variation within each the sciences and humanities themselves. In effect, in order to offer a broad overarching framework, the analysis presented in this chapter necessarily (though not necessarily uselessly) oversimplifies. The later chapters explore what kinds of insight we gain and how the analysis becomes more difficult as we dive deeper into more limited distinctions.

Chapter 5

Metadisciplinary span: the range of variation in the humanities and sciences

The framework for representing knowledge claims across the disciplines as presented in the analysis in chapter 4 offers a way of speaking broadly about differences in argument strategies, but necessarily glosses over a tremendous amount of detail. When we look closely back at Figure 4.1, especially looking vertically rather than horizontally, it becomes clear that the sciences and humanities alike have a huge amount of variation within them. Indeed, though the humanities and sciences split is clearly represented as the largest split statistically in the dendrogram in Figure 4.2, computer science and philosophy actually appear closer together than computer science and biology.

This chapter takes the starting points of the last chapter – the humanities and sciences – and elaborates and refines that framework by investigating what makes up the “span,” or range of variation in the humanities and sciences. If we break open the framework created in chapter 4, what kinds of new insights can we gain about the different kinds of rhetorical work being performed *within* the humanities and sciences, and how is that work being instantiated in recurring grammatical and linguistic form? Regarding large-scale textual interpretation, how does a transformation in the viewing angle of the humanities and sciences as not a monolith but a spectrum of variation change our understandings? How does it change the method of analysis? These are the questions that guide the analysis in this chapter.

5.1 Function words and the span of metadisciplinarity

Looking at the humanities and sciences each in Figure 4.1, the largest metadisciplinary (or, within the humanities and sciences each) spans occur between cell & molecular biology and computer science and between history and philosophy. Tables 5.1 and 5.2 illustrate the top four function words per discipline that contribute to the span between each of these pairs.

The analysis in this section details the usage of the words in these tables and the roles they play in the rhetorical action of each metadiscipline's span of arguments. To offer a brief synopsis of that analysis though, the span of the humanities is characterized by a prevalence of either analytical storytelling or humanistic reasoning. In history, *had* and *and* support complex narrative language by allowing for recounting events out of chronological order to serve an analytical purpose with the past perfect form, and pairing separate agents or events together as part of the same actions, respectively. *Was* and *did* help to create statements of being, or descriptions of events as something *was* or as someone *did not* do in order to support analysis of those events. Philosophy, while still engaging with the complexity of human action as demonstrated in chapter 4, operates fundamentally differently in its relation to people by engaging more explicitly in forms of reasoning. The prevalence of *is* and *does* signal the logical beginnings of reasoning patterns: *is* allows for definitions in terms of properties or categories of something, while *does* (especially when paired with *not*) allows authors to clarify definitional statements by addressing the audience's pre-existing beliefs, as in a phrase like, "This scholar's theory *does not* entail the instantiation that..."). These logical starting points ultimately lead to discussions of effects and implications that negotiate with the complexity of human reasoning. *If*-statements allow creations of hypothetical futures where authors can contemplate possible effects of logical choices, as does *be* in allowing for discussions of the states of beings that certain reasoning patterns might create. While history engages with the complexity of humanity in recounting and analyzing past stories and narratives in particular ways, philosophy navigates through ways of understanding the world in creating and reasoning about many possible hypothetical logical futures.

The span of the sciences ultimately breaks down into a distinction between empirical inquiry and problem-solving modes of science. This distinction it-

	Delta distance	Discipline	Log- likelihood	Effect size	N
<i>had</i>	42,177.3	History	4,333.73	3.255	5,756
<i>is</i>	42,985.7	Philosophy	21,199.36	2.789	67,687
<i>does</i>	38,259.9	Philosophy	2,182.62	3.300	2,668
<i>be</i>	33,909.2	Philosophy	4,222.08	1.629	27,882
<i>if</i>	33,492.5	Philosophy	4,092.32	2.518	12,665
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>was</i>	27,841.0	History	7,580.42	3.008	10,692
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>and</i>	26,511.1	History	5,026.80	0.862	47,524
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
<i>did</i>	26,225.3	History	948.47	2.520	1,685

Table 5.1: The top 8 function words that contribute most to the span in the humanities, or the distance between philosophy and history (note: the table is sorted by Delta Distance, but edited to only show the top 4 words per discipline). The Delta distance column shows the sum of the Delta distances between each philosophy and literature text for that particular function word (using the modified Delta formula described in chapter 3). The discipline column shows which discipline (philosophy and history) that the function word is most common in, based on which discipline has the highest average z-score for that word. The table also shows the log-likelihood measure of the word for the discipline it is most common in (using the less-common discipline as a reference corpus), the effect size of the word for the discipline it is most common in using log-ratio (Hardie, 2014), and the total number of instances of the word across all 125 texts for that discipline (N).

	Delta distance	Discipline	Log- likelihood	Effect size	N
<i>in</i>	34,981.3	Biology	1,687.72	0.609	88,439
<i>or</i>	31,118.9	Biology	1,194.60	1.329	33,531
<i>were</i>	31,084.8	Biology	4,326.72	3.804	4,279
<i>with</i>	30,477.7	Biology	837.98	0.698	9,516
<i>each</i>	29,970.9	Comp Sci	936.97	1.697	2,874
<i>can</i>	29,525.3	Comp Sci	1,687.39	2.031	4,539
<i>all</i>	29,108.1	Comp Sci	718.64	1.411	8,896
<i>every</i>	28,335.5	Comp Sci	803.67	4.296	799

Table 5.2: The top 8 function words that contribute most to the span in the sciences, or the distance between computer science and cell & molecular biology. The Delta distance column shows the sum of the Delta distances between each computer science and biology text for that particular function word (using the modified Delta formula described in chapter 3). The discipline column shows which discipline (computer science and biology) that the function word is most common in, based on which discipline has the highest average z-score for that word. The table also shows the log-likelihood measure of the word for the discipline it is most common in (using the less-common discipline as a reference corpus), the effect size of the word for the discipline it is most common in using log-ratio (Hardie, 2014), and the total number of instances of the word across all 125 texts for that discipline (N).

self is not new (see, for example, Carter (2007) and Flowerdew (2008)), but the analysis points to how these modes of science instantiate themselves in text through rhetorical moves supported by recurring grammatical forms. In cell & molecular biology, *with* and *in* signal heavy physical description in their modification of nouns with descriptive details and location of processes in the experiments, respectively. This description ultimately serves the statement of results, where *were* helps to describe states of being temporally and *or* helps to contrast the multiple scenarios under study and set up comparisons so that the results are meaningful. Computer science demonstrates a more constructivist approach, wherein the text builds and describes a logical system and makes claims about what that system can do. *Each*, *every*, and *all* each allow for complex (and in their separation, nuanced) references to multiplicity, which is a key element of logical systems in computer science and algorithms. That is, analysis of computing typically works with multiples (inputs, calls, programs, etc.) that must all be considered in different ways. The detailed discussion of how a system works in its multiplicity ultimately allows for statements about a system's capabilities. The prevalence of *can* in computer science supports statements about the capabilities of those systems, given their properties. Thus while biology participates in the physical description seen in the sciences through naming, locating, and comparing specific physical objects in experiments, computer science engages in a more abstract description of multiplicity in its building and analysis of logical systems.

5.1.1 Humanities

The span of the humanities is visually chunked into three groups in Figure 4.1: the more classical humanities disciplines of history, literature, and performing arts in the far top right corner; the social sciences with political science, sociology, and so forth toward the middle; and philosophy off on its own toward the bottom right.

History, perhaps unsurprisingly, is full of narrative and storytelling (on average, at least 4.7% of each text in history is explicitly narrative language, compared to just 2.7% in the rest of the humanities). This narrative is not a straightforward recounting of events though, but is instead carefully constructed for the purpose of analyzing those events as indicated by the prevalence of the words *had*, *and*, and *did*. *Had* appears on average about five

times more frequently in history texts than other humanities writing,¹ with the majority (approximately 70%) of those instances being in the auxiliary (rather than root verb) form. That is, *had* is frequently occurring in history to form the past perfect, or to form a linguistic past-within-a-past, which supports a recounting of events out of order. Take this example from a history paper that describes the role of film in humanitarian responses to World War I. After recounting various films that framed and responded to the humanitarian crises in the wake of the war, the author narrates how the film campaign eventually started coming to a close:

- (1) Despite best efforts to keep up with the need, the campaign faltered. Need had grown dramatically and quickly overwhelmed Anglo-American aid efforts, and the public's interest began to wane. (Tusan, 2017, p. 231) [History]

This example in (1) tells events slightly out of order: chronologically, need grew dramatically, the campaign tried to keep up, the campaign was overwhelmed, the public's interest began to wane, and the campaign faltered. By putting the events out of order though, the author highlights the campaign faltering as the important event and frames the second sentence as an explanation of the first. In this particular case, the first sentence in (1) is actually the first sentence of the paragraph, adding further emphasis to the campaign's faltering given its function as a topic sentence. Creating a past perfect form with *had* allows for the telling of these events out of order (the *had* creates a past before the "faltered" which is already in past tense), in effect allowing the author to choose what events in the narrative to highlight.

And too allows for a more analytical storytelling, where events are not recounted in a straightforward manner. Rather than allowing for events to be out of order though, the conjunction allows for very particular framings of events by allowing certain actors and events to be placed together as part of the same action. In this sentence from the beginning of a history text about investment practices and capital flow in slave trading from 1850-1866, the author recounts how the transatlantic slave trade came under assault at the beginning of the nineteenth century:

- (2) Saint Domingans landed a heavy first blow [to the slave trade] in

¹*Had* appears on average 46.0 times per paper in history, and 9.5 times in the other humanities disciplines as a whole.

1804 by successfully concluding a massive uprising and permanently closing the huge traffic to their shores. A few years later, the British and United States governments banned their trades, after long campaigns by Enlightenment thinkers, zealous evangelicals, and political reformers. (Harris, 2016, p. 409-410) [History]

Three groups are constructed by *and* in this excerpt: the massive uprising *and* closing of traffic are paired together as events which both did the same work of landing a heavy “blow” to the transatlantic slave trade; the British *and* United States governments both did the same action of banning their trades; and campaigns were launched by a list of people which all had the same cumulative effect of getting slave trading banned in these countries. Chronologically, each of these actions would have occurred separately: there was an uprising and then a closing of traffic in Saint-Domingue; the British and United States governments each individually banned trade; the Enlightenment thinkers, evangelicals, and reformers all each separately had various campaigns that the governments took notice of. The author of this text is engaging in an analytical grouping of these actors and events though, to place them as part of the same narrative together. This sort of grouping is more common in history than other humanities texts: *and* occurs on average 380 times per paper in history, and only 274 times in the other humanities. In doing so, the authors of these texts gain the ability to control the narratives they tell by showing which events belong together, integrating analysis by the very way the story is told.

And while *had* and *and* in narratives tell about various events that did happen, history texts also expand this analysis by pointing out what things did not happen. *Did* appears at nearly double the frequency in history texts than the other humanities,² and roughly half of those instances appear in the form *did+not*. In a paper tracing how African-American journalist William Worthy traveled to Cuba without a passport in 1962, returned to the US, and was prosecuted for unlawful entry into his own country, the author explains how Worthy’s protests to being prosecuted were received:

- (3) Worthy considered the protest and his NNPA speech to be successes. Robert Kennedy “got the message,” Worthy remarked in a *Baltimore Afro-American* article...The Justice Department, however, did not get

²On average, *did* appears 13.5 times per paper in history and 7.2 times in other humanities disciplines.

Worthy's intended message. In a July 1962 response to Roy Wilkins's telegram, J. Walter Yeagley, an assistant attorney general and director of the Internal Security Division, wrote, "While it is not the practice of the Department of Justice to comment on the grand jury's indictment in advance of trial, you may be assured that we would not have presented this matter to a grand jury for its deliberations unless we were convinced that there existed sufficient evidence to establish beyond a reasonable doubt that Mr. Worthy deliberately violated the law." (Lovelace, 2016, p. 120) [History]

The Justice Department here sits in contrast to Robert Kennedy, who seemed to support Worthy's mission of challenging Cold War censorship by traveling to Cuba as a journalist; that is, Kennedy "got the message" and the Justice Department "did not." To recount this situation more affirmatively, recounting what indeed *did* historically happen, the author may have instead written that, "The Justice Department was in disagreement with Worthy." But by stating what *did not* happen, the author is able to create an analytical contrast to Kennedy's reaction and create a narrative in which the absence of events or actions is more notable than what actually did occur. Grammatically speaking in English, *not* cannot negate a verb without an auxiliary verb preceding it (meaning for example, that we can say "I have work today" but not "I not have work today." To do this, we must add an auxiliary like, "I *might* not have work today" or "I *do* not have work today." A form of *do* is commonly used as this auxiliary verb when we do not want to add another more meaningful form, like a modal), meaning that these instances of *did not* in history appear to negate a simple past tense verb where there are no other auxiliaries, or to negate a simple declarative statement about an event occurring in the past to showcase an analytical absence in narrative.

Finally, while *had*, *and*, and *did* support the narrative part of analytical storytelling, sometimes the analysis itself takes center stage. Analysis becomes most obvious in the frequency of the verb *was*, which allows authors to integrate description into their stories. *Was*, as the past-tense of the *be*-verb, allows statements about states of being; the authors of history use these constructions to declare the state of being of an actor or object at the time, to define, place in context, or to comment on them. In the same history article about the trial of journalist William Worthy, the author explains why the trial was chosen to take place in Miami:

- (4) The city [Miami] in 1962, still two years before the Civil Rights Act of 1964, had a swelling population of Cuban expatriates, many of whom had fresh emotional wounds from the Bay of Pigs fiasco. Miami was thus at the crossroads of an escalating fight for the Third World and an intensifying civil rights movement. (Lovelace, 2016, p. 117) [History]

This example defines Miami as a critical location in the conflicts that Worthy was tied up in, using the description to give context to the choices made in the narrative of Worthy's trial. In this sense, the statement here about Miami is not a strict part of the narrative about Worthy and his trial in that it does not recount specific events in the story, but is instead a statement about a state of being that helps to illuminate the context and strengthen the analytical framing of the narrative. These kinds of constructions are common in history, with *was* as the root verb (rather than as an auxiliary in a passive voice construction) appearing on average 28.7 times per history paper.³ As a whole, these statements supplement the storytelling part of the narrative by linguistically forming part of the analysis through creating definitions, contexts, or comments. Together, the use of complex narrative to support analysis through these function words aligns with and provides linguistic evidence for what Stockton (1995) noticed when she explained that history students are often guided away from straightforward representations of events toward narrative complexity that allows for what Ricoeur (1984) calls "emplotment;" that is, history writing helps the transition from narration to explanation, which here occurs through grammatical forms that support strategic crafting of narratives that highlight events for the purpose of explaining.

Philosophy, at the far other end of the humanities spectrum, is instead characterized by a complex, humanistic reasoning rather than analytic narrative. While philosophy has more explicitly reasoning-based language than the rest of the humanities,⁴ complex philosophic reasoning occurs through a variety of linguistic forms. One of the key starting points of reasoning is in setting up definitions, which is signaled in philosophy by the prevalence of the word *is*. *Is* functions commonly as a definitional word: authors are giving properties, defining, putting things in categories. Take for example, this sen-

³ *Was* as the root verb appears an average of only 7.4 times per paper in the rest of the humanities.

⁴ At least 3.6% of philosophy papers on average are made up of explicitly reasoning-based language like *imply* or *because of the* (compared to 2.4% in the rest of the humanities)

tence drawn from a philosophy paper that argues for the importance of the concept of relationals, especially in relation to a theory about the identity of indiscernibles. In this particular sentence, the author is responding to two critiques (about Black's spheres and Kant's droplets) of a theory about the identity of indiscernibles, and arguing for why those critiques do not hold up under scrutiny:

- (5) I remark that Black's spheres case is the same case as Kant's droplets (namely, two absolutely indiscernible material objects), which is why we gloss over Kant's droplets altogether. (Muller, 2015, p. 209) [Philosophy]

The first instance of *is* in (5) is equating two cases: the author sets up these two cases as the "same," X is Y. The equation of the two is the author's own definitional and reasoning work, to argue that these cases have the same properties. The second instance of *is* defines the author's own metareasoning: that these two cases are the same *is* the reason that we will gloss over Kant's case. The focus and structure of this sentence – by using *is* as the root verb in these two verb phrases – is to do this kind of definitional work, rather than subsuming definitions to description through an extra clause or adjective phrase, as seen in this sentence from a history text, which defines "shaping" through a possessive and prepositional phrase and uses *is* instead to comment on the state of that shaping today, rather than define what it is:

- (6) The Eastern Front's shaping of the humanitarian imaginary during the war is largely forgotten today. (Tusan, 2017, p. 233) [History]

By foregrounding the definitions (in making the definitions the purpose of the subject-predicate form of the sentence through using *is* as the main verb), philosophy texts set up the definitions and terms needed to engage in the rest of the rhetorical reasoning work of the text.

In addition to setting up definitions, another starting point of reasoning texts is to respond to possible pre-existing beliefs of the audience. *Does* responds to these beliefs, especially when it is paired with *not*, to form a negative belief.⁵ In particular, *does not* allows for clarifications of definitional (or implicative) statements by dispelling beliefs that the author(s) think their audience might have. For example, this excerpt is from a paper that argues

⁵*Does* appears on average 21.3 times per paper in philosophy texts, with 60.2% of those instances being in the form *does not*.

against a previously-proposed “Best Bet Theory” (BBT) that was meant to replace truth-conditional semantics as a theory in generative linguistics:

- (7) In fact, until we get some account of the meaning relation, we have no reason to think that the BBT is better positioned than truth-conditional semantics when it comes to explaining our communicative abilities and vindicating GGH. Merely pairing sentence-types with different sorts of objects does not tell us anything about the uses (i.e., tokens) of those sentences, even if the objects are pairs of speech-act types and content forms. (Napoletano, 2017, p. 680) [Philosophy]

The author in this sentence is responding to and arguing against beliefs of the audience: because this BBT was previously proposed and perhaps believed by many of the readers of these texts, the author here can argue against that by explaining that BBT actually *does not* do what they might believe it does. That is, negation occurs primarily in response to something that already exists. These kinds of *does not* patterns are frequently responding to beliefs: the words immediately after *not* in these phrases tend to be reporting verbs like *does not exclude* (at 10.5% of words), reasoning verbs like *does not entail* or *does not tell* as in (7) (at 7.8% of words), or highly confident verbs like *does not prove* (at 4.4% of words). In stating what *does not* occur, the authors of these philosophy texts are able to strengthen their lines of reasoning by directly addressing the concerns or beliefs of their readers. This can ultimately allow for the “main path/faulty path” structure of philosophy writing that Geisler (1994) describes, wherein philosophy writers guide readers to a resolution of a problem by moving them along many “faulty” paths (of what is *not* true) in the pursuit of the main reasoning path.

The definitional type of work seen with *is* and *does* ultimately leads to discussions about the effects and implications of those beliefs, but they are complex and negotiate with the complexity of humanity in that logic. *If* is perhaps the most obvious word that precedes discussions of effects and implications, with the well-known logical *if-then* construct. *If* creates a hypothetical that can be used to reference assumptions we might want to make for various reasons. For example, these two sentences from a paper that argues against philosopher Sydney Shoemaker’s account of the mind-body problem use *if* in slightly different constructions when explaining a problem with Shoemaker’s logic:

- (8) That is, if P realizes Q, the instantiation of the backward-looking

powers of Q are instantiated. This means that the instantiation of P does not determine the instantiation of Q if P and Q are partly constituted by their backward-looking powers. (Pineda & Vicente, 2017, p. 103) [Philosophy]

Both of these sentences create a hypothetical with an *if* in order to state the effect that will come about by accepting that hypothetical, engaging in a sort of logical deduction that *if-then* is commonly used for in philosophy (Nesi & Gardner, 2012, p. 118). In these sentences, if P realizes Q, then the instantiation of the backward-looking powers of Q are instantiated; if P and Q are partly constituted by their backward-looking powers, then the instantiation of P does not determine the instantiation of Q. Neither of these constructions as written in (8) explicitly include a *then* in the sentence (just 9.6% of sentences in philosophy papers that contain an *if* also include a *then*), but the construction of an *if*-phrase linguistically implies another piece of the sentence to be the *then*-phrase. In creating these hypotheticals, authors of philosophy are able to imagine possible futures and effects of various choices in reasoning. In doing so, they can navigate through multiple possible situations of what we might accept as true or not (again, engaging in Geisler's (1994) "main path/faulty path" structure), and reason about the implications and effects of doing so.

Finally, *be* complements the hypothetical reasoning paths of *if* by creating possible futures in lines of reasoning. While *is* appears in philosophy texts to define things in the present moment, *be* as the same verb but a different tense allows this type of definitional rhetorical work to be done in hypothetical futures. *Be* shows up frequently in philosophy texts, at an average rate of 223 times per paper with just over one-quarter of those instances occurring in the form *modal+be* as demonstrated in (9):

- (9) However, both these responses involve commitments which many will want to avoid. The claim that there can never be sufficient reason to believe what's false—in effect, the claim that there can be no justified false beliefs—is highly, and rightly, controversial. Evidence provides reason to believe, and even very strong evidence can be misleading. (McHugh & Way, 2018, p. 162) [Philosophy]

In this paper, the authors develop and argue for an account of what makes for "good" reasoning. Here, they had just previously set up some problems with one account of good reasoning, imagined possible responses to those

problems, and in this passage are refuting those responses. In doing so, they construct three *can+be* phrases. These phrases are framed in the future rather present tense: they refer to what kinds of reasons, beliefs, and evidence might exist in a possible future. With these constructions, the authors can elide the question of whether there has been or even are currently sufficient reasons to believe what is false, justified false beliefs, or strong evidence in favor of imagining a hypothetical situation where these things might exist. In effect, this kind of pattern illustrates what believing or reasoning in certain ways might do, adding nuance to the “timeless dehistoricized present” that philosophy writing is known for (Geisler, 1994) by creating less of a “present” but more of a possible point in time. These constructions are interesting because they embed an uncertainty and a representation of lots of possible futures, by setting up the situation up as completely hypothetical and futuristic, rather than a statement of what *is*. *Modal+be*-statements like these bring reasoning patterns into a hypothetical future in order to imagine and think through many possible scenarios, adding a complexity to the reasoning by grappling with the many possibilities of what *can* or *might* or *[modal] be*.

5.1.2 Sciences

The span of the sciences is ultimately breaks down into a distinction between experimental and problem-solving science. Again, while this is perhaps not surprising or even novel, the function words that distinguish the disciplines do point to two new kinds of insights: that these basic paradigms of scientific study are not only overarching frameworks for thinking about novelty, but instantiate themselves throughout the entirety of texts; and that these paradigms are represented in text through regularly occurring rhetorical moves that are rooted in specific linguistic grammatical form. In particular, the cell & molecular biology texts are based upon heavy amounts of physical description that aid in the stating of results. On the other hand, computer science texts are based upon description through multiplicity that allows for a creation and analysis of a logical system and an explanation of that system’s capabilities.

Cell & molecular biology is characterized by more descriptive language than other sciences. Taken as a whole, biology texts have about 1.5x the amount of descriptive words and phrases that computer science has. They

are also very syntactically complex,⁶ meaning that the writing tends to pack a lot of information into single sentences. This is perhaps most notable in the two function words *with* and *in*, both of which appear in Table 5.2 as representative of biology and which together aid in creating complex descriptions of physical objects and experiments. In terms of describing physical objects, *with* frequently functions as a preposition to modify nouns with extra detail (almost one-third⁷ of all instances of *with* in biology take the form *noun phrase+with+noun phrase*). These instances of using a *with*-prepositional phrase to modify a noun occur most frequently in the large amounts of figure captions in biology texts, wherein authors name particular objects in the experiments in detail to describe how they looked in those experiments. The figure captions in these texts are interesting in that they are often doing the rhetorical work of describing the outcomes of the particular experiments, while the results sections themselves tend to do more comparison for the sake of understanding those results (see later discussion on *were* and *or*). For example, in these two figure captions from a biology article, the authors name the samples which are being illustrated in a figure:

- (10) A. Western blot analysis of nuclear extracts from indicated cell lines with antibodies for EZH2 and H3K27m2/3 mark. Lamin B1 and H3 are shown as loading controls (upper panel). qRT-PCR to test *HOTAIR* overexpression in the corresponding cell lines. Y-axis represents *HOTAIR* expression relative to U1 RNA (individual experiments and mean, $n = 2$) (lower panel).
- B. Heatmap showing expression intensity of the 1,000 genes with the higher interquartile range. Genes up- or downregulated from MDA-MB-231 EZH2++- and MDA-MB-231 EZH2— with (+*HOTAIR*) or without *HOTAIR* samples are shown. Red indicates high expression, and green indicates low expression. (Portoso et al., 2017, p. 982) [Biology]

The *with*-phrases in this sample each detail some feature of a subject that was involved in the experiment for the purpose of clarifying what made those subjects interesting or unique in their results: the authors tested cell lines

⁶At least 12.07% of all words in biology are function words or other types of connecting words that make sentences longer and more detailed.

⁷29.9%

with certain antibodies; the relevant 1,000 genes to notice in this particular experiment are the ones *with* the higher interquartile range; the experiment included samples both *with* and without *HOTAIR* for the sake of comparing effects. The prepositional phrases here add relevant details to the samples in the experiments that clarify what particular features are of interest. This pattern is incredibly common in these texts: of the *with*-prepositional phrases that modify nouns in biology texts, just over 40% of the nouns being modified are academic terms, meaning that they are discipline-specific or technical words like those that name objects in an experiment. In using this technique to modify nouns, the authors are able to describe the experiment more generally while also pointing to particular physical features of samples or results that are important for interpreting the results.

In addition to describing particular physical features of samples, biology texts also heavily describe the particular experiments, methods, and interactions involved in them. This is most frequently done not through a narrative of methods (indeed, long detailed methods sections are often relegated to a sort of appendix in many biology journals), but by modifying both verbs and nouns with prepositional phrases. Prepositional phrases with *in* also tend to take the *noun phrase+in+noun phrase* form, but these phrases tend to be used more for the purpose of describing where particular experimental processes did (or did not) take place. For example, in this sentence from a biology text, the authors describe the results of their experiments:

- (11) Coexpression of constitutively active RAC1 or CDC42 (Figure S2G, lanes 4-6), but not RHOA (Figure S2H), along with p110 β /p85 in COS7 cells strongly elevated cellular phospho-AKT and PIP₃ levels (Figure 2I), indicating that both GTPases activate p110 β in transfected cells. (Fritsch et al., 2013, p.1053) [Biology]

The sentence in (11) is overall structured to state a result and its interpretation: expression of certain chemicals in cells elevated other chemical levels, which means that something is active in those cells (to restate the sentence in a simplified form). Implicit in this result and interpretation statement is an indication of methods: certain chemicals were co-expressed in cells purposefully and other values were measured when this was done. These actions are not the main verbs in the sentence, but appear as a nominalization (“co-expression”) which allows the action to be discussed as the subject of the sentence, and as an action where the chemicals are the agents (“strongly

elevated”) which focuses the statement on the result of the elevation. The *in*-prepositional phrases are part of those method moves in that they give the location of the implicit actions that are occurring: RAC1, CDC42, and p110 β /p85 were expressed *in* COS7 cells, and GTPases were made to activate p110 β *in* transfected cells. These *in*-phrases thus both clarify the methodological steps that happened (but are not stated) so that the results can be more clearly put in context, and detail the results by indicating where certain measurements or actions occurred.

In, thus, adds descriptive detail to methods. And while *with* was previously described as being used to modify nouns with details, other times it functions similarly to *in* to modify certain types of actions with detail that is not locative and thus does not benefit from *in*-phrases. While about 30% of *with* instances modify nouns, another 57% follow verbs. These prepositional phrases especially help to describe how objects interact with and relate to each other in an experiment, which form the basis of statements of results. For example, these two sentences from results sections show the *verb+with* pattern:

- (12) RNA immunoprecipitation (RIP) with the same antibody further indicated that the fusion protein indeed interacts with the *MS2-HOTAIR* or *MS2-HOTAIR-Rev* RNAs (Fig 2C). (Portoso et al., 2017, p. 984) [Biology]
- (13) Furthermore, high PARP1 expression showed an impressive association with poor event-free and overall survival (Figures 1d and e). (Colicchia et al., 2017, p. 4683) [Biology]

The first *with* in (12) demonstrates the *noun phrase+with+noun phrase* discussed earlier, but the second instance follows “interacts” to indicate a relationship between the fusion protein and RNAs in this experiment. The sentence in (13) is also indicating a relationship demonstrated in the experiment, but is slightly more abstract to describe a trend or pattern (high PARP1 expression overall is associated with certain outcomes) that the authors have interpreted, rather than the specific details of the samples in the experiment. These distinctions are subtle, but both types ultimately provide details of actions around the experiments that help form a focused description of the experiment and its results.

Finally, the heavy description that is evident in biology texts is ultimately used in the service of stating results in terms of what was observed

in experiments, and drawing conclusions from those results through comparison. *Were*, in particular, helps to state results by aiding in description of the temporal state of something as it was observed in an experiment. Overwhelmingly, *were* is used in passive voice constructions (of all of the instances of *were* in biology texts, 83.5% are in passive voice). And while some of these are methodological descriptions (like, “these sites were mutated separately” (Rago et al., 2014, p. 909)),⁸ many are used to describe results and observations as in (14) below:

- (14) Moreover, although pole width was greater in the absence of Ncd, adjacent MTs were locally cross-linked more frequently (Fig. 2, A and B; some MTs exhibit a V shape by end connection), and the focal point had strong Asp-GFP signals (Fig. 2 D, fixed cell; F[right], living cell). (Ito & Goshima, 2015, p. 1000) [Biology]

Here, *were* is used to state results: adjacent MTs were observed to be cross-linked more often without Ncd. The results are framed as observations, as describing the state of being of MTs in various scenarios. These results are only meaningful, though, in comparison to other results. Drawing on a basic scientific method framework, results like these become meaningful because they differ from other controlled study results. Different samples may be tested in the same way, or the same sample tested in different ways, and their results compared, which is represented linguistically through the frequent use of *or*. *Or* occurs in biology at double the rate as it does in other sciences (on average, 32.6 times per paper in biology versus 16.6 times in the other sciences), because it sets up the basic paradigm of comparison that characterizes experimental science. The example in (15) shows how *or* creates comparisons in method and results statements:

- (15) The overexpression of miR543 or miR369-3p with miR496 results

⁸Biology writing in particular is frequently characterized as heavily using the passive voice to describe its method steps. In this corpus, though, “we” actually occurs in biology texts an average of 25.1 times per text (this is lower than the other sciences as a whole which use “we” on average 35.5 times per text, but is still a sizeable number). That is, the authors are frequently using first-person plural constructions to talk about what they did and observed. Thus, both forms of method description – passive and active voices – are common, and it appears to be a strategic rhetorical choice as to which makes the most sense for the purpose of focusing a particular sentence, much like the use of tenses and temporal shifts (Liddicoat, 2004).

in premature neuronal differentiation, as shown by a significant decrease in Pax6⁺ RGCs and an increase in Tbr1⁺ neurons (Supplementary Fig S5, Fig 5A and B). (Rago et al., 2014, p. 911) [Biology]

This sentence is framed as a statement of two results: the overexpression of miR543 with miR496 results in premature neuronal differentiation, as does the overexpression of miR369-3p. The two results are placed together into the same sentence, joined by an *or*, not simply for concision's sake, but for the sake of comparison: these two results look the same; it does not matter whether you use miR543 or miR369-3p in this situation. The statement alone implies that there might be some reason to believe that they could be different in its statement of the result at all, framing these observations as results that are being interpreted together. Ultimately, *or*, together with *were*, *with*, and *in*, together help constitute some of the basic grammatical forms that undergird biology texts' heavy description and statement of comparative results that characterize experimental science.

At the other end of the science spectrum from biology sits computer science, which overall demonstrates a more constructivist or problem-solving mode of science, as opposed to experimental. The function words that distinguish computer science largely aid in building and describing a logical system, and explaining what that system can do. That is, these papers are demonstrating that the authors have developed something new (an algorithm, a programming language, a system, etc.) and are describing it and arguing for its capabilities. Three of the top function words for computer science shown in Table 5.2 – *each*, *all*, and *every* – aid in an abstract description, one that describes not necessarily a physical object but an abstract system that is being constructed rather than tested and observed. These three words in particular all allow for complex references to multiplicity, which is a key element of logical systems like computer systems and algorithms: programs typically work with multiple queries, contexts, commands, calls, etc., and an analysis of that program must describe and evaluate all of those multiples in various ways. The three examples below demonstrate each of the three words in action, illustrating how the authors describe the workings of their creations:

- (16) For each procedure f , the set $K(f)$ specifies calling contexts that the analysis should differentiate while analyzing the procedure. (Oh et al., 2014, p. 477) [Computer Science]

- (17) To achieve this goal, we use an impact pre-analysis that aims at estimating the behavior of the octagon analysis under its fully relational setting. More specifically, like the fully relational octagon analysis, the pre-analysis tracks constraints of the form $\pm x \pm y \leq a$ for all variables x and y but approximately tracks the bound; we use one of two abstract values $\star$ and $\top$ as bound a , rather than all integers and ∞ . (Oh et al., 2014, p. 481) [Computer Science]
- (18) In this approach, the agent only receives a screen input every $k + 1$ frames, where k is the number of frames skipped between each step. The action decided by the network is then repeated over all the skipped frames. (Lample & Chaplot, 2017, p. 2143) [Computer Science]

The paper excerpted in (16) and (17) is presenting a new method for analyzing programs for security vulnerabilities. The paper introduces the difficulties in doing this analysis, briefly summarizes their approach to the problem, and then spends the majority of the paper formally defining their method. Both (16) and (17) are located in that formal definition. The *each* in (16) allows the authors to highlight that the context calling in their method happens for *each* individual procedure that is named, as opposed to an aggregate mass. This focuses the reader’s attention to the creation of *each* individual set of calling contexts that must be specified. In (17), though the *all* allows for a discussion in aggregate mass form: the pre-analysis tracks constraints for *all* variables, and the authors use only two values rather than *all* of them. The focus here shifts slightly to the pre-analysis and the form of the constraints being tracked, and to the two abstract values, and away from the individual elements that comprise each mass of *all*. Finally, the example of *every* in (18) is taken from an artificial intelligence paper that is augmenting the model behind bot performance for the video game *Doom*. The sentence in (18) is taken from a section where the authors describe how their model was trained, and write that as part of the training, the agent receives input *every* certain number of screen frames. This usage of *every* draws attention to the interval, pointing out specific frames as relevant.

The nuances between *each*, *all*, and *every* usage in computer science are subtle, but point to slight differences in focus when discussing the multiples of items in a system. As a whole, these three words occur at more than double

the average rate per paper in computer science than the other sciences.⁹ And when they occur, the words that follow them tend to be academic, description, and information terms,¹⁰ as shown in the three examples. As a whole, the frequency of these three terms helps to linguistically form an abstract description of a system that engages in multiples, where the authors can subtly shift focus to different elements of the processes of the system.

All of this detailed discussion of how systems work with their detailed multiplicity ultimately allows for a building up of systems that have certain capabilities. On a local level, authors might use *can* to describe how they are able to build or represent their system logically (as in, “The last three parameters can be partially derived from *TaskCode* on the development stage for a concrete Smart Connector.” (Spichkova et al., 2015, p. 398)). More broadly, *can* is used to signal the capabilities of their systems for users or other relevant parties. For example, the authors in (19) are presenting a new framework for computing and data management that enables high-power computing but for which the coder needs fewer specialized skills to use. In a case study of their framework at the end of their paper, they write the following:

- (19) In addition, the inclusion of automated graphing software means that the user can easily trace the flow of the calculations through the several sequential parallel executions that are often required to reach reasonable convergence to the experimental data. (Spichkova et al., 2015, p. 402) [Computer Science]

Can signals an ability and here, after having built up a system in most of the paper, the authors make a move to state the capabilities of their system more broadly for potential users. Paired together with the logical building up of systems that *each*, *all*, and *every* help to linguistically create, *can* brings the system to fruition by offering a way to reflect on what this newly built system is.

⁹*Each* occurs on average 19.7 times per paper computer science to 8.1 times in the sciences; *All* occurs on average 18.7 times per paper in computer science to 9.7 times in the sciences; *Every* occurs on average 6.4 times per paper in computer science to 0.9 times in the sciences

¹⁰Academic terms occur after these words at a rate of 13.6% for *each*, 12.5% for *all*, and 11.7% for *every*; Description terms occur after these words at a rate of 8.2% for *each*, 5.7% for *all*, and 5.5% for *every*; Information terms occur after these words at a rate of 5.8% for *each*, 5.8% for *all*, and 5.0% for *every*.

5.2 Reflections on complexity through span

While Figure 4.1 demonstrates a broad-level split between the humanities and sciences, each of those groups has a large span made up of distinct groups of disciplines. The function words outlined in Table 5.1 signal a distinction in the humanities between analytical storytelling and complex, humanistic reasoning. That is, history tells complex stories that are narrated in such a way as to analyze and make meaning out of them, engaging in what Ricoeur (1984) calls “emplotment,” rather than just recounting events chronologically. Philosophy engages in reasoning that navigates through textual space (Nystrand, 1982) and through faulty paths (Geisler, 1994) to hypothetical futures so that those futures can be considered carefully in regards to their implications. In the sciences, the function words outlined in Table 5.2 signal a distinction between empirical and problem-solving modes of inquiry that shape argument. Cell & molecular biology engages in syntactically-complex physical description in order to state and compare results to make meaning. Computer science, however, describes the details of logical systems that engage in complex multiplicity in order to make claims about the wider capabilities of those systems.

These distinctions in the humanities and sciences do not conflict with the analysis offered in chapter 4, but offer new insights and nuance. While a broad look at the humanities suggested that they grapple with the complexity of the human experience through stories about people, a more detailed look at the span of the humanities suggests that this is done in many ways. History texts embody the humanities view from chapter 4 most obviously in their narratives about people, but these narratives are complex and out of order, shaped carefully for the purpose of understanding the narratives in particular ways. Philosophy texts engage with the complexity of the human experience through reasoning about it, recognizing the complications of particular human experiences as they reason about the effects and implications of many hypothetical future scenarios.

In terms of the sciences, while a broad look suggested that they engage in complex description for the purposes of reasoning more abstractly, a more detailed look at the span of the sciences demonstrates the diversity of what descriptive language is and how it can be used for theory building. Biology texts engage in a description that is physical and localized, which allows for statements of results that are comparative to make meaning that can

be extended to other contexts. Computer science texts engage in a more abstract and kinetic description, detailing through the multiplicity of how logical systems operate and what capabilities those described details afford.

Regarding interpretive theory, how does this transformation in the viewing angle of the corpus change our understandings? The analysis in chapter 4 treated the humanities and sciences each as monoliths, characterizable in a single way. This chapter instead moved toward understanding those groups as a spectrum of variation by focusing on the end points of each group – the sets of disciplines that were furthest apart from each other. Comparison of corpora (or groups more broadly, statistically speaking) is a fundamental activity in computational text analysis, and this analysis demonstrates how to scale analyses that involve comparison by comparing subgroups (here, humanities and sciences). It also proposes that one way of investigating variation is to interrogate the *span* of a group. Doing so in this analysis produced some results that aligned with our previous understandings of disciplinarity in the sciences (and to some degree in the humanities given the social sciences’ distinctness from the other humanities), while also suggesting some new understandings of disciplinary difference with philosophy.

The analysis in this chapter (and in chapter 4) is still fundamentally comparative in nature though, comparing extreme ends of a span to draw some overarching insight. When looking more specifically at each individual discipline though, it is apparent that each discipline has their own intra-disciplinary variation. That is, each discipline exists as a sort of space of variation, as opposed to a clear spectrum. The next analysis chapter explores how to investigate this intra-disciplinary space to understand what kinds of variation might exist within texts that are relatively close to each other and when that kind of variation might become more “noise” than “signal.”

Chapter 6

Intradisciplinary variation: zooming in on computer science and history

As we keep drilling down further and further into the original visualization in Figure 4.1, we can investigate increasingly nuanced levels of variation. Chapter 5 built on chapter 4 by zooming in closer on interdisciplinary span, but if we look at the end points of either of those spans, each discipline on its own also has a level of intradisciplinary variation. That is, while individual disciplines seem to roughly cluster together and are largely distinguishable, each has its own nebulous space that it envelops.

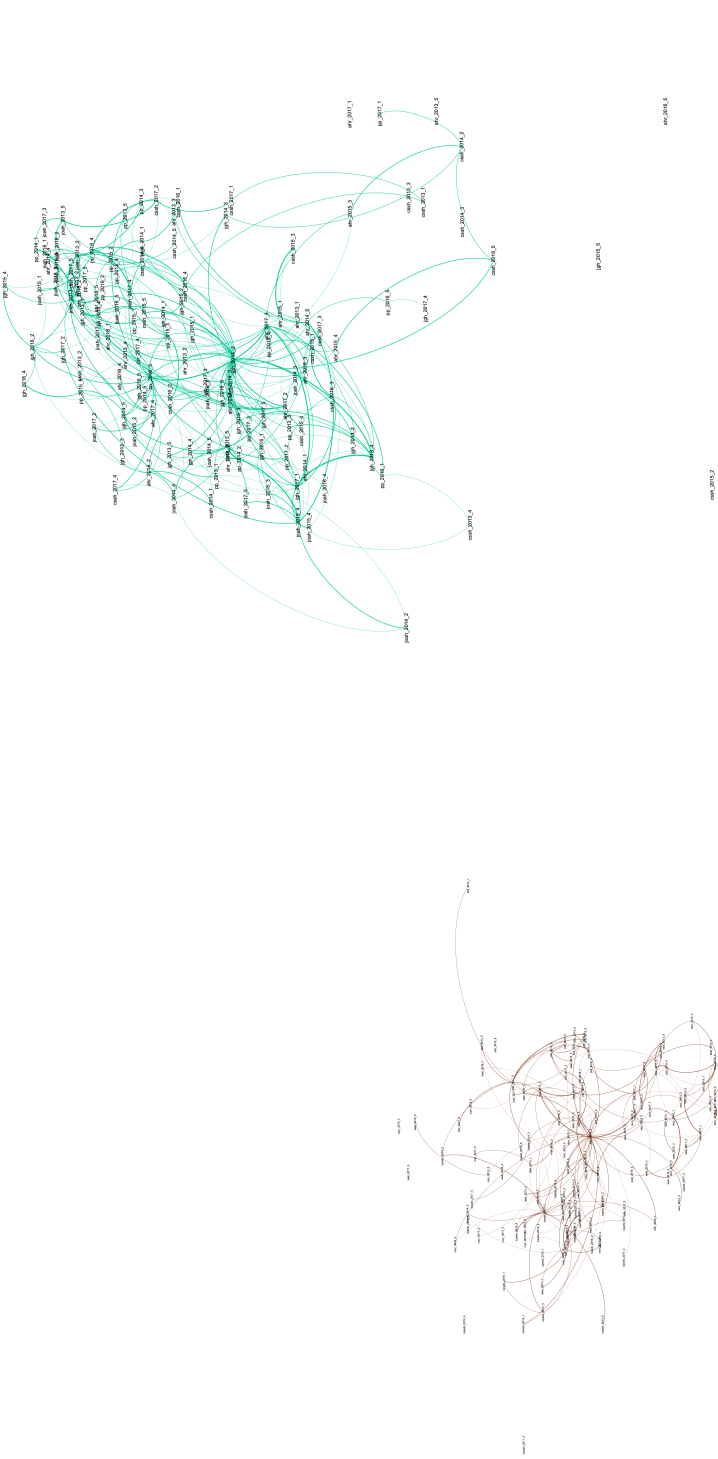
This chapter takes two of the disciplines from the last chapter – computer science and history – and elaborates the frameworks built for each by investigating what makes up the intradisciplinary variation in each discipline. What kinds of new insights can we gain about the different kinds of rhetorical work being performed within computer science and history as disciplines, and how is that work being instantiated in recurring grammatical and linguistic form? Progressing in to this fine level of detail also creates a bit of potential “fuzz”: are there really that many distinguishable rhetorical moves within the same discipline? How much of the variation is more likely to be a product of topic, or even just random? We will see in this chapter that while investigating differences in function words at this level of granularity does produce some dimensions of linguistic and rhetorical variation, the words under investigation also start to devolve into less meaningful patterns in terms of identifying

broad rhetorical moves across the discipline. A final question guiding this chapter, then, is how does this movement towards a very granular view of the disciplines change our understandings and method of analysis?

6.1 Function words and intradisciplinary variation

Within computer science and history as disciplines, there is a fair amount of intradisciplinary variation (see Figure 6.1): we see the disciplines spanning in many different directions, indicating several different dimensions of variation in function word usage. Tables 6.1 and 6.2 illustrate the top four function words per discipline that contribute most to the span within each discipline. More specifically, the tables show which words have the highest total sum of Delta distances for every possible pair of disciplinary texts. Tables 6.3 and 6.4 also show the correlations for each of these four words, in order to test if any of the words seem to be appearing together as part of the same rhetorical effect. In practice, only two words (*every* and *such that* in computer science) had any noticeable correlation, which suggests that all except these two seem to be pointing to somewhat different dimensions of variation.

The analysis in this section details the usage of the words in these tables and the roles they play in the rhetorical action of each discipline's span of arguments, explaining which words seem to indicate some dimension of rhetorical variation and which words seem to be fuzzier. To offer a brief synopsis of that analysis, one of the stronger dimensions of variation in computer science relates to a distinction between papers that focus on building theorems versus systems, where *every* and *such that* are heavily tied to proofs and theorem-related work. *Next* and *our* become fuzzier in interpretation and value for rhetorical move identification, but point to different kinds of metalanguage that are tied very loosely to the theorem/system distinction. In history, one of the stronger dimensions of variation relates to the focus of the narrative in storytelling, where the relative presence of *had* represents a narrower focus in topic. *Despite* points to another dimension of variation that has its roots in the analytical subject of the paper, as it allows for analyzing the complexity of people's thoughts as opposed to analyzing the complexity of more abstract entities or actions. *Against* and *anyone* both are primarily content-based distinctions that relate to simply what the paper is



(a) A visualization of the distances between 125 academic articles from computer science, with distances calculated using Delta (Burrows, 2002). Every point represents one unique article, and thicker lines and smaller distances indicate that the texts are more closely related in terms of their distributions of function word frequencies (for more detail, see Eder (2017)).

(b) A visualization of the distances between 125 academic articles from history, with distances calculated using Delta (Burrows, 2002). Every point represents one unique article, and thicker lines and smaller distances indicate that the texts are more closely related in terms of their distributions of function word frequencies (for more detail, see Eder (2017)).

Figure 6.1: Cluster visualization of distances within disciplines

	Delta dist.	Median freq.	St. dev.	Highest freq.	Lowest freq.	N
<i>every</i>	80.7	4.17	10.26	51.82	0.00	809
<i>such that</i>	66.1	1.99	8.89	47.71	0.00	592
<i>our</i>	57.3	31.84	19.48	24.83	0.00	3,096
<i>next</i>	56.4	3.38	3.44	4.11	0.00	381

Table 6.1: The top 4 function words that contribute most to the span in computer science. The Delta distance column shows the sum of the Delta distances between each pair of computer science texts for that particular function word (using the modified Delta formula described in chapter 3). The table also shows the following: the median normalized frequency (out of 10,000 words) for each word; the standard deviation of the normalized frequency (out of 10,000 words) for each word; the highest normalized frequency (out of 10,000 words) of one text in the discipline's subcorpus; the lowest normalized frequency (out of 10,000 words) of one text in the discipline's subcorpus; the total number of instances of the word across all 125 texts for that discipline (N).

	Delta dist.	Median freq.	St. dev.	Highest freq.	Lowest freq.	N
<i>had</i>	55.4	43.85	20.64	102.56	5.88	5,594
<i>despite</i>	49.5	2.77	2.83	13.13	0.00	404
<i>against</i>	48.8	6.60	6.12	27.54	0.00	1,018
<i>anyone</i>	46.7	0.00	0.96	5.08	0.00	64

Table 6.2: The top 4 function words that contribute most to the span in history. The Delta distance column shows the sum of the Delta distances between each pair of history texts for that particular function word (using the modified Delta formula described in chapter 3). The table also shows the following: the median normalized frequency (out of 10,000 words) for each word; the standard deviation of the normalized frequency (out of 10,000 words) for each word; the highest normalized frequency (out of 10,000 words) of one text in the discipline's subcorpus; the lowest normalized frequency (out of 10,000 words) of one text in the discipline's subcorpus; the total number of instances of the word across all 125 texts for that discipline (N).

	<i>every</i>	<i>such that</i>	<i>our</i>	<i>next</i>
<i>every</i>	1.0000	0.4912	0.1397	0.2126
<i>such that</i>	0.4912	1.0000	0.0248	0.2928
<i>our</i>	0.1397	0.0248	1.0000	0.2179
<i>next</i>	0.2126	0.2928	0.2179	1.0000

Table 6.3: Correlation table for the top four function words in computer science, based on word frequency (out of 10,000 words), using Pearson’s R.

	<i>had</i>	<i>despite</i>	<i>against</i>	<i>anyone</i>
<i>had</i>	1.0000	0.1557	0.1885	0.2783
<i>despite</i>	0.1557	1.0000	0.1161	0.0797
<i>against</i>	0.1885	0.1161	1.0000	0.1557
<i>anyone</i>	0.2783	0.0797	0.1557	1.0000

Table 6.4: Correlation table for the top four function words in history, based on word frequency (out of 10,000 words), using Pearson’s R.

about, rather than any noticeable dimension of rhetorical variation. Through both of these disciplines’ writing then, we see some insight into dimensions of rhetorical variation while also witnessing how the insight afforded by this particular methodology begins to break down at this level of granularity.

6.1.1 Computer Science

One of the strongest identifiable dimensions of rhetorical variation in computer science papers is a distinction between theorem-building and system-building as the activity taking up the primary volume of the text. Summaries of two sample papers can help to illustrate this distinction effectively: in one paper published in a conference on the theory of computing, the authors start by explaining that answering lots of counting queries in a database while maintaining differential privacy is quite difficult (Ullman, 2013). They then spend the vast majority of the paper proving that there is no efficient algorithm that can return a large number of queries while maintaining differential privacy, which is the main contribution of the paper. In another paper published in a conference on computer vision, the authors present a new approach for unsupervised learning of visual features in order to be able

to identify objects in a picture (Pathak et al., 2017). They develop an approach that uses video to train a network, and spend the majority of the paper explaining the structure of and evaluating their approach. In the first computing theory paper, the paper primarily works toward proving a theorem, and the bulk of the paper is spent constructing intricate mathematical proofs. By contrast, in the second computer vision paper, the paper primarily works toward building and evaluating a system, and the bulk of the paper is spent constructing the details of this system and how it represents and works with inputs.

These two sample papers showcase two fundamental activities in computer science (while the activities appear in separate papers in these examples, they are not necessarily exclusive in other papers though they may have different levels of focus attributed to them), and those activities take place through different linguistic resources. The relative presence (or absence) of these moves is indicated in part by the words *every* and *such that*, as two of the words that contribute to the span within computer science and also happen to correlate reasonably well with each other per Tables 6.1 and 6.3. *Every*, in particular, very frequently occurs in definitions and lemmas, which are building blocks of the larger argument toward proving a theorem. For example, in the theory of computing paper on counting queries in a database, the authors provide the following definition of a traitor-tracing scheme, which helps them to establish their final result:

- (1) DEFINITION 3.1. *Let Π_{TT} be an (n, k_{TT}) -traitor-tracing scheme. Let $\mathcal{P}$ be a (possibly randomized) algorithm. We say that $\mathcal{P}$ is a (k_{TT}) -available pirate decoder if for every $\kappa \in \mathbb{N}$, every set of user keys $\vec{sk} = (sk^{(1)}, \dots, sk^{(n)}) \in 0, 1^\kappa$, every $S \subseteq [n]$ such that $|S| \geq n - 1$, and every $c_1, \dots, c_{k_{\text{TT}}} \in \mathbb{C}^{(\kappa)}$, when $(\hat{b}_1, \dots, \hat{b}_{k_{\text{TT}}}) \leftarrow_R \mathcal{P}(\vec{sk}_S, c_1, \dots, c_{k_{\text{TT}}})$*

$$\Pr \left[\begin{array}{c} \exists j \in [k_{\text{TT}}], b \in \{0, 1\} \\ (\forall i \in S, \text{Dec}_{\text{TT}}(sk^{(i)}, c_j) = b) \wedge (\hat{b}_j \neq b) \end{array} \right] \leq o\left(\frac{1}{n(\kappa)^2}\right)$$

In other words, if every user key $sk^{(i)}$ (for $i \in S$) decrypts c to 1 (resp. 0), then $\mathcal{P}(\vec{sk}_S, \cdot)$ decrypts c to 1 (resp. 0), with high probability. (Ullman, 2013, p. 365) [CS]

The first three instances of *every* are in parallel, forming a large list of conditions that must be met in order for the algorithm $\mathcal{P}$ to be a pirate decoder.

To restate this part of the definition in very simplified terms, “We can give this algorithm a certain name, or quality, if for every instance of certain variables, certain other things are true.”

A different theorem, later in the paper, also uses *every* to refer to multiple instances of something, but in slightly different form:

- (2) THEOREM 5.5. ([18]). *For every function $\mathcal{E}_{\text{FP}} : \mathbb{N} \rightarrow [0, 1]$, there exists a $(o(1/n^2), O(n^2 \log n))$ -finger-printing code.* (Ullman, 2013, p. 368) [CS]

The nature of the statement in (2) is different from (1) in that the former is stating something that is true for *every* function, rather than using *every* instance of some variable to satisfy conditions for another statement. In both cases, *every* allows the author to reason about every instance of something that occurs in multiples (of all the sentences with *every* in computer science, 56% take also have a *for* in the sentence, thus taking a similar form to what is seen in the examples above). This kind of construction allows the authors to reason about things of which there are many, which becomes important in the context of computing where there are many possible inputs or scenarios impacting the same system. The choice to use *every* rather than *all* or *each* is again, somewhat interesting: *every* has the unique quality of allowing the author to refer to instances not as a uniform mass like *all* might, but without stressing the uniqueness of any given instance like *each* might. In this sense, it allows the author to recognize each instance as unique, but perhaps unique in uninteresting or unimportant ways. As we saw in chapter 5, this kind of reasoning about multiplicity is common in computer science, but we see here that it is actually uniquely positioned to aid in papers that are distinctly oriented around theorem-proving.

Example (1) also showcases an instance of *such that*, which also plays a role in proofs, theorems, and definitions. The example below is from an artificial intelligence paper that aims to prove some things about Stackelberg security games, which are common computational situations in many settings. This example comes just after a theorem in the “Main Results” part of the paper, and is the first part of the formal proof of that theorem:

- (3) *Proof.* Let there be $n = 2md$ targets, m resources, and a schedule set $\mathcal{D} = \{\mathcal{D} \subseteq T : |\mathcal{D}| = d\}$, such that any resource can defend any schedule. (Blum et al., 2014, p. 576) [CS]

Such that in this example allows a sort of world-building: the variables that are “let be” in the world are to be in such a way that “any resource can defend any schedule.” The grammatical structure of this sentence allows the authors to explain that what the variables being set in certain ways should accomplish, logically speaking. While *every* allows for references to multiplicity in logical structures, *such that* points to the building of the larger worlds that these authors are building with their definitions, lemmas, and proofs in order to show how their theorems are correct. These two words are closely associated with logical constructions then, even to the point that the presence of *such that* in a paper is closely correlated with the presence of conditional reasoning language at all (with a Pearson’s R value of 0.47).

By contrast, other types of papers in computer science – those low in *every* and *such that* – are engaged in a very different type of world-building, where they build up and evaluate systems. The computer vision paper mentioned earlier is a useful example of this type of rhetorical work. In this paper, the authors are describing a new approach to unsupervised feature learning. Take the below example, where they explain technically how they built their system, in particular explaining how they allowed for learning to segment with noisy labels:

- (4) As before, we feed the ConvNet cropped images, jittered in scale and translation, and ask it to predict the motile foreground object. Since the motion segmentation output is noisy, we do not trust the absolute foreground probabilities it provides. Instead, we convert it into a *trimap* representation in which pixels with a probability of <0.4 are marked as negative samples, those with a probability of >0.7 are marked as positives, and the remaining pixels are marked as “don’t cares” (in preliminary experiments, our results were found to be robust to these thresholds). (Pathak et al., 2017, p. 6029)

Neither *every* or *such that* occur in this example, and indeed, no proofs, definitions, or heavy mathematical sections occur in this paper at all. Instead, in order to fully describe the approach they have constructed, the authors write in the first-person about what they do with different bits of inputs in order to get to their final result. That is, rather than “letting be” certain variables, the authors here do various calculations and processes to the variables they work with as input. And rather than proving their theorem is correct with a formal mathematical proof, the authors in the computer vision paper include

an evaluation section where they benchmark their result against other state-of-the-art approaches to the same task. This section includes graphs and tables (Pathak et al., 2017, pp. 6029-6031), demonstrating how well their system performs.

In some sense, the computer vision paper is a more extreme example, including no formal mathematics as part of their paper. In a different paper which was also quite low on frequencies of *every* and *such that*, the authors propose a new technique to maximize computer systems’ utility and fairness when distributing tasks across multiple cores, which they call FRAME (Mohtasham et al., 2015). This paper takes a roughly similar form to the computer vision paper in describing how the system is built and evaluating it, but they describe the system more formally in ways that includes large mathematical sections. Even still, the paper has just one instance of *every* and none of *such that*, as seen in this example where the authors rephrase a key optimization problem of their work in mathematical terms:

- (5) The goal of the resource allocation is to maximize the system’s overall utility. Since the maximization should be done over the feasible region, which is characterized by (4) and (3), the optimization problem can be formulated as:

$$\begin{aligned} & \underset{\vec{\tau}}{\text{maximize}} \quad \prod_{i=1}^{\mathcal{P}} \mathcal{U}_i(\tau_i) \\ & \text{subject to} \quad \vec{1}^\top \vec{\tau} \leq \mathcal{C} \text{ and } \vec{\tau} \leq \vec{\mathcal{M}}, \end{aligned}$$

where $\mathcal{U}_i$ denotes the i th process’s utility function. (Mohtasham et al., 2015, p. 603) [CS]

In example (5), we see the authors making a move to represent the problem in mathematical terms, but not in the same way that earlier authors represented their problem in mathematical terms through formalized definitions – the authors here simply state how the problem “can be formulated” rather than defining certain things explicitly with phrases like “Let there be...” as seen in (3). The fundamental goals behind the papers are different, which leads to a difference in this kind of definitional work: the authors of this FRAME paper are looking primarily to problem-solve. That is, they know of a problem about distributing tasks across cores, represent it mathematically (which is what we see happening in (5)), and approach the problem in a specific way through that is unique and the main contribution of the paper.

By contrast, the authors of the theorem-proving papers built their own logical worlds, letting certain variables “be” definitionally, and proving various things about the multiples of variables that passed through those definitions. *Every* and *such that* are ultimately signals of moves to establish relationships between variables as part of definitional work for how various things entail each other and relate. In their absence, we see instead papers which might engage mathematically with their problems, but are using math simply to represent the problem as part of their own system that does processes on those problems, instead of building logical worlds where defining how every variable is engaged with in certain ways becomes important. While previous literature has established that definitional work is important and somewhat varied in linguistic form in computer science (e.g., Triki (2019)), we see a distinction here in the functions of definitions as related to different fundamental rhetorical activities in the discipline.

While *every* and *such that* point to distinctions in theorem- versus system-building in computer science as one dimension of linguistic variation in the discipline, another dimension occurs in the types of metalanguage that papers engage in. *Next* and *our* both represent primarily forms of metalanguage that roughly map onto the theorem- and system-building dimension, but they also show where the insight afforded by this close level of data interpretation becomes fuzzier and perhaps less meaningful. *Next* has overall fairly low frequency in papers (its median occurrence per 10,000 words is only 3.38), but of the times it is used, it tends to be primarily in metalanguage. Almost two-thirds (64%) of *next* instances in the entire computer science corpus are in the adjective form, with the most common nouns it modifies being “section” (17% of the total nouns modified by *next*), “lemma” (10%), and “step” (8%). The artificial intelligence paper with the proof about Stackelberg games from (3), for example, contains the following two uses of *next* as metalanguage:

- (6) The next lemmas are required for proving Theorem 1. (Blum et al., 2014, p. 575) [CS]
- (7) We next show that when m and d are considerably larger than k , there is a security game in which the approximation gap between playing S_{∞}^* and S_{∞}^k is constant. (Blum et al., 2014, p. 577) [CS]

Both instances in this paper serve as metalanguage, telling the reader what will occur *next* in the paper or how it fits within the larger purpose of the text. This kind of metalanguage is especially helpful for very mathematically-

oriented papers, where it may not be readily apparent to the reader how one piece of math relates to the next. Take as contrast this transition to a new subsection from a paper where the authors present a new framework for computing and data management that enables high-power computing, but which handles specialized tasks automatically and thus does not require certain specialized skills from users (and which has only one instance of *next*, in a figure):

- (8) Despite all the benefits of cloud computing, failure while setting up a cloud-based execution environment or during the execution itself is arguably inevitable:

- some or all of the requested VMs may not be successfully created/instantiated;
- communicating with an existing VM may fail (due to either permanent or intermittent network failure, or VM failure);

An SC at most recovers a failed execution, at least prevents the failed execution from causing a failure in the entire system.

A. Failed VM instantiation

Of n required VMs, only m are created, where $m < n$. The SC... (Spichkova et al., 2015, p. 400) [CS]

In this example, rather than writing explicitly what will come next and the purpose it will serve in the paper, the writers can lean on repetition from the first bullet point and section heading to create coherence. This type of coherence is more difficult to create when the majority of the paper takes the forms of logic, numbers, and equations; despite this, *next* is only one of many possible instances of metalanguage that create coherence. Notably, we do not see other cohesion-based metalanguage terms like “thus” or “following” in the top words that create variation in computer science texts, indicating that papers do, by and large, all contain roughly similar types of cohesive metalanguage, with the exception of *next* which has a unique role in creating cohesion for mathematically-oriented papers. In this sense, *next* points to some degree of variation in the texts, but does not necessarily reveal significantly different larger rhetorical moves.

Finally, *our* is another metalanguage marker used frequently in system-

building types of papers because it allows the authors to refer to their own system when evaluating it. Out of all of the instances of *our* in computer science, the head nouns of the noun phrases it precedes are overwhelmingly academic terms like “approach” (7%), “method” (6%), “algorithm” (5%), and “results” (3%). For example, in the evaluation section of the computer vision paper where they create and test a new approach for unsupervised learning of visual features, the authors write the following:

- (9) The quality of our representation (measured by Fast R-CNN performance on VOC 2007 with all conv layers frozen) grows roughly logarithmically with the number of frames used. With 396K frames (50K videos), it is already better than prior state-of-the-art [8] trained on a million ImageNet images, see Figure 8(b). With our full dataset (1.6M frames) accuracy increases substantially. If this logarithmic growth continues, our representation will be on par with one trained on ImageNet if we use about 27M frames (or 3 to 5 million videos, the same order of magnitude as the number of images in ImageNet). (Pathak et al., 2017, pp. 6030-6031) [CS]

The authors use *our* as a modifier here to refer back to their approach, specifically to distinguish it in comparison to other systems that they are evaluating it against. In this sense, it occurs primarily in papers where evaluation is present. However, the ties of *our* to evaluation are thin. Notably, there are other papers with evaluation sections that simply use other techniques to distinguish their approach from others. Take this example from a paper which proposes a new approach to parallel traffic simulation (called, the “Elliptically-shaped Workload Estimation Algorithm (ES)”) and evaluates it in comparison to another approach called the “LL algorithm”:

- (10) By using the OD Matrix, the ES algorithm can reduce the Sum Error by 36%, compared with the LL algorithm. (Xu & Tan, 2013, p. 429) [CS]

In this example, the authors simply use the name of their new algorithm (and of the one they compare it to), rather than modifying it with *our*. While there is a slight distinction in how the authors from (9) and (10) refer to their work that creates slight differences in shades of meaning, the overall effect as tied to evaluation is quite similar. In this sense, then, *our* is primarily tied to evaluation as a rhetorical move in computer science papers, but is not a

unique indicator of the activity.

6.1.2 History

Moving to history writing, which engages heavily in narrative and story-telling, one of the most identifiable dimensions of variation relates to the focus of the narrative. In chapter 5, it was explained that *had* is used in history writing in order to tell events out of order, given that it allows for grammatically creating the past present form. While that is primarily true (*had* takes the past perfect form 72% of the time in history writing), this dimension of variation relates to *when* telling history out of order is advantageous. This brief passage is taken from a history text that explains how a UN relief group, UNRRA, was established just after World War II, and how the group wound up being at the crossroads of differing views of the world that were developing at this time. This particular passage describes how UNRRA grew out of and had both continuities and differences with past relief work after World War I:

- (11) Undoubtedly there were continuities in the relief project, as Keeny observed, but the world had also changed since the end of the First World War. Many commentators thought it had shrunk as a result of radical improvements in communications and transport. Arthur Salter, Deputy General of UNRRA, had observed already in 1933 that cheap sea transport, motor-cars, aeroplanes, telegraphs, telephones, and wireless had changed the political geography of the world. (Reinisch, 2013, p. 76) [History]

This passage looks similar to the examples seen in chapter 5, where *had* creates the past present form in order to give contextual information. In this paper, the author is focused primarily on the development of UNRRA over time, explaining how relief workers' and foreign policy officials' worldviews changed their perception of the group over time. In that sense, the primary narrative focuses around UNRRA, and statements about UNRRA appear primarily in the simple past form (for example, "there were continuities in the relief project," at the beginning of example (11)). By contrast, when the author wants to bring in events that happened before this primary narrative about UNRRA, these must be placed in a more distant past, and thus use the past present form with *had*, as we see here.

By contrast, other papers in history are not as narrowly focused on one particular entity as the main character of their narrative, and thus background information is weaved into the narrative in different ways. In one paper that compares protests in Cairo and Damascus, focusing specifically on the role of urban subalterns in revolutionary action, the author presents background information about Cairo as follows:

- (12) Army soldiers, volunteers, and conscripts were settled in the city's periphery. This pattern of settlement was tied to inhabitants' relations with the regime. Thus, the quarter of Mezza, built into the rocky terrain of Mezza Mountain, emerged in the mid-1970s for the purpose of settling the members of Rif'at al-Asad's "86 Division." Its inhabitants were low-level army volunteers (*saf dabit*), below officer rank. (Ismail, 2013, p. 882) [History]

This paper describes protests in Cairo and Damascus, but that comparison is ultimately part of a larger analysis about the role that urban subalterns play in revolutions. As such, the paper is not wholly organized chronologically around the events of these protests and thus providing background information that might appear in past perfect; rather, the paper is organized around analytical concepts in these protests, like housing quarters and subaltern communities. Because there is no prominent past-tense narrative that runs throughout the whole paper (as the development of UNRRA does in example (11)), contextual information can also be stated in the simple past present as part of a larger analysis of the situation. That is, the use of *had* and the past present is primarily dependent on the presence of an overarching focused narrative, while its absence is more closely tied to a paper organized around analytical concepts that can thus depict the majority of events in the simple past without needing to focus on how they relate to each other chronologically. This explains why the frequency of *had* in history is positively correlated with narrative and character language, and negatively correlated with descriptive language that depicts states of being (such as the location of soldiers and settlements in (12)).¹ Stockton (1995) explains that history papers deeply engage with narrative complexity and that the validity of their historical progression depends on the narrative structure and evi-

¹*Had* is positively correlated with narrative past (0.41), narrative verbs (0.33), and character pronouns (0.30) language. It is negatively correlated with information about states (-0.37), general academic (-0.34), and descriptions of activity (-0.32) language.

dence, and we see in this distinction how that narrative structure can take different forms depending on focus, while still creating complexity in both cases.

Apart from narrative focus, another dimension of rhetorical variation in history papers has roots in the analytical subject. *Despite* is a contrastive that allows authors to set up actions that might seem in conflict with each other, but is uniquely tied to analyzing how people act in ways that seem contradictory to their thoughts or other actions. Cortes (2004) explains that contrastives are generally quite important in history writing, and indeed we see that with *despite* in how it allows for analytical complexity of characters' actions and thoughts. In one paper about how people during the American Revolution had ambiguous, shifting, and sometimes disingenuous allegiances during the revolution, the author describes how one pro-revolution woman ultimately married a Loyalist man with little trouble:

- (13) A few months after condemning predatory British officers, [Hannah] Lawrence married Jacob Schieffelin, an American officer serving in the king's army. Despite her earlier strong feelings against the British cause, Lawrence apparently felt few qualms about marrying into the occupying force. (Johnson, 2017, p. 616) [History]

Despite here creates a contrast, indicating the things are in conflict: Hannah Lawrence seemed to feel one way, but act in another. A paper about Nixon's introduction of a new economic policy that controlled wages in an effort to curb inflation has a similar construction, explaining how a variety of forces prompted Nixon to act in ways that seemed contradictory to this thoughts:

- (14) By June 1971 a combination of economic and political factors had convinced Nixon, despite his ideological misgivings, to follow the advice of Federal Reserve chairman Arthur Burns and consider "a strong wage and price policy." (Waterhouse, 2013, p. 466) [History]

Similarly in this example about Nixon, the use of *despite* helps to set up the contradiction in his thoughts and actions. This type of move creates analytical complexity in these history papers, where the author can put actions and evidence together as part of the same story, indicating how complexities instantiated themselves in these actors in their particular historical contexts. Yet while *despite* seems to be primarily tied to developing contradictions and complexity when people are the focus of the analysis, that is not to say that

other texts do not do similar moves in other ways. For example, in the text on UNRRA, which has zero instances of *despite*, the author still creates a similar type of contrast:

- (15) In 1943, American policy-makers had argued that Americans aims were best served within a multilateral international relief organization. By spring 1947, however, they came to the conclusion that American agendas would only be fulfilled in specific and targeted aid projects. (Reinisch, 2013, p. 87) [History]

In this example, the two different arguments of American policy-makers are contrasted by the use of *however*. But, these two sentences could easily be rewritten to read, “By spring 1947, American policy-makers came to the conclusion that American agendas would only be fulfilled in specific and targeted aid projects, despite their arguments in 1943 that Americans aims were best served within a multilateral international relief organization” with little real change in meaning. In this sense, while *despite* may indicate a rhetorical move to contrast the thoughts and actions of actors, and this may be a unique move to certain papers and not others, the interpretability and utility of the explanation of *despite* is a bit fuzzy.

The distinctions made within history writing at this point become even noisier in the sense that they become primarily content-based, rather than being clearly tied to differing rhetorical strategies across the field. *Against*, in particular, is heavily tied to discussions of politics that create an “us versus them” narrative. In fact, frequencies of “against” are well correlated to language about negative acts of resistance (0.66), negative relations (0.42), aggressive negative acts (0.41), and discussions of the public (0.34). It appears in a discussion of the effects of Nixon’s new economic policy as the following:

- (16) In 1965 three senior vice presidents – from the General Electric Company, Macy’s, and the American Smelting and Refining Company – organized the Labor Law Study Committee, a collection of public affairs attorneys from sixty industrial companies, to fight against labor’s attempt to repeal the so-called right-to-work provision of the 1947 Labor Relations Act, better known as the Taft-Hartley Act. (Waterhouse, 2013, p. 462) [History]

And again in a discussion of subalternity in Damascus, and how some people

have been able to mobilize to protest the ruling regime:

- (17) In turn, the mobilization against the regime in quarters like Qaboun and Barzeh is driven by the people's experience of social, economic, and political exclusion. (Ismail, 2013, p. 890) [History]

In both of these cases, *against* is used to place actors in conflict with each other, particularly drawing on a war-like metaphor of relations. *Against*, notably, does not appear at all in a paper discussing investment practices and capital flow in slave trading (Harris, 2016) nor a paper that explains what ancient trading diasporas in the Indian ocean tell us about how networks operated in that time (Seland, 2013), neither of which focus in any substantial way on actors who are pitted against each other.

Anyone, too, seems to be primarily content-dependent. It occurs very infrequently even in the papers with the highest frequency in the corpus, and allows the authors to refer to a hypothetical member of the public, especially doing so to reference how some character in their narrative is discussing others. For example, in this paper on the importance of immigration acts in establishing identities of nation states, the author uses *anyone* as part of the description of Canada's immigration laws:

- (18) Fifty years before the Malaysian act, in 1919, Canada amended its already-detailed list of prohibited immigrants to include anyone who sought to overthrow the government 'by force or violence'. (Bashford, 2014, p. 43) [History]

The same UNRRA paper uses the word in a quote about relief workers regarding what they had been told when beginning their relief work abroad:

- (19) One group of UNRRA workers visiting Warsaw from Germany, she noted, were 'scared stiff of Poland after all the rumours they've heard. Our sense of humour getting the better of us we advised them to stop instantly if anyone challenged them, and to drop to the ground immediately if they heard shots ... after all, one of them had told us of his exploits in the Burma campaign.' (Reinisch, 2013, p. 83) [History]

In both of these examples, *anyone* is very heavily tied to the particularities of the content: it allows for reference to an abstract individual who might immigrate to Canada or attack relief workers in Poland. Notably, too, these

instances are tied to others' speech – Canada's laws or the aid workers' perceptions. In practice, *anyone* allows for discussions of hypothetical others as they appear in the imaginaries of characters in the history narratives these authors write, allowing the authors to reflect the thinking of their topics while still maintaining specificity in their own writing (even if their topics lack it). In this sense, *anyone* too is not a clear signal of rhetorical variation within history, but rather a product of topical differences.

6.2 Reflections on high-granularity variation

While the analyses in chapter 5 offered broad depictions of computer science and history as individual disciplines, each discipline also has some degree of intradisciplinary variation. The function words shown in Table 6.1 show a variation in computer science between theorem- and system-building as two fundamental activities of computer science knowledge production, while the function words shown in Table 6.2 show variations in history related to the focus of the analysis and the types of analytical rhetorical moves they afford. These dimensions of variation certainly do not conflict with the analyses offered in chapter 5, but rather offer nuance and new types of distinctions. Computer science in chapter 4 was largely represented as working with the multiplicity of logical systems, but in this chapter it becomes clear that there are distinctions in what that multiplicity looks like and the types of logical systems being engaged with. Here, computer science is not simply working through various multiples, but is doing so in service of particular proving theorems and describing and evaluating systems. Likewise, in chapter 5, history writing was represented as providing complex narratives about people. In this chapter's analysis, it becomes clear that the analysis only sometimes focuses explicitly on people (even if people are somehow present in all sorts of analyses), and that the relative scope and focus of a history narrative changes the type of rhetorical work needed to build that analytical story effectively. Both of these types of insight expand our understandings of the rhetorical action occurring in each discipline, for history adding nuance to the types of narrative complexity and contrastives that occur in history, and expanding what we know about computer science writing at all as a generally-understudied form of writing.

What do we learn from zooming so far into a corpus and looking at distinctions within one discipline? This analysis in this chapter operated in

a fundamentally different way than past chapters, in that it was not strictly comparative in nature. Chapters 4 and 5 directly compared two groups of texts given their distances, which were most clearly visible along one clear dimension each (horizontally in chapter 4, and vertically in chapter 5). The analysis in this chapter looked instead at a whole space of variation, rather than a clear dimension that moved in one direction. As such, each function word pushed at typically different dimensions of that variation, and as such the relationship between explanations of individual words became somewhat less cohesive. While *every* and *such that* correlated together to some degree and seemed to work toward a similar larger rhetorical goal, the rest of the words had looser connections. This type of analysis, then, led to many different narratives about the types of variation in these disciplines, none of which conflicted, but each of which offered different kinds of insight that were less closely connected than we saw with words in previous chapters. Thinking more broadly, this chapter demonstrates the value of viewing a corpus from a variety of different angles, and using different subcorpora to do so: by taking different slices and viewpoints of a corpus, the analysis can expand to show not simply different, but different kinds of rhetorical insights.

Additionally, the analysis in this chapter also demonstrated how interpretation began to become “fuzzy.” Working with function words in particular necessarily points to differences in grammatical structures of the texts. Very consistent differences in grammatical structure (as we saw in the first two analysis chapters) pointed relatively clearly toward sweeping differences in rhetorical moves across the disciplines as they related to representations of knowledge production in academic writing. That is, because a grammatical pattern was distinctive of an entire discipline, it seemed to have less to do with topic and more to do with fundamental rhetorical moves of the discipline’s knowledge-representation rhetorical strategies. However, in this chapter as we dove deeper within disciplines, some of the words became less rhetorically insightful: *against*, for example, seemed to be simply closely related to discussions of politics where actors are adversarial. It became clear, then, that as differences in grammatical patterns become smaller and less regular, they are more likely to be a product of similar topics that might occur across small chunks of disciplines, or perhaps even somewhat random. Ultimately, given the nature of computational text analysis, the computer is likely to be able to find some kind of differences between groups of texts, and perhaps even differences that appear (or, are) statistically significant. At this fine level of granularity, though, the differences become less interesting because they are

less closely tied to categories of interest (here, disciplines), and we sense that noise begins to take over. In computational work, when zooming in, there appears to be a law of diminishing returns – when the analysis becomes so zoomed in and fine-grained to the point that the differences no longer relate to the categories of interest, textual patterns may become less interesting for the research at hand. The scale of differences then, even if those differences are relatively regular or large in a small group, becomes an important factor when thinking about interpretability itself (certainly, a word that appears infrequently like *against* is more difficult to analyze) as well as the types of insights we can gain from interpreting computational analyses of texts.

Chapter 7

Conclusion

This dissertation has spanned a variety of issues and topics, from the grammatical intricacies of how *the* is used in the sciences to overarching interpretive frameworks for computational analyses of texts. In this final chapter, I want to offer some reflections on all of those pieces – starting from the most specific and working up to the broader issues of concern – as well as some thoughts on future work and applications.

At its most minute level of detail, this dissertation speaks to the power of function words and grammatical form in creating the rhetorical action of a text, particularly in regards to disciplinarity and genre. The majority of attention to function words in language studies falls within the field of linguistics. Indeed, the *Longman Grammar* (Biber et al., 1999) and other similar resources provide huge amounts of detail about the particularities of how certain function words are used in different grammatical forms. While there has been some work paying attention to the utility of function words in creating rhetorical effects (e.g., Pennebaker’s (2011) study on pronouns), relatively little is known about their ability to contribute to rhetorical effects in texts, which may explain why stylometric studies have had a difficult time explaining their findings (Burrows, 2002; Kestemont, 2014; Rybicki et al., 2016).

One of the more provocative results of the analyses in this dissertation is not simply that function words can distinguish disciplinary writing, but that the reasons they do so are identifiable and interesting. That is, it becomes clear how particular grammatical form undergirds specific rhetorical moves that are tied to disciplinary and generic rhetorical actions. In chapter 4, for

example, *the* as a definite article preceding noun phrases was shown to be an integral part of the heavy physical description that takes place in the sciences in order to describe the particularities of their experiments and creations. When Carter (2007) talks about disciplines as “ways of knowing and doing,” part of the way of knowing in the sciences is recounting the physical details of experiments or creations, which involves heavy use of particular and definite noun phrases that get preceded with *the*. Similarly, when Miller (1984) talks about genre as performing different kinds of “social action” in the world, we learn through studying academic writing as genres that the action of arguing for knowledge claims in the world involves, for the sciences, the repeated rhetorical move of physical description that is made up of many definite noun phrases. Function word frequency is a somewhat surprising way to get to these results, and this speaks to the hidden and still-undiscovered power of computational text analysis for rhetorical studies. Ultimately, the analyses in chapters 4 - 6 offer rhetorical theories that explain how function words facilitate the creation of meaning and knowledge representation at various levels of (meta)disciplines. Focusing specifically on linguistic form as tied to rhetorical action, especially the linguistic forms that are repeated and statistically distinguish groups of texts, helps us to better flesh out theoretical concepts like disciplinarity and genre by explaining how those broader types of social action are accomplished through linguistic resources.

The analyses in this dissertation also expand our understandings of how scholars argue for knowledge claims in disciplines in regards to the English for Academic purposes and Writing in the Disciplines communities. There has been a fair bit of research that makes explicit connections between language-level features and rhetorical action, especially in academic disciplinary writing. For example, MacDonald (1992) analyzes disciplinary texts to elucidate patterns in their ways of knowing, such as compactness versus diffuseness and explanatory versus interpretive writing. MacDonald’s (2010) later work even analyzes grammatical structures in disciplinary writing to make claims about how disciplines warrant their epistemological claims. In a focus on metadisciplines, Walsh (2010), too, identifies specific *topoi* common in STEM research articles that those authors use to talk about their work, giving specific examples of each. The analyses in this dissertation offer a view of disciplinary writing conventions focused on particular grammatical form, especially as constructed through function words. This type of low-level language analysis helps us to better articulate disciplinary writing conventions both at the level of rhetorical move, as well as how those moves actually appear in spe-

cific written form. In doing so, we can better understand conventions and purposes of disciplinary writing (seeing, for example, how the sciences value description through noun phrases as part of their evidentiary moves) as well as potentially articulate those conventions to students as part of our Writing in the Disciplines and English language learner pedagogies.

Methodologically speaking, this dissertation offers a sketch for interpreting results from stylometry methods in particular, which has historically been a particularly difficult task. While close statistical reading as a process can be applied more broadly, adapting a theoretical framework to a particular method is complex. The steps outlined in chapter 3 explained how traditional distance measurements could be modified in order to isolate which words were contributing the most to the overall distance between subcorpora. And while close statistical reading proposes using various other forms of data to build a more stable argument about texts, the analyses here show some particular other forms of data and measurements that are helpful for studying function words in particular. More specifically, drawing on information about grammatical function of particular words, using data from DocuScope about texts, and drawing on tools from spaCy for text analysis were all frequently used throughout the chapters to further investigate the usages of particular function words and better understand how they supported rhetorical action in these texts. Finally, the progression of the analysis chapters from metadisciplines to intradisciplinary variation demonstrated how to investigating the same corpus through different views leads to different kinds of insights, but also how “noise” can start to overtake “signal” as those views become increasingly paper-thin. The early chapters showed how function words were tied to larger rhetorical moves that were meaningful for understanding the social action that these texts performed, while some of the words in chapter 6 primarily pointed to minor differences in content that were less insightful for understanding how authors argued for their knowledge claims. As a whole, the progression of the chapters indicates how stylometric results might have differing scopes of meaning at differing levels of textual variation.

Finally, close statistical reading offers a new hermeneutic for interpreting the outputs of computational analyses of textual corpora that allows for defensible and insightful interpretation, especially for rhetorical analysis. One of the key challenges for computational text analysis work in the humanities has been answering a “so what?” question, or explaining what kinds of humanistic insights we gain from performing computational text analysis. Digital humanities has frequently grappled with this question, with some

arguing that, “[i]f text analysis is to participate in literary critical endeavor in some manner beyond fact-checking, it must endeavor to assist the critic in the unfolding of interpretive possibilities” (Ramsay, 2011, p. 10). Doing so has been a challenge, in part because it has been unclear what kinds of hermeneutics and methods of interpretation we need (Rockwell & Sinclair, 2015). While some other work in digital humanities spaces has begun to theorize how we can interpret computational models (e.g., Piper (2015)), close statistical reading offers a rhetorical, argumentative approach to this problem by offering an interpretive theory for how to read and analyze computational results that foregrounds the importance of synthesis and triangulation of data, as well as an argumentative strategy for justifying claims about textual corpora. In this sense, it draws upon a rhetorical hermeneutics approach (Mailloux, 1985, 1991), to offer a way of interpreting texts that is based upon how we can argue for those interpretations in a robust and defensible way. This theory has wide-reaching applicability in both literary and rhetorical studies for textual interpretation, as well as possible applicability in the area of interpretable machine learning more broadly.

Moving forward with this work, there are several exciting areas of future study. Pedagogically, it would be exciting to explore how these findings might be able to shape writing instruction. There are several lines of work that currently have been exploring the utility of computational text analysis to writing instruction in higher education (e.g., Aull (2015), Cotos (2011), Helberg et al. (2018)), but many of these applications are relatively recent and are still exploring how to best integrate knowledge about writing and disciplinary norms into the pedagogies that are rooted in tools. One potentially fruitful first place of intervention would be to integrate some of the distinctions made in function words’ rhetorical actions illustrated in this dissertation into dictionary-based text analysis tools like DocuScope (Kaufer & Ishizaki, 2020). This integration would help the analytical tools on which some of these pedagogical applications are based to be more sensitive to the nuances of grammatical form, which could then in turn shape analyses of student texts (as in Ringler et al. (2018)) and ultimately move into pedagogy.

Theoretically, there are also some interesting new directions for the theory of interpretation in rhetorical analysis, as well as computer science and statistics. Scholars working with natural language processing, machine learning, and artificial intelligence more broadly have in recent years struggled with the question of interpreting computational models (on arXiv, an open-access archive for research in some STEM areas, the search hits for “interpretable

machine learning” have nearly doubled every year from 2015 to 2020). This work offers an interdisciplinary step toward those challenges by drawing on the humanities’ expertise and long history with hermeneutics and textual interpretation. Moving forward, it will be useful to collaborate with computer scientists to continue to address these challenges and potentially expand close statistical reading to better accommodate more complex data analysis techniques. How can close statistical reading be adapted into a methodology for other types of analytical techniques?¹ While the methods drawn on in this dissertation were relatively straightforward mathematically (distance measurements in stylometry are not much more than a simple addition), the more complex models developed by things like neural networks might require different interpretive frameworks.

Finally, regarding rhetorical analysis, this dissertation offers a first step toward understanding the types of rhetorical insights we might gain through computational text analysis, but there are a host of further questions that might be asked. Exactly what all kinds of insight can we expect to potentially gain from computational text analysis, and what can we not? What types of rhetorical patterns can be identified through recurring patterns that can be identified automatically, and what types cannot? How might this type of analysis be shaped by feminist, critical race, queer, or other minority-based theories of rhetoric (indeed, D’Ignazio and Klein (2020) remind us that *who* does data analysis is a crucial question)? Rhetorical analysis is a varied and theoretically complex field, and more fully integrating computational text analysis into the methods of rhetorical studies will require careful thinking about the affordances and ethics of this form of text analysis. However, computational text analysis does offer a new exciting lens for rhetorical studies, and this dissertation provides a step toward learning to understand the outputs of that analysis fruitfully for rhetoric as a field.

¹Turning close statistical reading into a more full-fledged methodology is work currently being undertaken in an interdisciplinary project (Ringler & Argamon, in submission).

Appendix A

The table below includes a full listing of the five journals used for each discipline in constructing the corpus. Each journal was randomly sampled to choose five unique research articles for each year from 2013 to 2017. See chapter 3 for more detail on the selection criteria, and <https://github.com/hannahringler/function-words> for the entire metadata for the corpus.

Discipline	Journals
Cell & Molecular Biology	<i>Cell</i> <i>The EMBO Journal</i> <i>Journal of Cell Biology</i> <i>Molecular Cell</i> <i>Oncogene</i>
Chemical Engineering	<i>Applied Catalysis B: Environmental</i> <i>Chemical Engineering Journal</i> <i>Fuel</i> <i>Industrial & Engineering Chemistry Research</i> <i>Journal of Catalysis</i>
Computer Science	<i>AAAI Conference on Artificial Intelligence</i> <i>ACM SIGACT Symposium on Theory of Computing</i> <i>ACM SIGPLAN Conference on Programming Language</i> <i>Design and Implementation</i> <i>IEEE/CVF Conference on Computer Vision and</i> <i>Pattern Recognition</i> <i>IEEE International Conference on Parallel and</i> <i>Distributed Systems</i>
Education	<i>American Educational Research Journal</i> <i>Early Childhood Research Quarterly</i>

	<i>Educational Researcher</i> <i>Higher Education</i> <i>Teaching and Teacher Education</i>
History	<i>The American Historical Review</i> <i>Comparative Studies in Society and History</i> <i>The Journal of American History</i> <i>Journal of Global History</i> <i>Past & Present</i>
Linguistics	<i>Applied Linguistics</i> <i>Language</i> <i>Lingua</i> <i>Linguistic Inquiry</i> <i>Journal of Phonetics</i>
Literature & Literary Theory	<i>American Literary History</i> <i>English Literary History</i> <i>Modern Language Quarterly</i> <i>Narrative</i> <i>New Literary History</i>
Performing Arts	<i>Asian Theatre Journal</i> <i>Dance Research</i> <i>Dance Research Journal</i> <i>The Drama Review</i> <i>Theatre Journal</i>
Philosophy	<i>Erkenntnis</i> <i>Mind</i> <i>Noûs</i> <i>Philosophical Studies</i> <i>Philosophy and Phenomenological Research</i>
Physics	<i>Journal of Physics D: Applied Physics</i> <i>Philosophical Transactions of the Royal Society A</i> <i>Physical Review A</i> <i>Physical Review E</i> <i>Reviews of Modern Physics</i>
Political Science	<i>American Journal of Political Science</i> <i>American Political Science Review</i>

	<i>British Journal of Political Science</i> <i>The Journal of Politics</i> <i>World Politics</i>
Sociology	<i>American Journal of Sociology</i> <i>American Sociological Review</i> <i>Annual Review of Sociology</i> <i>European Sociological Review</i> <i>Journal of Marriage & Family</i>
Statistics	<i>Annals of Applied Statistics</i> <i>The Annals of Statistics</i> <i>Biometrika</i> <i>Journal of the American Statistical Association</i> <i>Journal of the Royal Statistics Society: Series B</i> <i>(Statistical Methodology)</i>

Appendix B

Below is the full list of the 211 function words used for the analysis. This list is derived from the *Longman Grammar* (Biber et al., 1999), and excludes personal pronouns given their close connection to subject matter.

a, a couple, a few, a little, a lot, a number, a good deal, a good many, a great deal, a great many, about, above, across, after, against, all, along, also, although, am, among, an, and, another, any, anybody, anyone, anything, are, around, as, as long as, aside, at, away, back, be, because, been, before, behind, being, below, beneath, beside, besides, between, beyond, both, but, but that, by, can, certain, concerning, could, despite, did, do, does, down, during, each, each other, eight, either, every, everybody, everyone, everything, except, except that, excepting, fewer, fewest, five, for, former, forth, four, from, had, has, have, home, how, however, if, in, in case, in front of, in order that, in spite, in that, inside, instead, into, is, it, its, itself, last, latter, least, less, lest, like, little, lots, many, may, might, more, most, much, must, near, neither, nevertheless, next, nine, no, no one, nobody, none, nor, not, nothing, now that, of, off, on, on the other hand, once, one, one another, onto, or, other, others, our, out, outside, over, past, plenty, rather than, regarding, round, same, save that, seven, several, shall, should, since, six, so, some, somebody, someone, something, such, such as, such that, ten, that, that is, that's, the, there, these, this, those, though, three, through, throughout, till, to, toward, towards, two, under, underneath, unless, until, up, upon, was, were, what, when, whenever, where, whereas, wherever, whether, which, while, whilst, who, whose, why, will, with, within, without, would, yet, zero

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